
Supplementary information

***peri*-Fused polyaromatic molecular contacts
for perovskite solar cells**

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Supplementary Materials for

Peri-fused polyaromatic molecular contacts for perovskite solar cells

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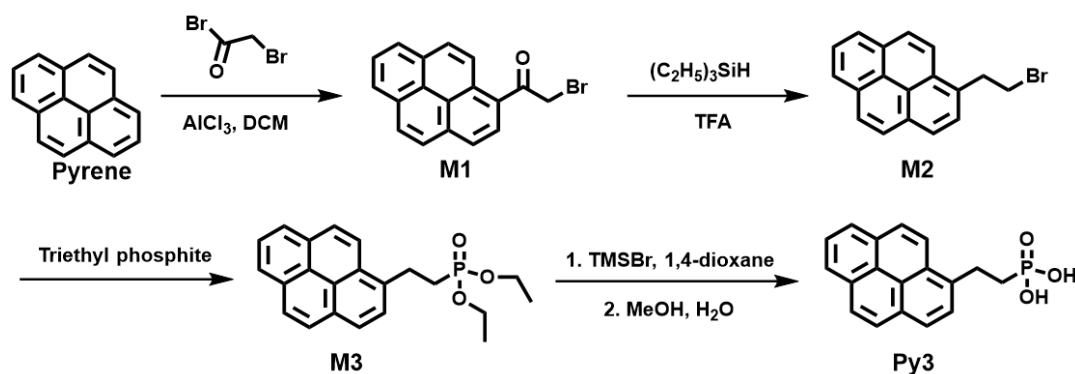
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Supplementary Text S1

Synthesis and characterizations of Py3 and its naphthalene-, anthracene-, and perylene-based variants

We also synthesized the naphthalene-, anthracene-, and perylene-based variants of Py3 and assessed their performances in the perovskite cells. The reagents and starting materials are commercially available and were used without any further purification. Compound pyrene (97%), perylene (95%), Pd(OAc)₂ (47.5% Pd), diethyl vinylphosphonate (97%), dichloromethane (DCM, Super dry, 99.9%), 1,4-dioxane (Super dry, 99.5%), triethylsilane (99%), bromotrimethylsilane (TMSBr, 98%), chloroform-*d* (CDCl₃, 99.8%) and dimethyl sulfoxide-*d*₆ (DMSO-*d*₆, 99%) were purchased from J&K Scientific. 1-Bromonaphthalene (98%), 9-bromoanthracene (98%) and aluminium trichloride (99%) were purchased from Energy Chemical. HCOOK (97%) and tri-*o*-tolylphosphine (P(*o*-tol)₃, 99%) were purchased from Bidepharm. Triethylamine (99%) and trifluoroacetic acid (TFA, 99%) were purchased from TCI. NaBH₄ (AR) and CoCl₂ (99%) were purchased from Sinopharm. Triethyl phosphite (98%), 2-bromoacetyl bromide (97%), and methanol (MeOH, 99.7%) were purchased from Innochem.

¹H NMR, ¹³C NMR and ³¹P NMR spectra were measured on Bruker AVANCE NEO 600 MHz and Bruker AVANCE NEO 500 MHz spectrometers. High resolution mass spectrometry (HRMS) was performed on a Waters I-Class Plus/Syanpt-XS spectrometer.



Scheme S1. Synthetic route of Py3¹⁻².

M1 Under argon atmosphere, pyrene (607 mg, 3.0 mmol), aluminum trichloride (480 mg, 3.6 mmol) and dichloromethane (10 mL) were added in Schlenk flask. The mixture cooled to 0 °C and 2-bromoacetyl bromide (666 mg, 3.3 mmol) was added dropwise. The mixture was warmed to room temperature and stirred for 4 h, then diluted with dichloromethane and washed with water. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain M1 (775 mg, 80%) as light-yellow powder.

^1H NMR (600 MHz, CDCl_3) δ 8.99 (d, $J = 9.3$ Hz, 1H), 8.37 (d, $J = 8.0$ Hz, 1H), 8.30 – 8.26 (m, 3H), 8.22 – 8.18 (m, 2H), 8.08 (t, $J = 7.9$ Hz, 2H), 4.73 (s, 2H).

^{13}C NMR (151 MHz, CDCl_3) δ 194.61, 131.00, 130.63, 130.24, 127.03, 126.78, 126.51, 124.61, 123.93, 34.16.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for ($\text{C}_{18}\text{H}_{12}\text{BrO}$): 323.0071, found 323.0067.

M2 Under argon atmosphere, **M1** (323 mg, 1.0 mmol) and trifluoroacetic acid (5 mL) were added in Schlenk flask. The mixture cooled to 0 °C and triethylsilane (244 mg, 2.1 mmol) was added dropwise. The mixture was warmed to room temperature and stirred overnight, then diluted with dichloromethane and washed with a saturated aqueous solution of sodium bicarbonate. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain **M2** (263 mg, 85%) as white powder.

^1H NMR (600 MHz, CDCl_3) δ 8.23 (d, $J = 9.2$ Hz, 1H), 8.20 – 8.18 (m, 2H), 8.16 – 8.12 (m, 2H), 8.07 – 7.99 (m, 3H), 7.89 (d, $J = 7.8$ Hz, 1H), 3.90 (t, $J = 8.0$ Hz, 2H), 3.77 (t, $J = 8.0$ Hz, 2H).

^{13}C NMR (151 MHz, CDCl_3) δ 132.69, 131.38, 130.79, 130.64, 128.83, 127.98, 127.62, 127.46, 127.25, 126.06, 125.31, 125.10, 124.92, 122.62, 37.21, 32.46.

M3 Under argon atmosphere, **M2** (309 mg, 1.0 mmol) was dissolved in triethyl phosphite (5 mL) in a Schlenk flask, and the mixture was heated at 160 °C for 12 h. After the reactant was cooled to room temperature, the organic solvent was removed by rotary evaporation. The crude product was purified by column chromatography to obtain **M3** (330 mg, 90%) as colorless oil.

^1H NMR (600 MHz, CDCl_3) δ 8.25 (d, $J = 9.2$ Hz, 1H), 8.17 (t, $J = 7.3$ Hz, 2H), 8.14 – 8.10 (m, 2H), 8.02 – 7.98 (m, 3H), 7.89 (d, $J = 7.8$ Hz, 1H), 4.20 – 4.12 (m, 4H), 3.67 – 3.61 (m, 2H), 2.31 – 2.24 (m, 2H), 1.36 (t, $J = 7.1$ Hz, 6H).

^{13}C NMR (151 MHz, CDCl_3) δ 135.09, 134.98, 131.38, 130.83, 130.24, 128.36, 127.77, 127.45, 126.93, 126.75, 125.95, 125.12, 124.95, 122.76, 61.73, 28.38, 27.47, 26.41, 16.60.

^{31}P NMR (243 MHz, CDCl_3) δ 30.56.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for ($\text{C}_{22}\text{H}_{24}\text{O}_3\text{P}$): 367.1463, found 367.1454.

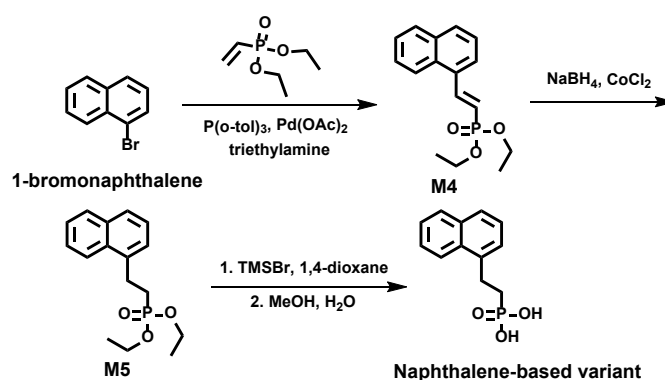
Py3 Under argon atmosphere, **M3** (366 mg, 1.0 mmol) was dissolved in dry 1,4-dioxane (10 mL) in a Schlenk flask and bromotrimethylsilane (1.53 g, 10.0 mmol) was added dropwise. The mixture was warmed to room temperature and stirred for 12 h at room temperature. The solvent was removed by rotary evaporation under reduced pressure to afford the intermediate. After the intermediate was dissolved in methanol (10 mL), distilled water (20 mL) was added dropwise to the solution until the solution became opaque which was then stirred for an additional 12 h. The solid was collected by filtration and washed with water to obtain **Py3** (280 mg, 90%) as white powder.

^1H NMR (600 MHz, $\text{DMSO-}d_6$) δ 8.32 – 8.21 (m, 5H), 8.17 – 8.12 (m, 2H), 8.07 (t, J = 7.6 Hz, 1H), 7.99 (d, J = 7.8 Hz, 1H), 3.58 – 3.49 (m, 2H), 2.08 – 1.98 (m, 2H).

^{13}C NMR (151 MHz, $\text{DMSO-}d_6$) δ 136.28, 130.75, 130.26, 129.35, 127.60, 127.45, 127.34, 126.92, 126.51, 126.08, 124.99, 124.84, 124.12, 122.73, 30.42, 29.54, 26.60.

^{31}P NMR (243 MHz, $\text{DMSO-}d_6$) δ 24.55.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for $(\text{C}_{18}\text{H}_{16}\text{O}_3\text{P})$: 311.0837, found 311.0829.



Scheme S2. Synthetic route of naphthalene-based variant³⁻⁴.

M4 Under argon atmosphere, 1-bromonaphthalene (618 mg, 3.0 mmol), Pd(OAc)_2 (34 mg, 5 mol%), P(o-tol)_3 (91 mg, 10 mol%), triethylamine (1.5 g, 15.0 mmol) and DMF (15 mL) were added in a Schlenk flask. The mixture was heated to 130 °C and diethyl vinylphosphonate (492 mg, 3.0 mmol) was added dropwise. The mixture was heated at 130 °C for 4 h, then cooled to room temperature, diluted with dichloromethane and washed with water. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain M5 (435 mg, 50%) as colorless oil.

^1H NMR (600 MHz, CDCl_3) δ 8.33 (dd, J = 22.5, 17.3 Hz, 1H), 8.19 (d, J = 8.4 Hz, 1H), 7.89 (t, J = 8.6 Hz, 2H), 7.73 (d, J = 7.2 Hz, 1H), 7.59 – 7.47 (m, 3H), 6.42 – 6.33 (m, 1H), 4.24 – 4.16 (m, 4H), 1.39 (t, J = 7.1 Hz, 6H).

^{13}C NMR (151 MHz, CDCl_3) δ 133.64, 132.58, 132.43, 131.10, 130.45, 128.71, 126.89, 126.26, 125.42, 124.74, 123.32, 117.89, 116.64, 61.97, 16.46.

^{31}P NMR (243 MHz, CDCl_3) δ 18.84.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for $(\text{C}_{16}\text{H}_{20}\text{O}_3\text{P})$: 291.1150, found 291.1153.

M5 Under argon atmosphere, M4 (580 mg, 2.0 mmol), CoCl_2 (273 mg, 2.1 mmol), tetrahydrofuran (10 mL) and methanol (10 mL) were added in a Schlenk flask and NaBH_4 (454 mg, 12.0 mmol) was added in portions at 0 °C. Afterwards, the mixture was warmed to room temperature and stirred for 24 h, then diluted with dichloromethane and washed with water. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary

evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain M5 (526 mg, 90%) as colorless oil.

^1H NMR (600 MHz, CDCl_3) δ 8.02 (d, $J = 8.4$ Hz, 1H), 7.86 (d, $J = 8.1$ Hz, 1H), 7.74 (d, $J = 8.1$ Hz, 1H), 7.54 (t, $J = 7.6$ Hz, 1H), 7.49 (t, $J = 7.4$ Hz, 1H), 7.40 (t, $J = 7.6$ Hz, 1H), 7.36 (d, $J = 6.9$ Hz, 1H), 4.18 – 4.11 (m, 4H), 3.38 (q, $J = 9.1$ Hz, 2H), 2.21 – 2.15 (m, 2H), 1.35 (t, $J = 7.1$ Hz, 6H).

^{13}C NMR (151 MHz, CDCl_3) δ 137.13, 137.02, 133.91, 131.36, 128.93, 127.25, 126.20, 125.69, 123.32, 61.65, 27.46, 26.54, 25.88, 16.50.

^{31}P NMR (243 MHz, CDCl_3) δ 30.62.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for $(\text{C}_{16}\text{H}_{22}\text{O}_3\text{P})$: 293.1306, found 293.1309.

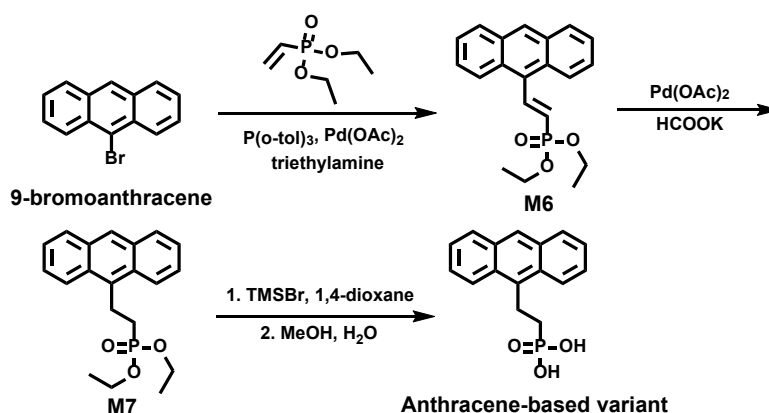
Naphthalene-based variant Under argon atmosphere, M5 (292 mg, 1.0 mmol) was dissolved in dry 1,4-dioxane (10 mL) in a Schlenk flask and bromotrimethylsilane (1.5 g, 10.0 mmol) was added dropwise at 0 °C. The mixture was warmed to room temperature and stirred for 12 h. The solvent was removed by rotary evaporation under reduced pressure to afford the intermediate. After the intermediate was dissolved in methanol (10 mL), distilled water (10 mL) was added dropwise until the solution became opaque which was then stirred for an additional 12 h. The solid was collected by filtration and washed with water to obtain naphthalene-based variant (201 mg, 85%) as white powder.

^1H NMR (600 MHz, $\text{DMSO}-d_6$) δ 8.01 (d, $J = 8.4$ Hz, 1H), 7.94 (d, $J = 7.3$ Hz, 1H), 7.79 (d, $J = 7.9$ Hz, 1H), 7.58 (t, $J = 7.6$ Hz, 1H), 7.53 (t, $J = 7.4$ Hz, 1H), 7.46 – 7.37 (m, 2H), 3.26 – 3.21 (m, 2H), 1.94 – 1.84 (m, 2H).

^{13}C NMR (151 MHz, $\text{DMSO}-d_6$) δ 138.39, 138.28, 133.95, 131.42, 129.23, 127.18, 126.67, 126.22, 126.16, 126.06, 123.64, 30.06, 29.18, 26.66.

^{31}P NMR (243 MHz, $\text{DMSO}-d_6$) δ 24.62.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for $(\text{C}_{12}\text{H}_{14}\text{O}_3\text{P})$: 237.0681, found 237.0685.



Scheme S3. Synthetic route of anthracene-based variant³⁻⁴.

M6 Under argon atmosphere, 9-bromoanthracene (771 mg, 3.0 mmol), Pd(OAc)₂ (34 mg, 5 mol%), P(o-tol)₃ (91 mg, 10 mol%), triethylamine (1.5 g, 15.0 mmol) and DMF (15 mL) were added in a Schlenk flask. The mixture was heated to 130 °C and diethyl vinylphosphonate (492 mg, 3.0 mmol) was added dropwise. The mixture was heated at 130 °C for 4 h, then cooled to room temperature, diluted with dichloromethane and washed with water. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain M6 (459 mg, 45%) as yellow powder.

¹H NMR (600 MHz, CDCl₃) δ 8.46 – 8.37 (m, 2H), 8.21 (d, *J* = 8.7 Hz, 2H), 8.02 (d, *J* = 7.6 Hz, 2H), 7.53 – 7.48 (m, 4H), 6.34 – 6.24 (m, 1H), 4.34 – 4.23 (m, 4H), 1.45 (t, *J* = 7.1 Hz, 6H).

¹³C NMR (151 MHz, CDCl₃) δ 146.00, 131.25, 128.97, 128.87, 128.12, 126.35, 125.39, 125.31, 125.07, 124.07, 62.10, 16.59.

³¹P NMR (243 MHz, CDCl₃) δ 17.20.

HRMS-ESI (*m/z*): [M+H]⁺ Calcd. for (C₂₀H₂₂O₃P): 341.1306, found 341.1308.

M7 Under argon atmosphere, M6 (681 mg, 2.0 mmol), Pd(OAc)₂ (45 mg, 10 mol%), HCOOK (841 mg, 10.0 mmol) and DMF (10 mL) were added in a Schlenk flask. The mixture was heated to 80 °C for 8 h, then cooled to room temperature, diluted with dichloromethane and washed with water. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain M7 (616 mg, 90%) as colorless oil.

¹H NMR (600 MHz, CDCl₃) δ 8.38 (s, 1H), 8.26 (d, *J* = 8.9 Hz, 2H), 8.02 (d, *J* = 8.3 Hz, 2H), 7.57 – 7.51 (m, 2H), 7.51 – 7.45 (m, 2H), 4.25 – 4.16 (m, 4H), 3.94 – 3.89 (m, 2H), 2.24 – 2.17 (m, 2H), 1.39 (t, *J* = 7.1 Hz, 6H).

¹³C NMR (151 MHz, CDCl₃) ¹³C NMR (151 MHz, Chloroform-*d*) δ 133.07, 132.95, 131.62, 129.36, 129.22, 126.42, 126.05, 125.00, 123.78, 61.80, 27.45, 26.54, 20.74, 16.56.

³¹P NMR (243 MHz, CDCl₃) δ 30.55.

HRMS-ESI (*m/z*): [M+H]⁺ Calcd. for (C₂₀H₂₄O₃P): 343.1463, found 343.1467.

Anthracene-based variant Under argon atmosphere, M7 (342 mg, 1.0 mmol) was dissolved in dry 1,4-dioxane (10 mL) in a Schlenk flask and bromotrimethylsilane (1.5 g, 10.0 mmol) was added dropwise at 0 °C. The mixture was warmed to room temperature and stirred for 12 h. The solvent was removed by rotary evaporation under reduced pressure to afford the intermediate. After the intermediate was dissolved in methanol (10 mL), distilled water (10 mL) was added dropwise until the solution became opaque which was then stirred for an additional 12 h. The solid was collected by filtration and washed with water to obtain anthracene-based variant (243 mg, 85%)

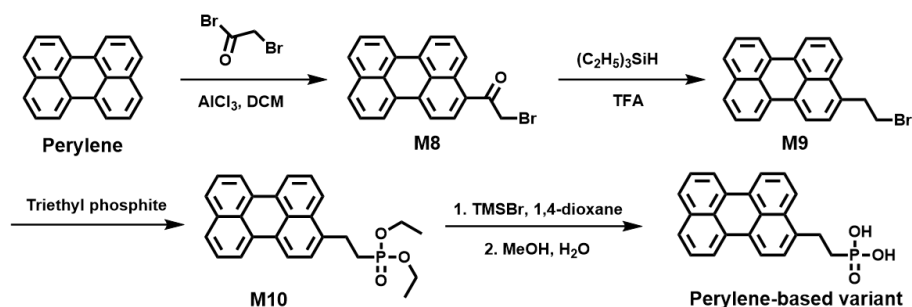
as white powder.

^1H NMR (600 MHz, $\text{DMSO-}d_6$) δ 8.51 (s, 1H), 8.26 (d, J = 8.7 Hz, 2H), 8.11 (d, J = 8.1 Hz, 2H), 7.64 – 7.50 (m, 4H), 3.83 – 3.72 (m, 2H), 1.92 (t, J = 17.3 Hz, 2H).

^{13}C NMR (151 MHz, $\text{DMSO-}d_6$) δ 133.93, 133.81, 131.06, 129.09, 128.53, 126.05, 125.69, 125.07, 123.56, 29.69, 28.82, 21.18.

^{31}P NMR (243 MHz, $\text{DMSO-}d_6$) δ 24.11.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for $(\text{C}_{16}\text{H}_{16}\text{O}_3\text{P})$: 287.0837, found 287.0838.



Scheme S4. Synthesis route for perylene-based variant¹⁻².

M8 Under argon atmosphere, perylene (994 mg, 3.0 mmol), aluminum trichloride (480 mg, 3.6 mmol) and dichloromethane (10 mL) were added in Schlenk flask. The mixture cooled to 0 °C and 2-bromoacetyl bromide (666 mg, 3.3 mmol) was added dropwise. The mixture was warmed to room temperature and stirred for 4 h, then diluted with dichloromethane and washed with water. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure. After concentration, the crude product M8 was used directly for the next step without purification.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for $(\text{C}_{22}\text{H}_{14}\text{BrO})$: 373.0228, found 373.0229.

M9 Under argon atmosphere, M8 (373 mg, 1.0 mmol) and trifluoroacetic acid (5 mL) were added in Schlenk flask. The mixture cooled to 0 °C and triethylsilane (244 mg, 2.1 mmol) was added dropwise. The mixture was warmed to room temperature and stirred overnight, then diluted with dichloromethane and washed with a saturated aqueous solution of sodium bicarbonate. The organic layer was collected and dried with anhydrous sodium sulfate. The dichloromethane was removed by rotary evaporation under reduced pressure, and the crude product was purified by column chromatography to obtain M9 (180 mg, 50%) as yellow powder.

^1H NMR (500 MHz, CDCl_3) δ 8.23 – 8.13 (m, 4H), 7.82 (d, J = 8.4 Hz, 1H), 7.69 (d, J = 8.2 Hz, 1H), 7.55 (t, J = 7.9 Hz, 1H), 7.51 – 7.45 (m, 1H), 7.39 (d, J = 7.7 Hz, 1H), 3.70 (t, J = 7.9 Hz, 2H), 3.58 (t, J = 7.9 Hz, 2H).

^{13}C NMR (126 MHz, CDCl_3) δ 134.59, 132.84, 132.08, 131.20, 131.11, 130.80, 129.23, 128.49, 128.00, 127.84, 126.91, 126.65, 122.95, 120.45, 120.32, 120.19, 119.96, 37.06, 31.87.

M10 Under argon atmosphere, M9 (359 mg, 1.0 mmol) was dissolved in triethyl phosphite (5 mL) in a Schlenk flask, and the mixture was heated at 160 °C for 12 h. After the reactant was cooled to room temperature, the organic solvent was removed by rotary evaporation. The crude product was purified by column chromatography to obtain M10 (330 mg, 90%) as yellow solid.

^1H NMR (500 MHz, CDCl_3) δ 8.21 – 8.12 (m, 3H), 8.08 (d, $J = 7.7$ Hz, 1H), 7.82 (d, $J = 8.4$ Hz, 1H), 7.69 – 7.63 (m, 2H), 7.52 (t, $J = 7.9$ Hz, 1H), 7.49 – 7.43 (m, 2H), 7.34 (d, $J = 7.7$ Hz, 1H), 4.20 – 4.12 (m, 4H), 3.33 – 3.28 (m, 2H), 2.23 – 2.13 (m, 2H), 1.36 (t, $J = 7.1$ Hz, 6H).

^{13}C NMR (126 MHz, CDCl_3) δ 136.87, 136.73, 134.63, 132.58, 131.96, 131.26, 130.21, 129.12, 128.46, 127.88, 127.63, 126.74, 126.56, 123.15, 120.31, 119.96, 61.69, 27.22, 26.12, 16.52.

^{31}P NMR (203 MHz, CDCl_3) δ 30.86.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for ($\text{C}_{26}\text{H}_{26}\text{O}_3\text{P}$): 417.1620, found 417.1629.

Perylene-based variant Under argon atmosphere, M10 (416 mg, 1.0 mmol) was dissolved in dry 1,4-dioxane (10 mL) in a Schlenk flask and bromotrimethylsilane (1.53 g, 10.0 mmol) was added dropwise. The mixture was warmed to room temperature and stirred for 12 h at room temperature. The solvent was removed by rotary evaporation under reduced pressure to afford the intermediate. After the intermediate was dissolved in methanol (10 mL), distilled water (20 mL) was added dropwise to the solution until the solution became opaque which was then stirred for an additional 12 h. The solid was collected by filtration and washed with water to obtain perylene-based variant (324 mg, 90%) as yellow-green powder.

^1H NMR (500 MHz, $\text{DMSO}-d_6$) δ 8.41 – 8.36 (m, 2H), 8.34 – 8.29 (m, 2H), 7.88 (d, $J = 8.3$ Hz, 1H), 7.82 – 7.74 (m, 2H), 7.62 (t, $J = 7.9$ Hz, 1H), 7.58 – 7.51 (m, 2H), 7.46 (d, $J = 7.6$ Hz, 1H), 3.22 – 3.17 (m, 2H), 1.99 – 1.87 (m, 2H).

^{13}C NMR (126 MHz, $\text{DMSO}-d_6$) δ 138.39, 134.73, 132.66, 131.64, 131.06, 129.53, 128.79, 128.42, 128.08, 127.48, 127.36, 127.17, 123.83, 121.22, 120.79, 29.69, 28.63, 26.89.

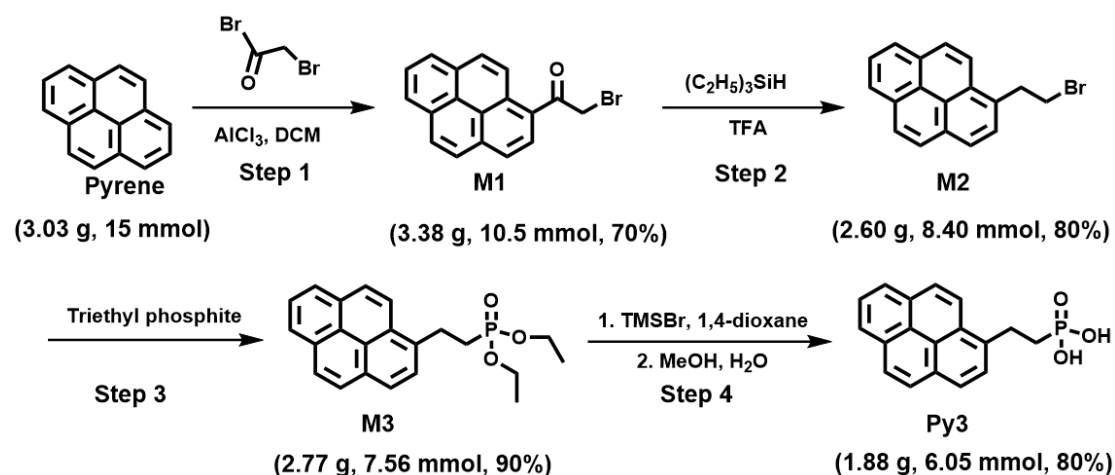
^{31}P NMR (203 MHz, $\text{DMSO}-d_6$) δ 24.64.

HRMS-ESI (m/z): $[\text{M}+\text{H}]^+$ Calcd. for ($\text{C}_{22}\text{H}_{18}\text{O}_3\text{P}$): 361.0449, found 361.1003.

Supplementary Text S2

Upscaled production of Py3 and cost analysis

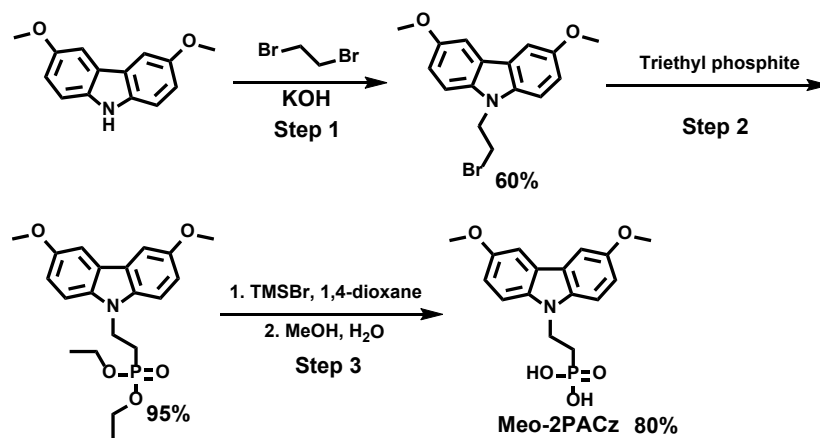
We assessed the scalability of the synthetic route in making high-purity Py3 by conducting a scaled-up production of Py3 in our lab. We found that the synthesis of Py3 can be effortlessly scaled up and the production yield of each step is shown in Scheme S5. The entire synthetic route does not involve heavy metal or noble metal catalysts, and the substrates are readily available and inexpensive. The detailed cost analysis is presented in Supplementary Table S1^{5,6}. The result indicates that the major cost in producing Py3 was caused by product purification. Based on the upscaled results, we calculated the cost of Py3 to be US\$7.2/g, approximately 2% of the cost of the commercial MeO-2PACz (US\$ 470/g, from TCI). It noteworthy that Py3 is a new chemical compound has not been previously synthesized or reported, to the extent that it does not even have a CAS number, and thereby its production has yet to be tested in a real industrial level. Once the structure becomes published recognized by commercial vendors and its production can be scaled up to a real industrial level, we anticipate that the cost of Py3 can be further reduced.



Scheme S5 Synthetic route of the upscaled production of Py3 with the production yield of each step presented.

Since the cost of MeO-2PACz shown above is sourced from commercial vendors and may exceed its actual production cost, we have also evaluated the cost of MeO-2PACz using the same calculation methodology employed for Py3. This may ensure a more equitable comparison of their respective costs. The synthetic route for MeO-2PACz was shown in Scheme S6⁷, in which the preparation of phosphonate in step 2 and the hydrolysis in step 3 are similar to those that are concerned in producing Py3. Therefore, the major cost difference between MeO-2PACz and Py3 preparation lies in their first and second steps. We conducted a cost analysis of the first step in MeO-2PACz preparation (Supplementary Table S2). Due to the relatively high price of the substrate 3,6-dimethoxy-9H-carbazole (US\$ 14.5/g, from Macklin), the estimated cost of 9-(2-

bromoethyl)-3,6-dimethoxy-9H-carbazole, the product of the first step in MeO-2PACz preparation, is US\$ 26.16/g, which is a significantly higher price compared to first and second step cost of Py3 (US\$ 4.0/g). Given the anticipated price drops once the economies of scale apply for Py3, we believe that Py3 would become a much more cost-effective option than MeO-2PACz.



Scheme S6 Synthetic route of MeO-2PACz.

Supplementary Text S3

Sub-1 cm⁻¹ High Resolution Broadband Sum Frequency Generation Vibrational Spectroscopy (HR-BB-SFG-VS)

This work employed a novel tabletop sub-1 cm⁻¹ high-resolution broadband sum-frequency generation vibrational spectroscopy (HR-BB-SFG-VS) system developed at Westlake University. This system synchronizes a 6.5 W, ~35 fs, 800 nm Ti: Sapphire femtosecond laser (Astrella, Coherent Co., Ltd., USA) with a 2 W, ~70 ps, 1064 nm picosecond laser⁸. The synchronization involves a master-slave configuration between their mode-locked oscillators (~80 MHz) using a phase lock electronics module (Spectra-Physics Lasers, model 3930). Amplification timing relies on independently delayed 1 kHz electronic trigger pulses.

The femtosecond laser output (2.6 mJ) pumps an optical parametric amplifier (OPA, TOPAS-Prime, Light Conversion) with a collinear AgGaS₂ difference frequency generation (DFG) stage, providing ~35 μ J/pulse at 3.3 μ m board band infrared pulse laser. The infrared (IR) beam is focused slightly above the sample surface with a plano-convex CaF₂ lens (80 mm focal length), yielding a spot size of ~300 μ m. The IR beam is p-polarized with respect to the sample surface and propagates at angle $\beta_{\text{IR}} = 55^\circ$.

The picosecond laser's energy (2 mJ) is doubled through an KDP crystal, resulting in 1.2 mJ of 532 nm pulse. The energy is attenuated to 60 μ J using a half-wave plate and a polarized beam splitter for HR-BB-SFG-VS experiments. A motorized delay stage (FMS120-300, Feinixs Inc.) adjusts timing between the 532 nm and IR pulses. The sum-frequency generation (SFG) beam is collected by an achromatic lens (200 mm focal length) and isolated from the 532 nm by notch filters.

The SFG signal is focused onto the spectrometer slit (open to 50 μ m) using an achromatic lens (80 mm focal length) on a translation stage. The spectra are recorded by a thermoelectric-cooled (-60 °C) CCD (ProEM, Princeton Instruments) after dispersion by a 1800 lines/mm grating (Acton SpectraPro SP-2750, Princeton Instruments). Spectra are averaged over 10 sites on the sample, each with a 1000-second exposure. Background spectra are obtained by blocking the femtosecond IR pulses during exposure. All spectra are normalized with the SFG signal from the top surface of a 10 mm thick z-cut α -quartz crystal under the same conditions.

We performed the SFG-VS measurement on the as-deposited Py3 layer. At an interface between two different materials or media, the inversion symmetry is broken, i.e. with ordered molecular arrangement, and an SFG-VS signal can be generated. Therefore, the SFG-VS technique is sensitive to an ordered monolayer surface. As shown in Supplementary Fig. 14, SFG-VS signals were observed in the sample, with the most evident sharp peak centered at around 3050 cm⁻¹, representing the C-H stretching mode in the pyrene ring. This verifies that the sample contained an ordered self-assembled monolayer at the interface, which justifies our DFT model. Furthermore, the intensity of this 3050 cm⁻¹ C-H peak is only about four times of the free O-H peak at the air/water interface⁹, which is one of the weakest signal of molecular monolayer interfaces. This

indicates that the thickness of the Py3 layer that contributes to the observed SFG-VS signal is most likely not more than a single monolayer thick.

Supplementary Text S4

DFT calculations

We performed the computational part of this study using the density functional theory (DFT) and first-principles calculations. The VASP package was used with a plane-wave basis set and the projected augmented wave (PAW) method^{10,11}. The Perdew-Burke-Ernzerhof (PBE) functional, which incorporates a generalized-gradient approximation for the exchange-correlation functional, was employed for both the self-consistent field calculations and geometry optimization¹². For the geometry optimizations, a $4 \times 4 \times 1$ Γ -centered k-mesh and a plane wave basis with an energy cutoff of 500 eV were utilized. To ensure relaxation, the atomic positions and cell volumes were adjusted using a conjugate gradient algorithm until all residual forces were below 0.01 eV/Å. For surface calculations involving both ITO and FAPbI₃, a surface/vacuum structure with a vacuum thickness of 15–20 Å were considered.

Interaction between the molecular layer and its neighboring layers:

We first calculated electrostatic potential surfaces (EPS) of 2PACz and Py3 molecules in the gas phase (Supplementary Fig. 15). Our DFT results indicated that phosphonate moieties of both systems are clearly holding rich negative potentials whereas the carbazole and pyrene groups appear to be at low potentials. The latter should be due to the delocalization of charges. Therefore, in order to model a strong and efficient molecule-ITO interaction, we selectively placed the phosphonate ends of both molecules on the ITO surfaces.

Regarding 2PACz/FAPbI₃ and Py3/FAPbI₃ interface models, we investigated the interaction energy between molecules and perovskite surface. The interaction energy between a molecule and the FAPbI₃ slab is calculated from

$$\Delta E_{int} = E_{ML-FAPbI_3} - (E_{ML} + E_{FAPbI_3})$$

where $E_{ML-FAPbI_3}$, E_{ML} and E_{FAPbI_3} are the electronic energies of the whole molecular layer and perovskite complex, molecular layer only and laterally periodic FAPbI₃ slab only, respectively. From the relaxed geometries, the molecule-surface distances are calculated from the average of multiple nearest hydrogens of the molecular layer normal to the surface in the supercell and the surface plane formed by the upmost perovskite slab sites.

The DFT calculated interaction between Py3 and perovskite is found to be stronger (-1.97 eV) than that of 2PACz (-1.30 eV). However, both interactions are still weak (Supplementary Fig. 16). Based on these DFT results, we anticipated a physical contact between these molecular layer and perovskite, rather than a chemical bonding. We also calculated the distance between the Pb-I surface of perovskite and the hydrogens of carbazole of 2PACz and pyrene of Py3, respectively. Since several hydrogens are in

contact with the surface, we obtained an average distance of 3.52 Å and 3.38 Å for 2PACz and Py3, respectively. The relatively large distances suggest a very weak interaction between these molecules and perovskite surface. Besides, it is also evident from EPS maps in Supplementary Fig. 15 that the carbazole and pyrene groups are at low potential.

To experimentally verify the above-mentioned theoretical results from DFT calculations, we used FTIR to study the interaction between the pyrene ring of Py3 and the perovskite, either in a manner of precursor or full lattice. Supplementary Fig. 17 shows the FTIR spectra of pure FAI, pyrene, and pyrene with FAI. The characteristic peaks in pyrene and FAI did not show any noticeable shifts upon their interaction. The pyrene-perovskite sample displayed peak patterns that were exactly a superposition of the signals from pure FAI and pyrene, suggesting negligible interaction between pyrene in Py3 and FAI. The comparison of the FTIR spectra of pure PbI₂, pyrene, and pyrene combined with PbI₂ (Supplementary Fig. 18) likewise indicates minimal interaction between pyrene in Py3 and PbI₂, as evident from the absence of peak shifts. Consistent with the observations made in the case of perovskite precursors, we did not observe any noticeable peak shifts when testing the interaction between pyrene and perovskite (Supplementary Fig. 19). These results corroborated by the FTIR measurements confirm the relatively weak interaction of the perovskite layer with both the Py3 and 2PACz layer. This can also be evidenced by the structural characteristics of the conjugation units exposed on the surface of these molecular layers, both featuring benzene units that can hardly interact with perovskite beyond a relatively feeble physical contact. Therefore, the comparably weak interaction, and the similar contact mode between the perovskite layer and both the Py3 and 2PACz layers suggest that this interfacial interaction would not make a big difference in the carrier transport between these two cases. If anything, the shorter interaction distance in the case of Py3 could potentially enhance the carrier hopping across the interface.

Molecular stacking energy:

We utilized the optimized geometry of the molecule-ITO complex to extract the molecular stacking energy, $E_{stacking}$. We first calculated the total energy of the molecular layer system, E_{ML} , consisting of 4 molecules within the supercell. Then, we calculated the energy of isolated molecules, E_{mol} . The molecular stacking energy is calculated by the difference of the total energy of molecular layer system and the energies of the isolated molecules (Supplementary Fig. 20).

$$E_{stacking} = E_{ML} - 4E_{mol}$$

Work function:

We computed the impact of the molecular absorption on the work function of ITO using DFT. Our results presented below are based on the theory of molecular absorptions at the metal-organic systems. We first calculated planar averaged electrostatic potentials from

$$\bar{V}(z) = \frac{1}{A} \iint V(x, y, z) dx dy$$

and work function is calculated with $W = V(\infty) - E_F$; $V(\infty)$ is the asymptotic EP. The modification of W due to the molecular adsorption is quantified by work function change^{13,14}.

$$\Delta W = W_{ML-ITO} - W_{ITO}$$

where W_{ML-ITO} is the work function of molecule-ITO side and W_{ITO} of the ITO side. The appearance of ΔW stems from the charge accumulation (or depletion) to the ITO surface. Besides, the noticable (and realistic) “discontinuity” of asymptotic EP across the z direction (giving rise to non-zero ΔW) was computationally made by dipole corrections to total energy. Here, the dipole corrections are calculated from the energy difference between a dipole in the supercell and the same dipole in a supercell with the corresponding lattice vector approaching infinity. As presented in Supplementary Fig. 21, our DFT results showed in both cases the molecular adsorption modified work function of ITO, and the change in work function ΔW is determined to be 0.63 eV and 0.23 eV for 2PACz and Py3, respectively. The calculated results are consistent with our experimental results obtained through UPS measurements (Supplementary Fig. 22). The 2PACz delivered a work function of -4.88 eV, differing by 0.50 eV from ITO, while Py3 delivered a work function of -4.65 eV, differing by 0.27 eV from ITO (Supplementary Fig. 23A). As a result, when contacting with the perovskite layer, Py3 demonstrated a better band alignment with the perovskite layer than 2PACz (Supplementary Fig. 23B), thereby facilitating the hole extraction.

Supplementary Text S5

Analysis of device physics

To gain insights into the improved performance of Py3-based solar cell devices, we measured the V_{OC} as a function of the light intensity for devices with varying molecular contacts, resulting in ideality factors of 1.56 and 1.19 for 2PACz and Py3, respectively (Supplementary Fig. 24). The ideality factor of Py3-based device was determined to be closer to 1, verifying that the interfacial non-radiative charge recombination was reduced compared to 2PACz-based device. The difference in charge recombination was also reflected in the reverse saturation current as revealed by dark $J-V$ measurements. The comparison of the dark $J-V$ curves of 2PACz and Py3-based devices is presented in Supplementary Fig. 25. The Py3-based device featured a lower reverse saturation current, indicative of reduced interfacial trap states and thus suppressed non-radiative charge recombination. To gain insights into the improvement in FF, we extracted the shunt resistance R_{sh} and series resistance R_s of the devices. The R_{sh} value for 2PACz and Py3 was determined to be 8751 Ω and 79308 Ω , respectively. The increased R_{sh} in Py3-based device suggests lower current leakage than that in 2PACz-based device. This can be attributed to the enhanced molecular stacking stemming from the strong π - π interaction in Py3 indicated above, which can lead to a more robust interlayer preventing the perovskite from directly contacting ITO. The R_s value for 2PACz and Py3 was determined to be 29.5 Ω and 13.2 Ω , respectively. The decrease in R_s indicates a lower interfacial resistance in Py3, which aligned well with the higher conductivity as revealed by the c-AFM results.

Supplementary Text S6

Comparison of the device performances of Py3 and its naphthalene-, anthracene-, and perylene-based variants

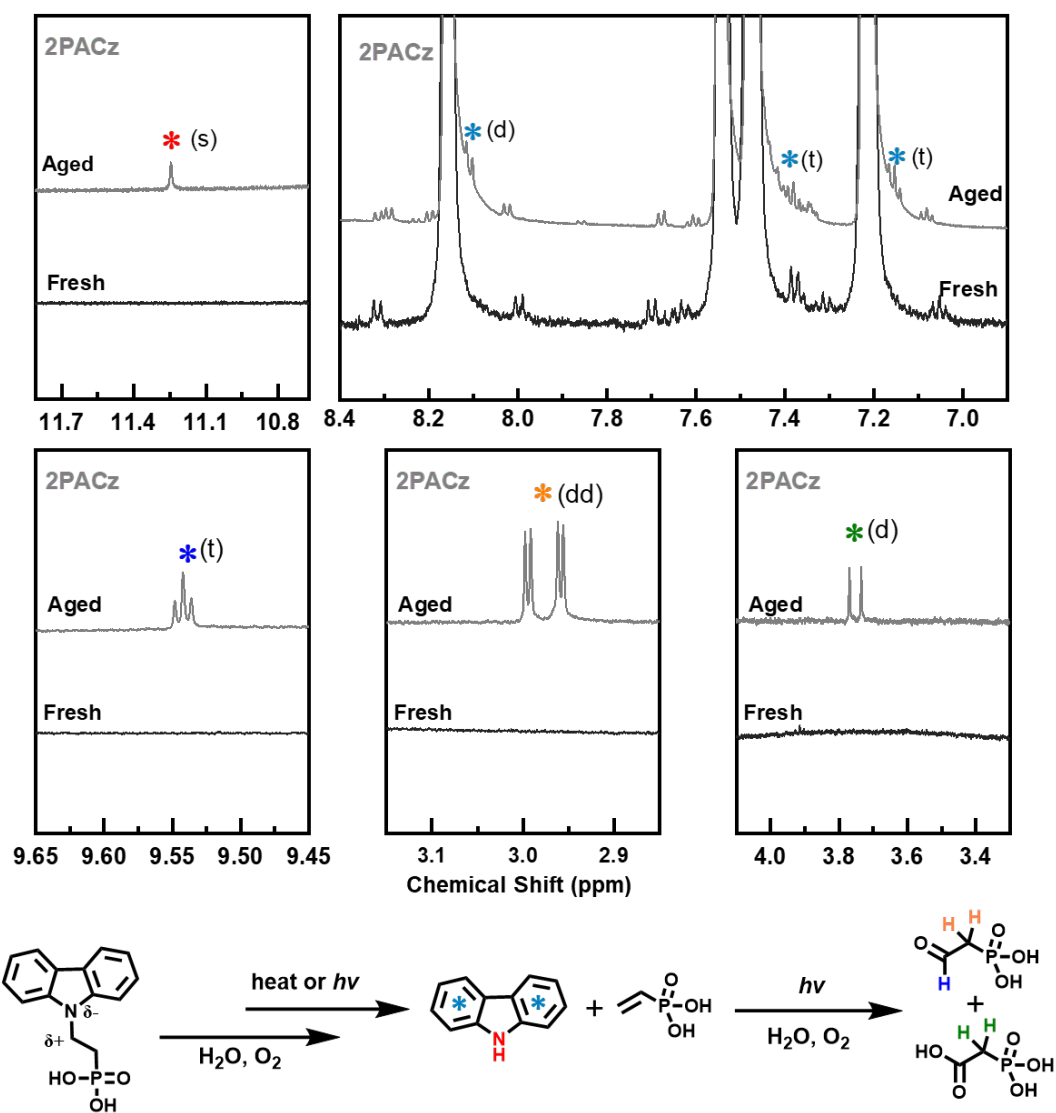
We used the synthesized Py3 molecule and its variants as the hole-selective contacts of perovskite solar cells and the statistical photovoltaic parameters are compared in Supplementary Fig. 26. We noticed that as the conjugation extended, the J_{SC} of the devices in general increased. The improved J_{SC} can be attributed to better charge transporting capability of the molecules with increased π -conjugation. However, as the number of the fused rings increased from 4 to 5 (i.e. from pyrene to perylene), the V_{OC} and the FF started to decrease. The level of the π -conjugation not only can impact the charge transporting capability, but also can change the energy level. Hence, the greatly extended π -conjugation in perylene could push the occupied frontier orbital to a higher energy level, making the interfacial energetics less aligned with that of the perovskite layer. As a result, more carrier and energy losses at the interface compromised the V_{OC} and the FF. Moreover, we observed that the perylene, which features the largest conjugation unit, exhibited a high tendency to self-aggregate because its π - π stacking is too strong. This added difficulties to its solution-processing into a uniform layer, which could also result in the reduced FF.

We performed the ultraviolet photoelectron spectroscopy (UPS) to verify the surface energetics for these molecules. The band diagrams in Supplementary Fig. 9 extracted from the UPS results (Supplementary Fig. 10) show the energy levels of these molecules deposited on the ITO substrates in comparison to the perovskite layers. As the number of the fused rings increased, the difference between the Fermi level (E_f) and the valence band maximum (VBM) gradually decreased, indicating enhanced p-type feature and thus improved hole transporting capability. As the number of the fused rings increased from 2 to 4, the VBM of the molecular layer shifted upwards and became better aligned with that of the perovskite layer. However, when the number further increased to 5, the VBM was too high to match well with that of the perovskite layer and thus would increase the non-radiative recombination loss at the interface, which was consistent with the photovoltaic parameters. These results suggest when designing this peri-fused polyaromatic family, it is crucial to consider the extent of π -conjugation, which should neither be too large nor too small. If it is too small, the charge transporting capability would be compromised; if it is too large, the band alignment and the solution-processability would be adversely affected. As a result, pyrene-based structure achieved the optimal performance by balancing all these factors.

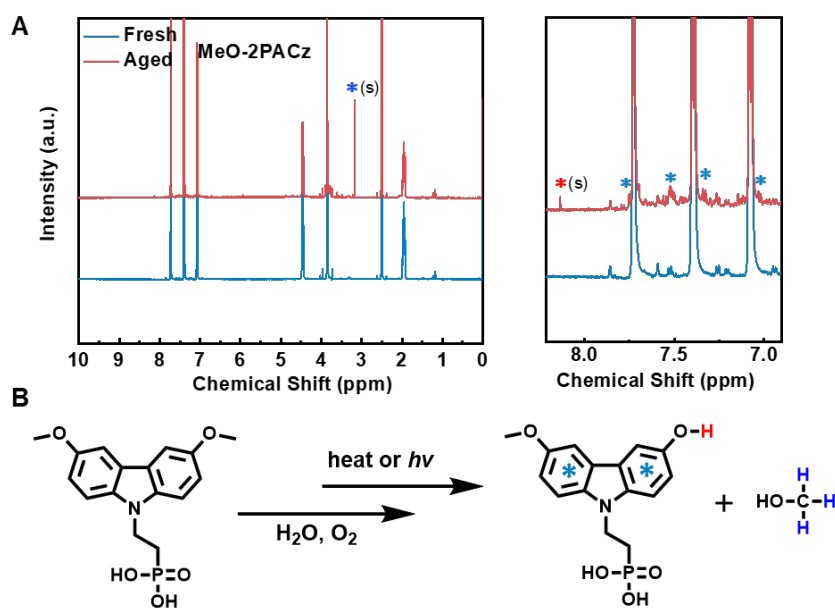
Supplementary Text S7

Suppressed ion migration across the molecule-based selective contacts

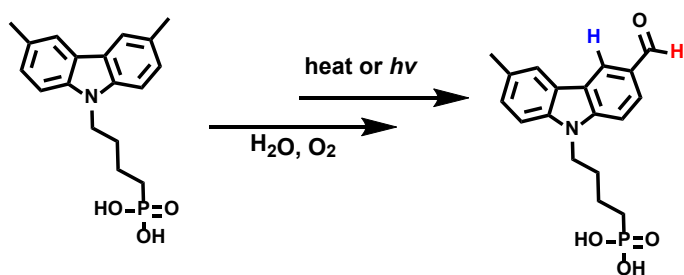
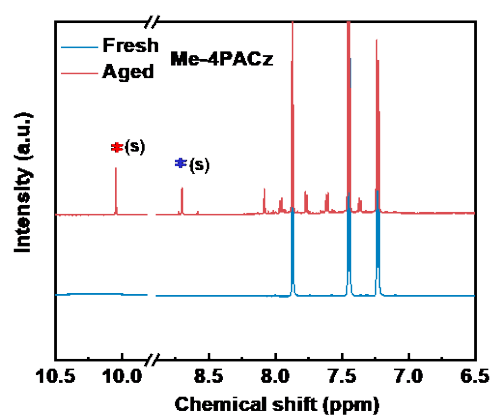
To understand the perovskite degradation resulting from the different hole-selective contacts, we used cross-sectional scanning transmission electron microscopy (STEM) and energy-dispersive X-ray (EDX) spectroscopy mapping to examine the interfaces pertinent to the molecular layers of the aged devices. The examined devices were the ones after the aging tests under continuous one-sun illumination at 65 °C under open-circuit for 3300 h and they were then stored under dark in nitrogen atmosphere before the measurements. The cross-sectional high-angle annular dark-field (HAADF) images obtained from STEM and the corresponding EDX mapping of the aged devices with 2PACz and Py3, respectively are presented in (Supplementary Fig. 27). In the device with 2PACz, iodide ions, the most mobile species in perovskite films, migrated across the interface to the ITO layer. In contrast, significantly lower amounts of iodide species were observed in the ITO layer of the Py3-based device and the interface remained clearly outlined by the EDX mapping. Therefore, replacing 2PACz by Py3 can greatly strengthen the interface between the perovskite and ITO, suppressing the ion migration across the interface.



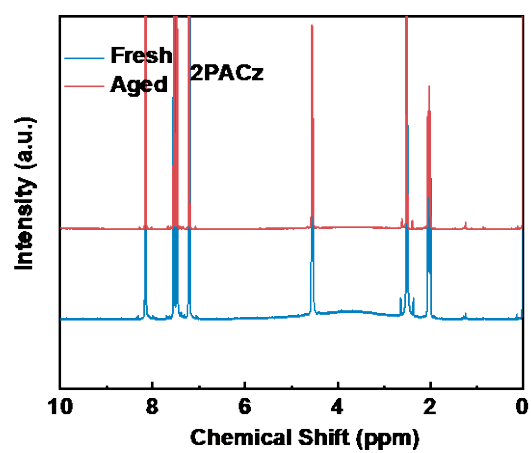
Supplementary Fig. 1 Peak assignments of the NMR spectra of 2PACz after aging and the possible degradation pathways of the molecular layer.



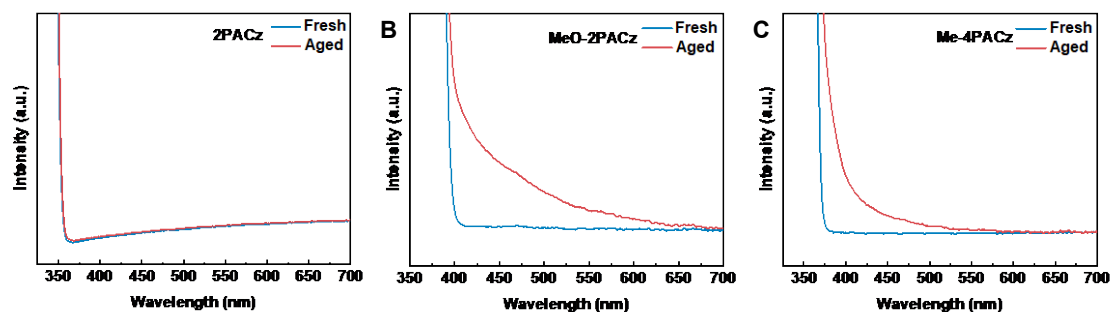
Supplementary Fig. 2 (A) NMR spectra of MeO-2PACz before and after aging and (B) its degradation pathway.



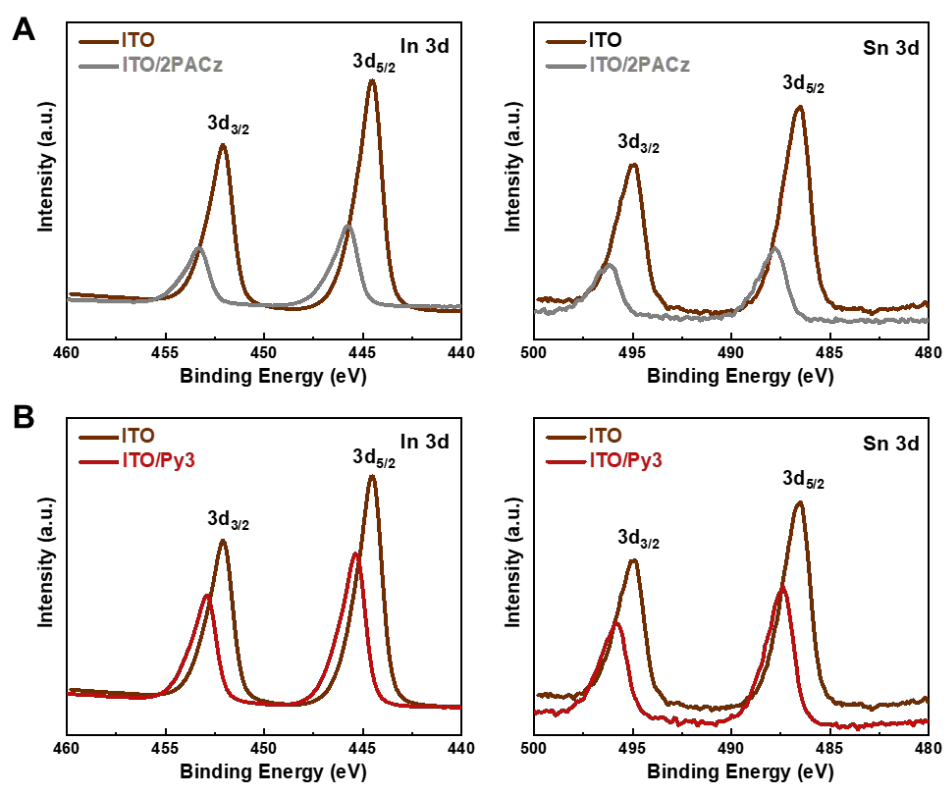
Supplementary Fig. 3 (A) NMR spectra of Me-4PACz before and after aging and (B) its degradation pathway.



Supplementary Fig. 4 NMR spectra of 2PACz before and after the aging for a short course of 48 h.



Supplementary Fig. 5 UV-Vis absorption spectra of (A) 2PACz, (B) MeO-2PACz and (C) Me-4PACz before and after the aging test.



Supplementary Fig. 6 XPS data of In 3d and Sn 3d for (A) ITO/2PACz and (B) ITO/Py3.

Measurement Report

Report No. 23TR082201

Client Name Westlake University
Client Address No.18 Shilongshan Road, Xihu District, Hangzhou, Zhejiang
Sample Inverted Perovskite solar cell
Manufacturer Westlake University, Rui Wang Group
Measurement Date 22nd August, 2023
Performed by: Qiang Shi *Qiang Shi* 
Reviewed by: Wenjie Zhao *Wenjie Zhao* 
Approved by: Yucheng Liu *Yucheng Liu* 
Date: 24/8/2023

Address: No.235 Chengbei Road, Jiaxing, Shanghai
 E-mail: solarcell@mail.sim.ac.cn
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Report No. 23TR082201

Sample Information	
Sample Type	Inverted Perovskite Solar Cell
Serial No.	9-1#
Lab Internal No.	23082201-1#
Measurement Item	I-V characteristic
Measurement Environment	25.7 ± 2.0°C, 44.5 ± 5.0%RH

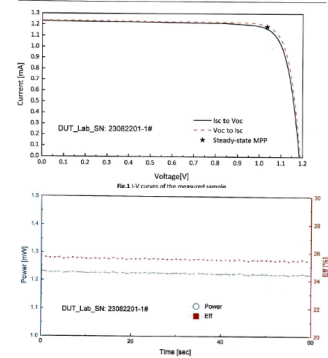
Measurement of I-V characteristic	
Reference cell	PVM 1121
Reference cell Type	mono-Si, WPVS, calibrated by NREL (Certificate No. ISO 2075)
Calibration Value/Date of Calibration for Reference cell	144.53mA/V Feb. 2023
Measurement Conditions	Standard Test Condition (STC): Spectral Distribution: AM1.5 according to IEC 60904-3 Ed.3, Irradiance: 1000 ± 50W/m ² , Temperature: 25 ± 2°C
Measurement Equipment/ Date of Calibration	AAA Steady State Solar Simulator (YSS-155S-2M) / July 2022 IV test system (ADCMT 6246) / June, 2022
Measurement Method	I-V measurement: Logarithmic sweep based on IEC 60904-1:2020 MPP-Tracking: Tracking for 60 seconds by P&O method, the reported Pmax represents the average value of total data
Measurement Uncertainty	Isc: 2.0%(k=2); Voc: 1.0%(k=2); Pmax: 2.5%(k=2)



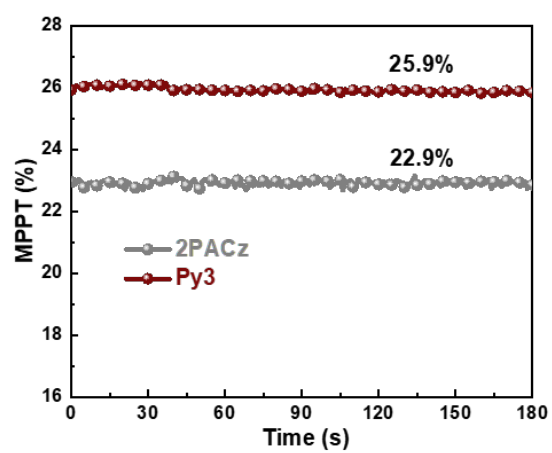
Report No. 23TR082201

Measurement Results			
	Forward Scan (Isc to Voc)	Reverse Scan (Voc to Isc)	MPP-Tracking
Area	0.048 cm ²		
Isc	1.232 mA	1.235 mA	/
Voc	1.185 V	1.188 V	/
Pmax	1.209 mW	1.236 mW	1.225 mW
Ipm	1.173 mA	1.189 mA	1.183 mA
Vpm	1.031 V	1.039 V	1.035 V
FF	82.84 %	84.17 %	/
EF	25.19 %	25.74 %	25.57 %

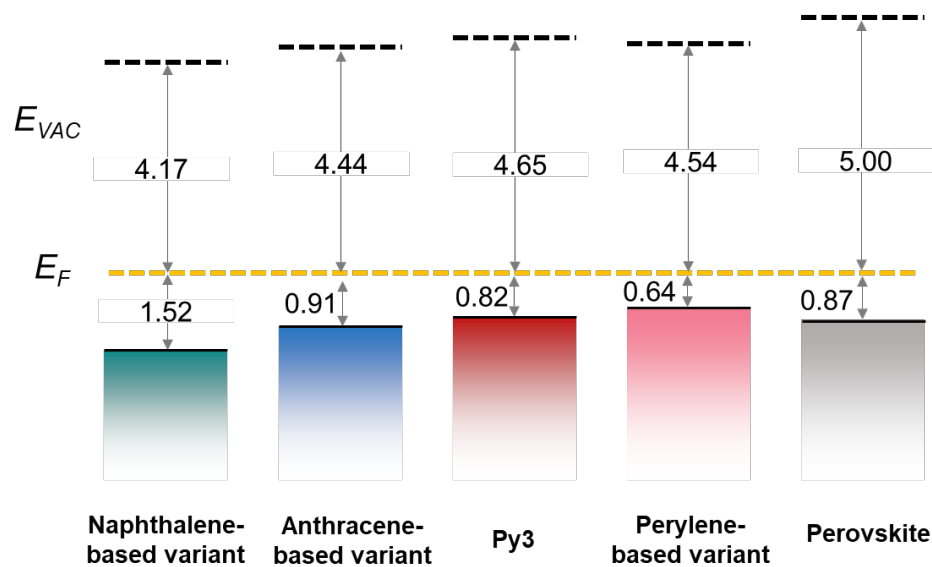
- Designated Illumination area defined by a thin metal mask was provided by the client.
 - Room temperature control during MPP-Tracking.
 - Test results listed in this measurement report refer exclusively to the mentioned measured sample.
 - The results apply only at the time of the test, and do not imply future performance.



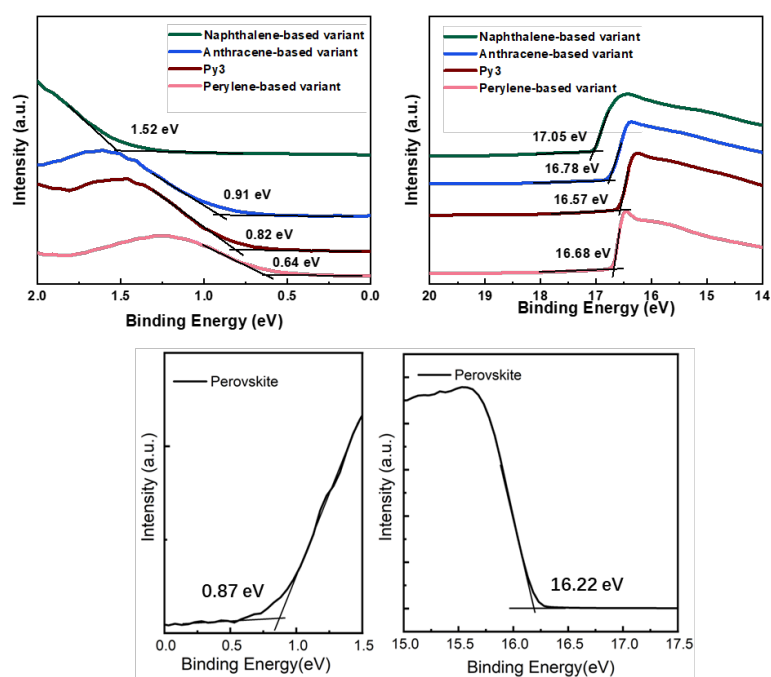
Supplementary Fig. 7 Certification of the perovskite solar cell device with optimal bandgap by the Shanghai Institute of Microsystem and Information Technology (SIMIT), Chinese Academy of Sciences.



Supplementary Fig. 8 The stabilized efficiency measured by maximum power point (MPP) tracking.



Supplementary Fig. 9 Band alignment at the interface of the ITO with different molecules and the perovskite.



Supplementary Fig. 10 UPS results of the ITO modified with different molecules and the perovskite layer.

Measurement Report

Report No. **23TR082202**

Client Name Westlake University
Client Address No.18 Shilongshan Road, Xihu District, Hangzhou, Zhejiang
Sample Inverted Perovskite solar cell
Manufacturer Westlake University, Rui Wang Group
Measurement Date 22nd August, 2023

Performed by: Qiang Shi *Qiang Shi* **Date:** 22/08/2023
Reviewed by: Wenjie Zhao *Wenjie Zhao* **Date:** 22/08/2023
Approved by: Yucheng Liu *Yucheng Liu* **Date:** 22/08/2023

Address: No.235 Chengbei Road, Jiading, Shanghai **Post Code:** 201800
E-mail: solarcell@mail.sim.ac.cn **Tel:** +86-021-69976921

The measurement report without signature and seal are not valid.
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Report No. 23TR082202

Sample Information	
Sample Type	Inverted Perovskite Solar Cell
Serial No.	29-2#
Lab Internal No.	23082201-2#
Measurement Item	I-V characteristic
Measurement Environment	25.7±2.0°C, 44.5±5.0%RH
Measurement of I-V characteristic	
Reference cell	PVM 1121
Reference cell Type	mono-Si, WPV5, calibrated by NREL (Certificate No. ISO 2075)
Calibration Value/Date of Calibration for Reference cell	144.53mA / Feb. 2023
Measurement Conditions	Standard Test Condition (STC): Spectral Distribution: AM1.5 according to IEC 60904-3 Ed.3, Irradiance: 1000±50W/m ² , Temperature: 25±2°C
Measurement Equipment/ Date of Calibration	AAA Steady State Solar Simulator (V55-T355-2M) / July 2022 IV test system (ADCMT 6246) / June. 2022
Measurement Method	I-V measurement: Logarithmic sweep based on IEC 60904-1:2020 MPP-Tracking: Tracking for 60 seconds by P&O method, the reported P _{max} represents the average value of total data
Measurement Uncertainty	Isc: 2.0%(k=2), Voc: 1.0%(k=2); P _{max} : 2.5%(k=2)



Report No. 23TR082202

	Forward Scan (Isc to Voc)	Reverse Scan (Voc to Isc)	MPP-Tracking
Area		0.048 cm ²	
Isc	1.075 mA	1.079 mA	/
Voc	1.199 V	1.200 V	/
P _{max}	1.059 mW	1.088 mW	1.079 mW
I _{pm}	1.013 mA	1.034 mA	1.031 mA
V _{pm}	1.045 V	1.052 V	1.047 V
FF	82.14 %	84.07 %	/
EFF	22.05 %	22.46 %	22.48 %

- Designated illumination area defined by a thin metal mask was provided by the client.
- Room temperature control during MPP-Tracking.
- Test results listed in this measurement report refer exclusively to the mentioned measured sample.
- The results apply only at the time of the test, and do not imply future performance.

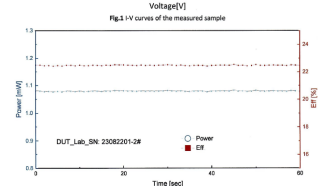
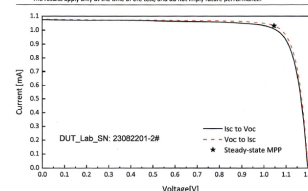
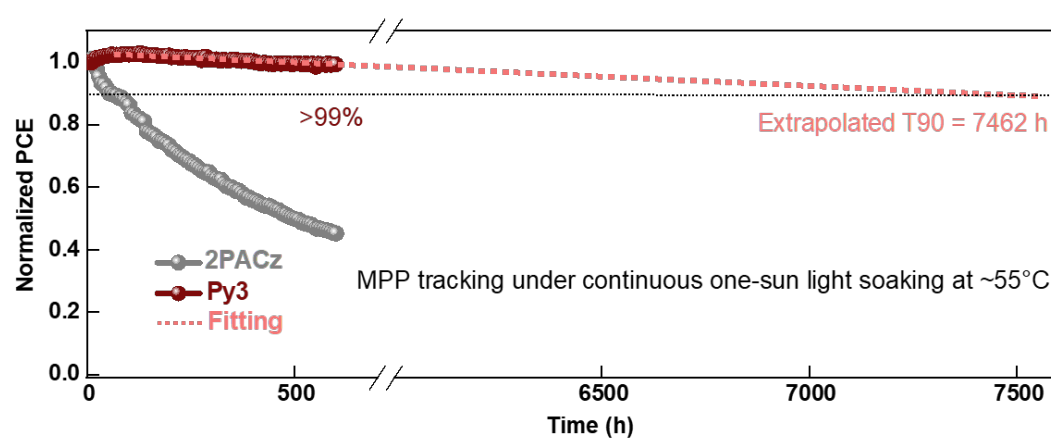
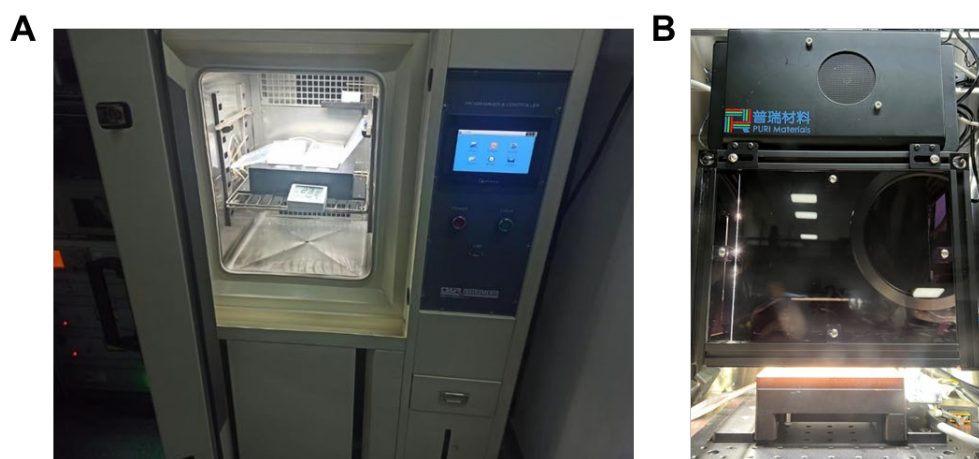


Fig.2 Steady-state maximum power output of the measured sample

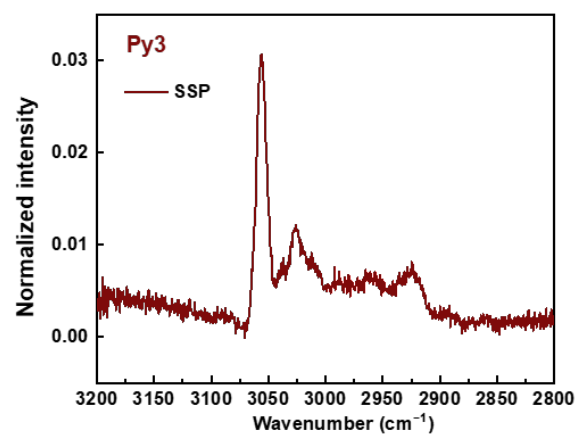
Supplementary Fig. 11 Certification of the perovskite solar cell device with wide bandgap by the Shanghai Institute of Microsystem and Information Technology (SIMIT), Chinese Academy of Sciences.



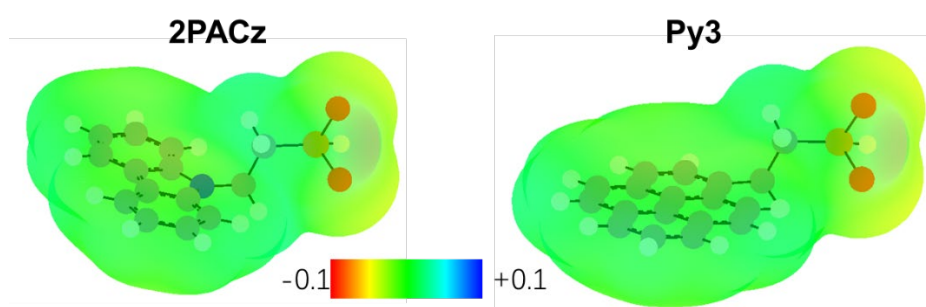
Supplementary Fig. 12 Maximum power point tracking of the PSCs under continuous 1-sun light soaking at ~55°C.



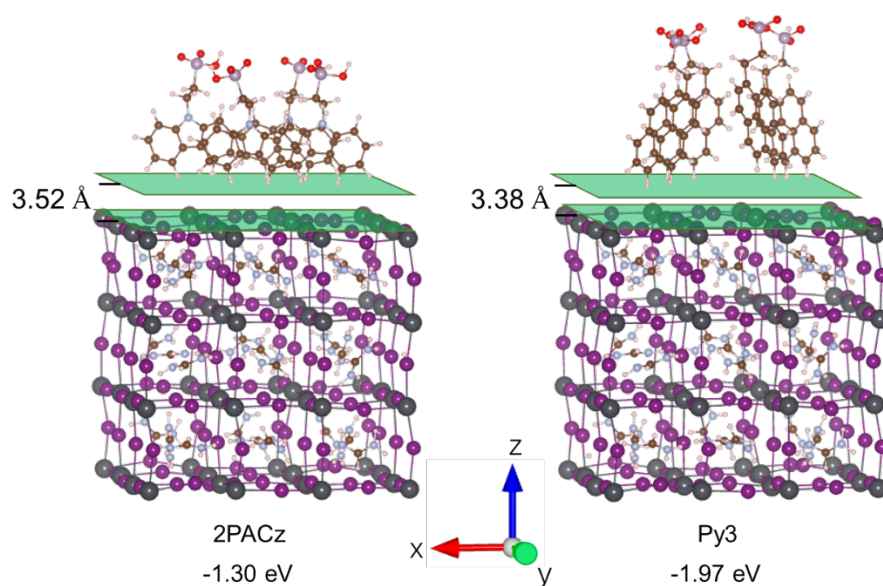
Supplementary Fig. 13 The image of device stability test instrument ((A)150 L, SuZhou D&R Instrument Co., Ltd and (B) Solar cell stability test system, PR-SCCS, PURI Materials).



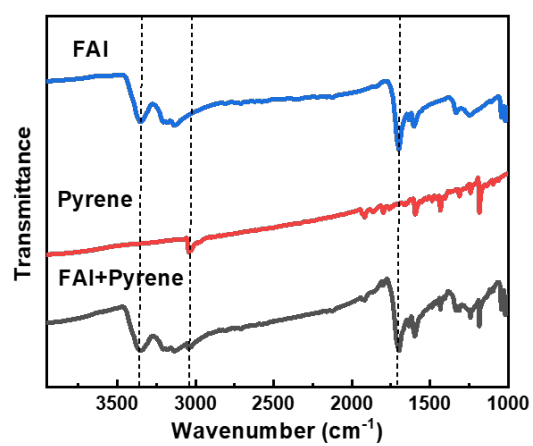
Supplementary Fig. 14 SFG-VS measured in the *ssp* polarization combination for the Py3 sample.



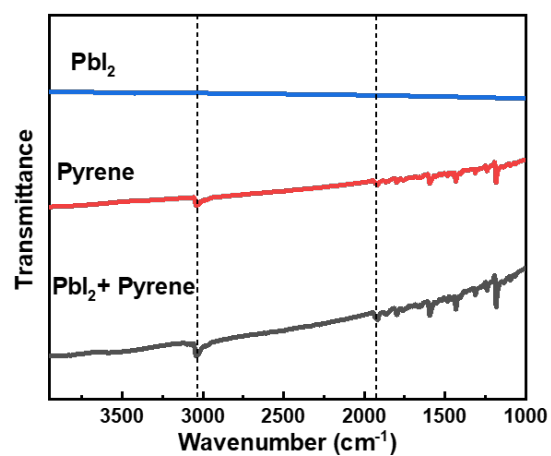
Supplementary Fig. 15 Electrostatic potential surfaces of 2PACz and Py3 molecules in the gas phase. Dipole moments of each structure is also indicated.



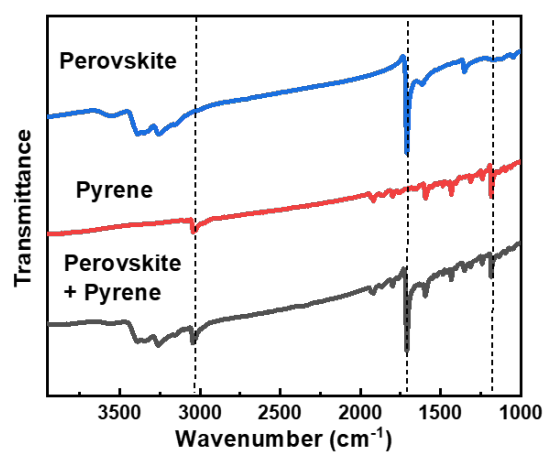
Supplementary Fig. 16 Interaction between perovskite and the molecular layers. The numbers on top show the interaction energy between the molecules and perovskite surface. The average distances between the molecules and perovskite surfaces are also indicated.



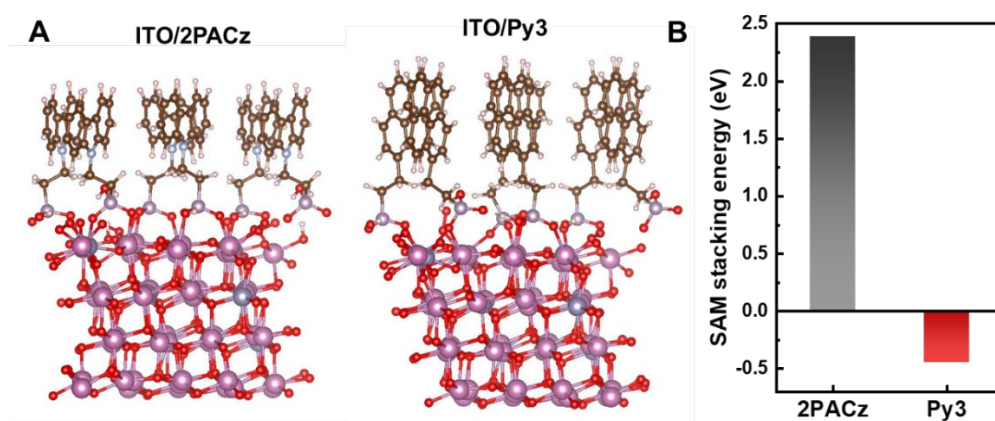
Supplementary Fig. 17 FTIR spectra of of pure FAI, pyrene, and pyrene with FAI.



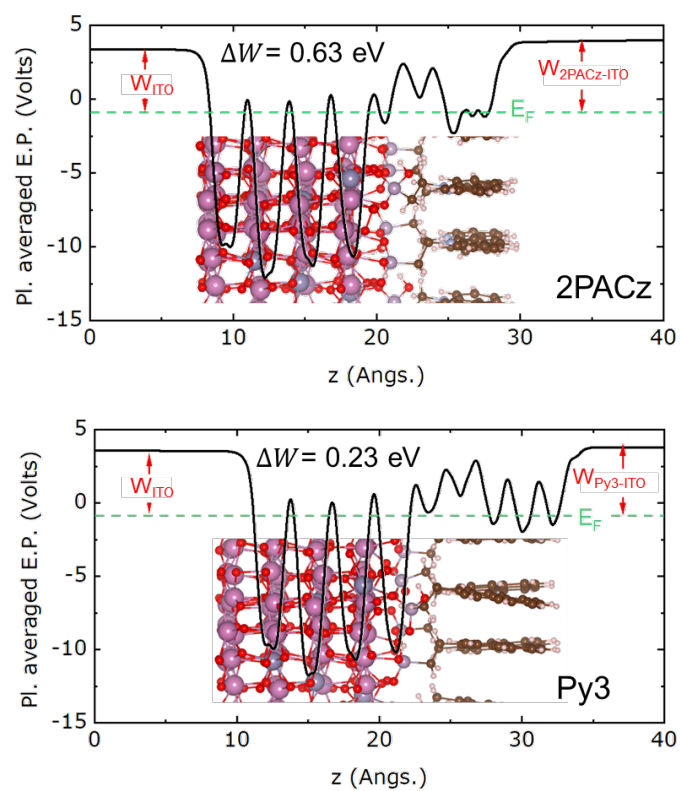
Supplementary Fig. 18 FTIR spectra of pure PbI_2 , pyrene, and pyrene with PbI_2 .



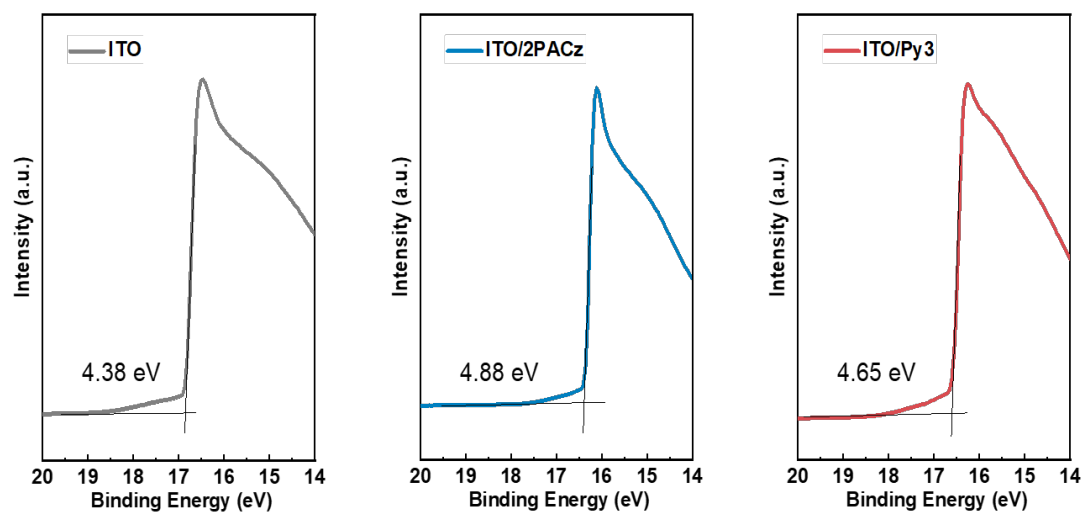
Supplementary Fig. 19 FTIR spectra of pure perovskite, pyrene, and pyrene with perovskite.



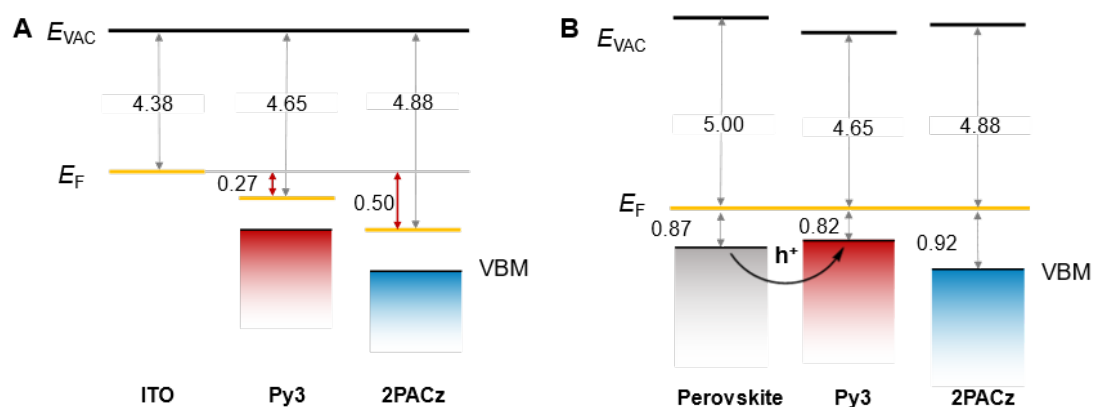
Supplementary Fig. 20 (A) Side view of intercalation configuration modeled by density functional theory (DFT) for ITO/2PACz and ITO/Py3 film. (B) π - π stacking energy calculated for 2PACz and Py3.



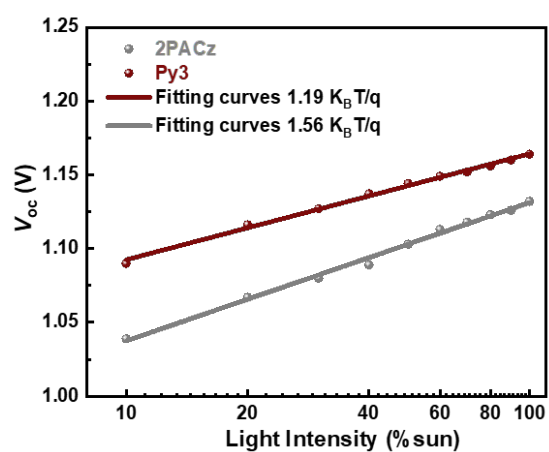
Supplementary Fig. 21 Planar-averaged electrostatic potentials $\bar{V}(z)$ of 2PACz and Py3 assembled on the ITO surface. The asymptotic potentials, fermi energies (E_F) and work functions (W_{ITO} or W_{ML-ITO}) are indicated.



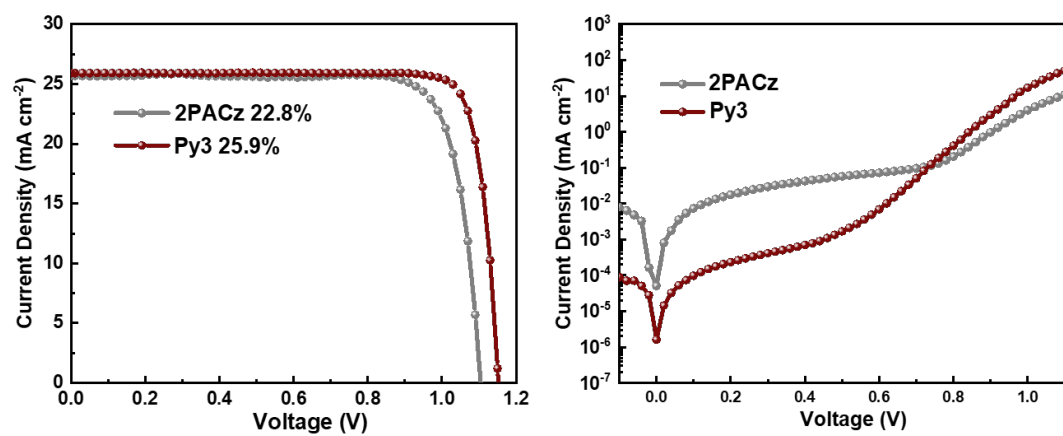
Supplementary Fig. 22 UPS results of the work function of bare ITO, ITO with 2PACz, and ITO with Py3.



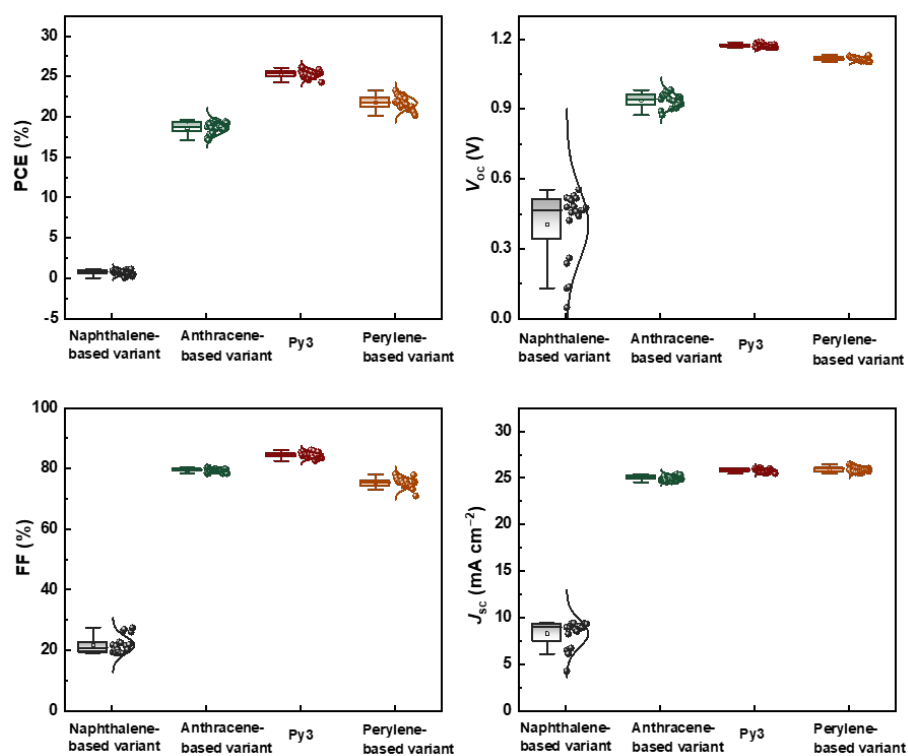
Supplementary Fig. 23 (A) Band structures of bare ITO, ITO with 2PACz and ITO with Py3 extracted from the UPS results; (B) Band alignment at the interface of perovskite layer and the ITO with 2PACz and Py3.



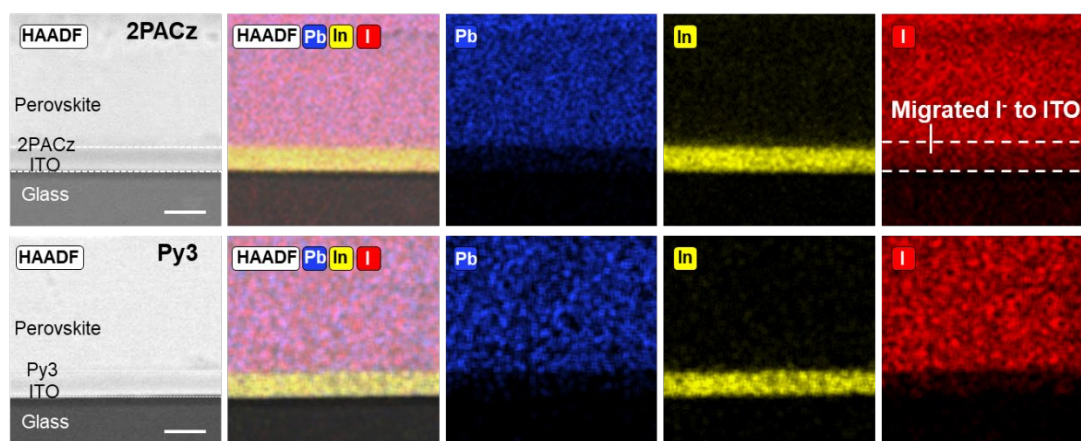
Supplementary Fig. 24 V_{oc} as a function of illumination intensity of the 2PACz and Py3-based devices.



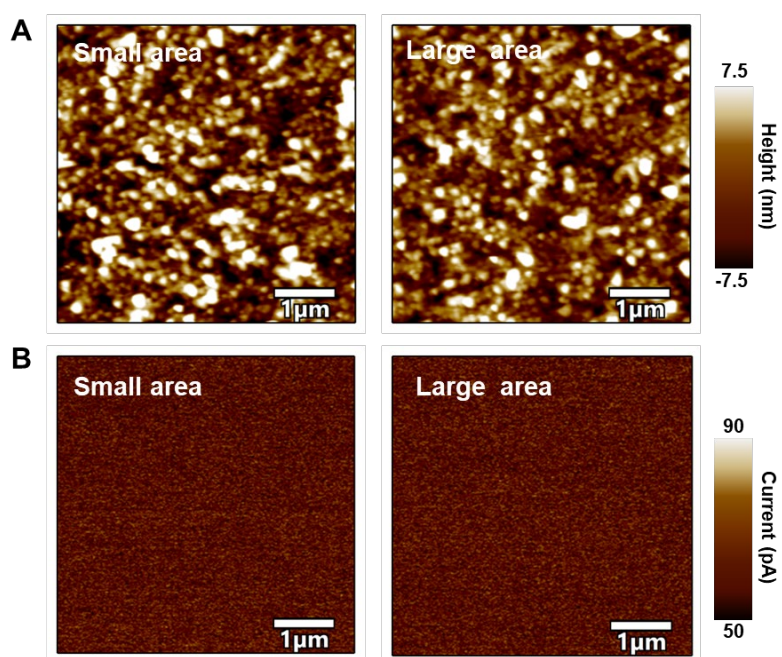
Supplementary Fig. 25 Light and dark J - V curves of perovskite solar cells with 2PACz and Py3 respectively.



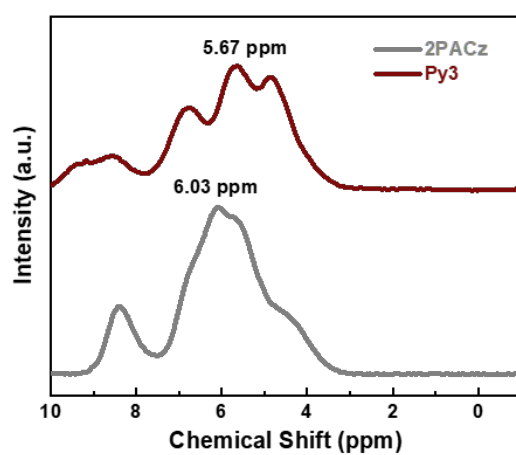
Supplementary Fig. 26 Statistical photovoltaic parameters of the solar cells made by Py3 and its naphthalene-, anthracene-, and perylene-based variants. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



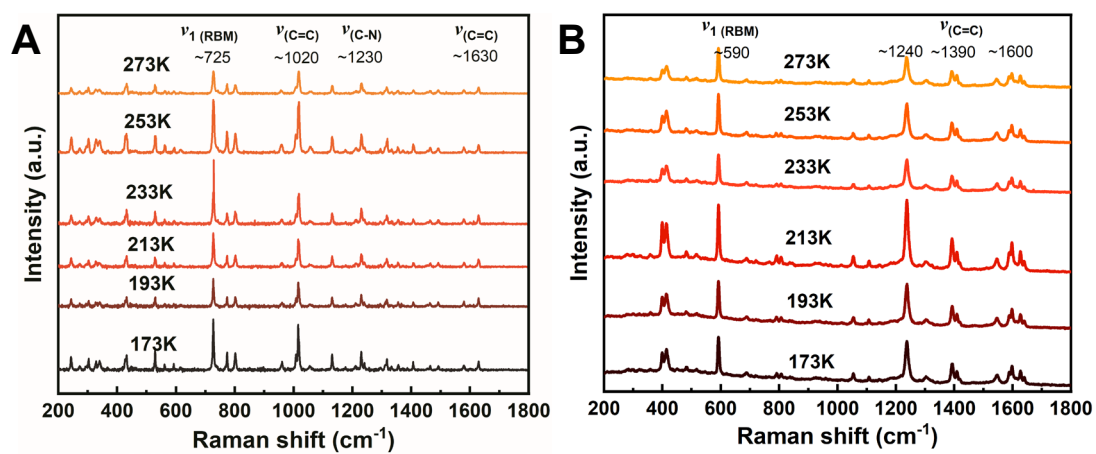
Supplementary Fig. 27 Cross-sectional HAADF images and EDX mapping of the aged devices fabricated with (A) 2PACz and (B) Py3, respectively.



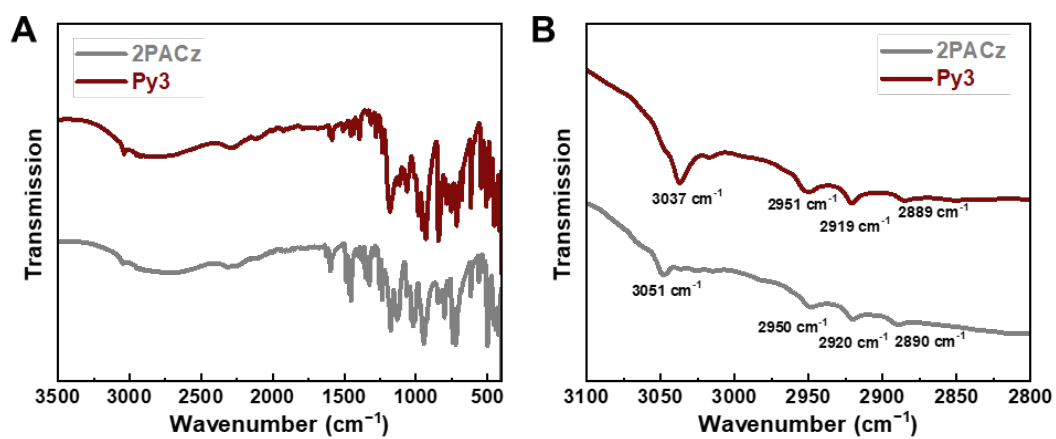
Supplementary Fig. 28 (A) AFM images of the surface morphologies and (B) surface current signals of Py3 films deposited onto ITO substrates with small and large areas.



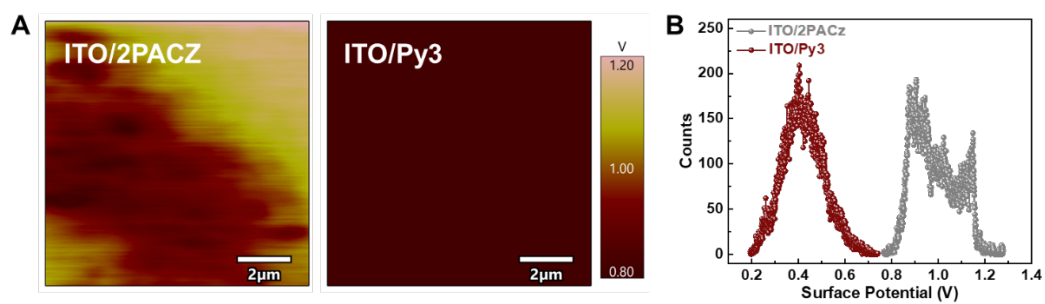
Supplementary Fig. 29 The ^1H solid-state NMR spectra of 2PACz and Py3.



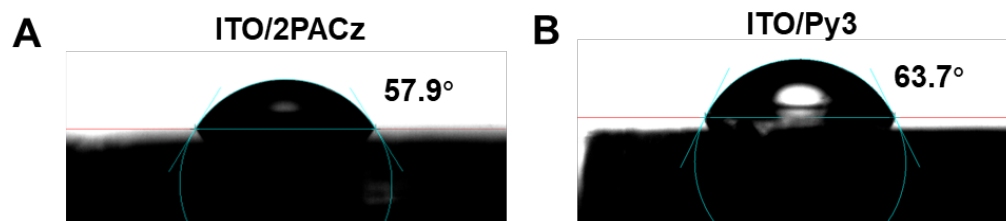
Supplementary Fig. 30 The full-band Raman spectra of (A) 2PACz and (B) Py3 at different temperature.



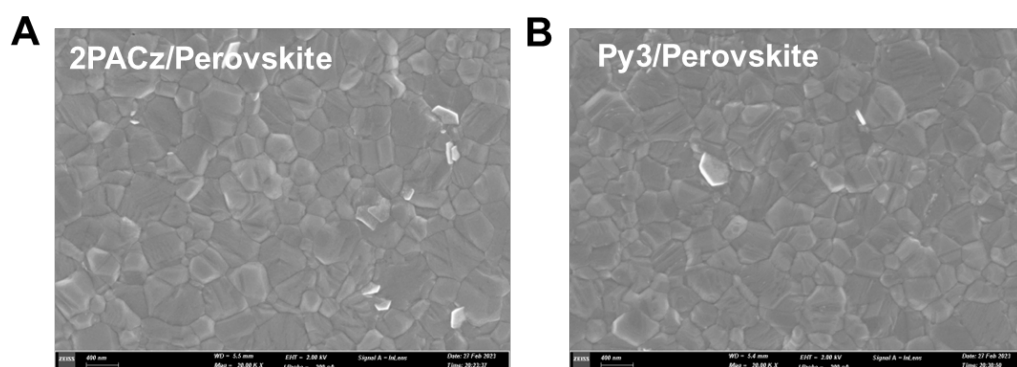
Supplementary Fig. 31 (A) Full-band and (B) magnification FTIR spectra of 2PACz and Py3 solids.



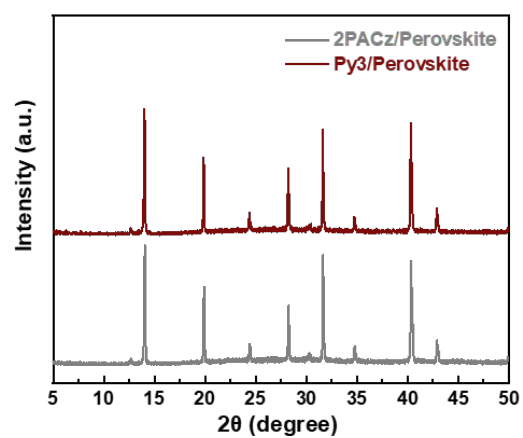
Supplementary Fig. 32 (A) Kelvin probe force microscopy (KPFM) images and (B) the corresponding surface potential signals of ITO substrates covered with 2PACz and Py3.



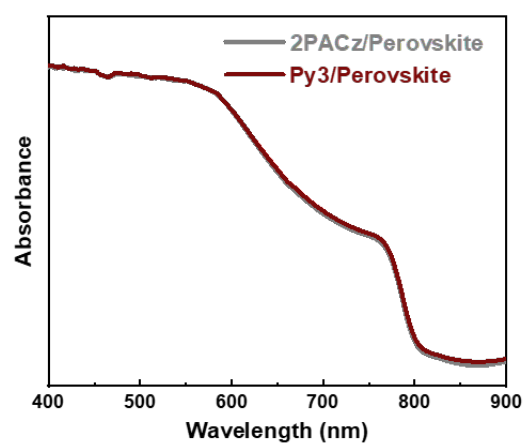
Supplementary Fig. 33 Contact angle of (A) ITO/2PACz and (B) ITO/Py3.



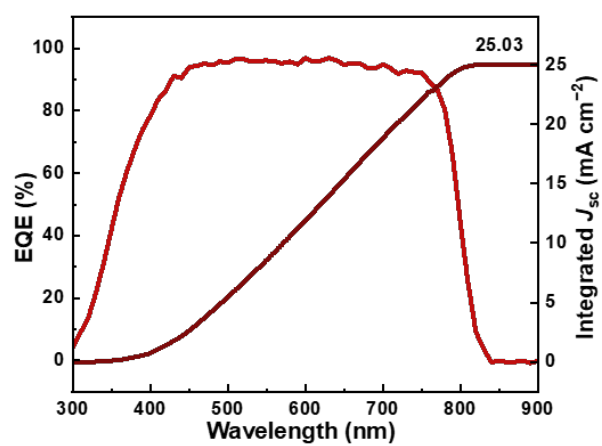
Supplementary Fig. 34 SEM images of perovskite films deposited on **(A)** 2PACz and **(B)** Py3 film.



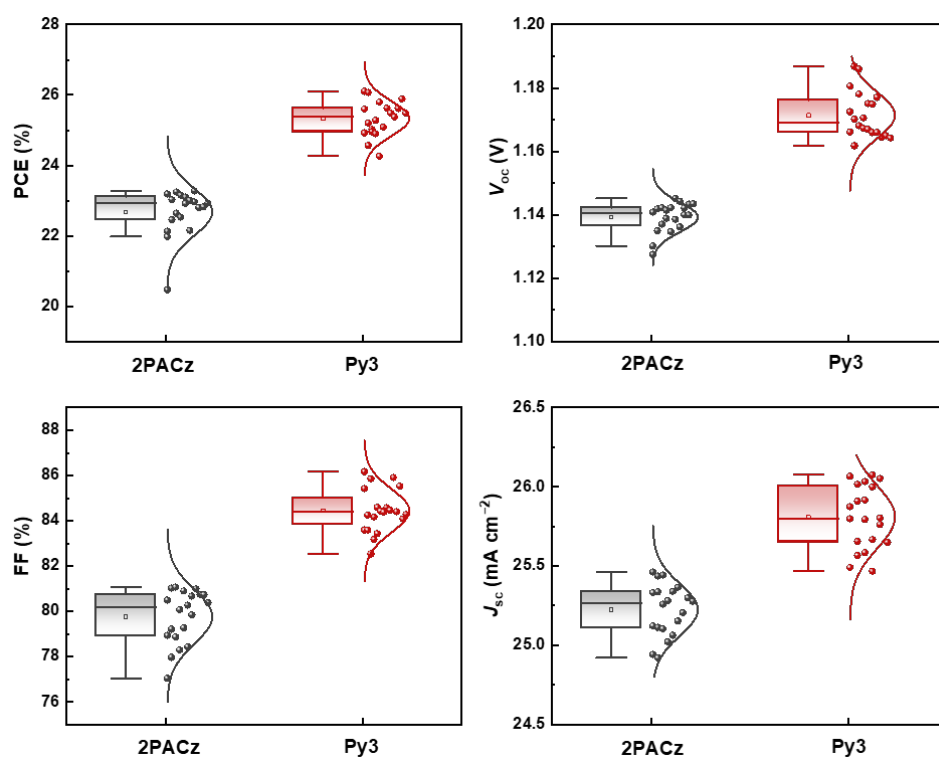
Supplementary Fig. 35 XRD patterns of perovskite films deposited on 2PACz and Py3.



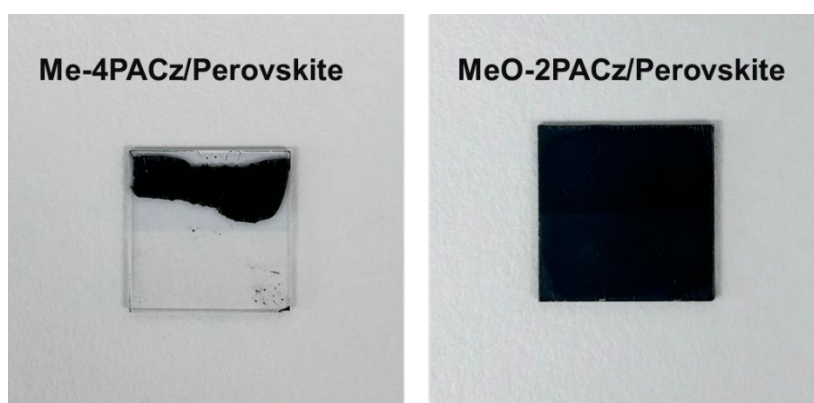
Supplementary Fig. 36 UV-Vis spectra of perovskite films deposited on 2PACz and Py3.



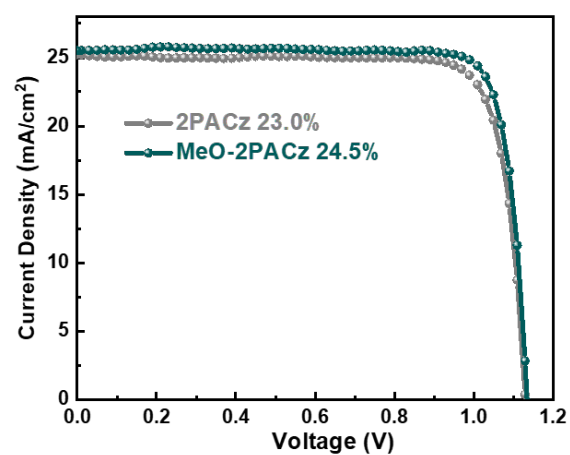
Supplementary Fig. 37 The EQE spectra of the PSC based on Py3.



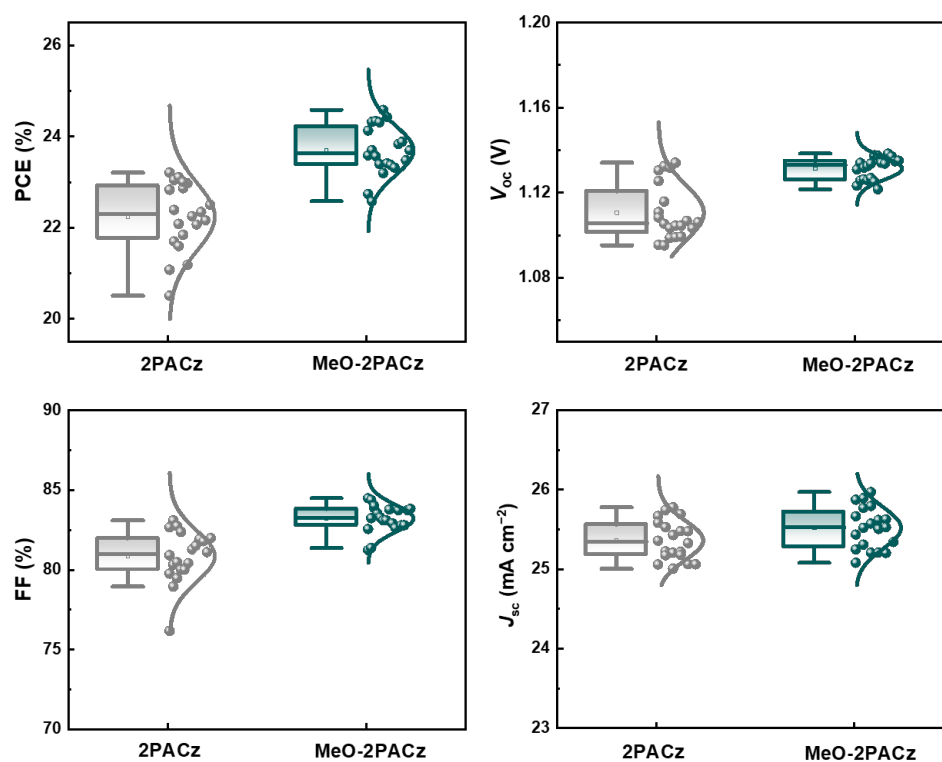
Supplementary Fig. 38 Statistics of photovoltaic parameters (PCE, V_{oc} , FF and J_{sc}) obtained from the J - V characteristic of 20 devices ((FA_{0.98}MA_{0.02})_{0.95}CS_{0.05}Pb(I_{0.98}Br_{0.02})₃ PSCs). The average value and standard deviation of each parameter is summarized in Table S3. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



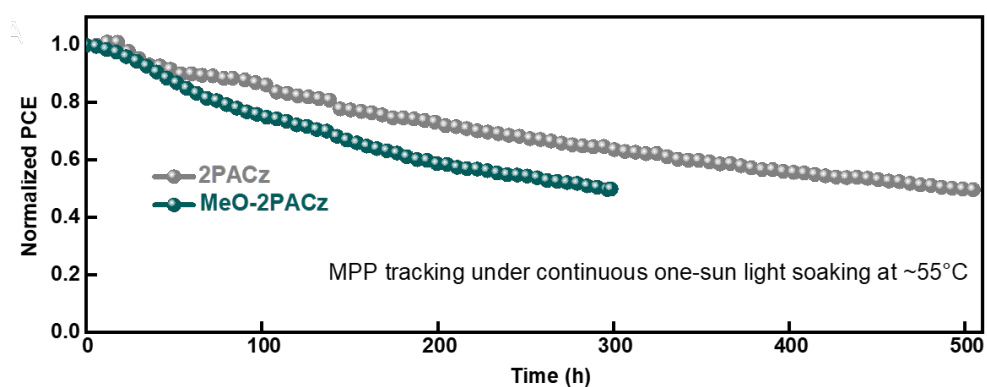
Supplementary Fig. 39 Photos of perovskite layers deposited on the ITO substrates with Me-4PACz and MeO-2PACz respectively.



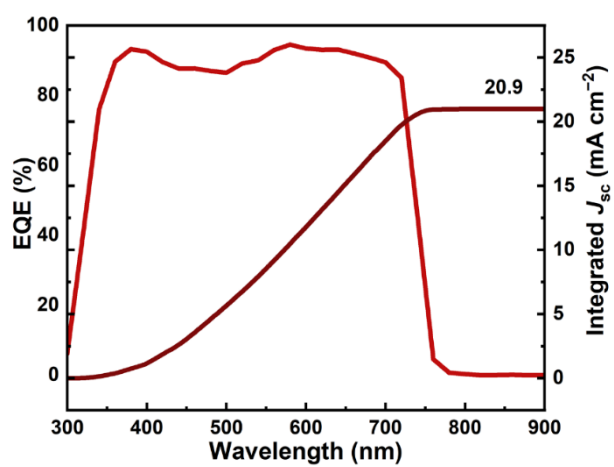
Supplementary Fig. 40 *J-V* curves of champion perovskite solar cells fabricated based on 2PACz and MeO-2PACz within the same batch of devices.



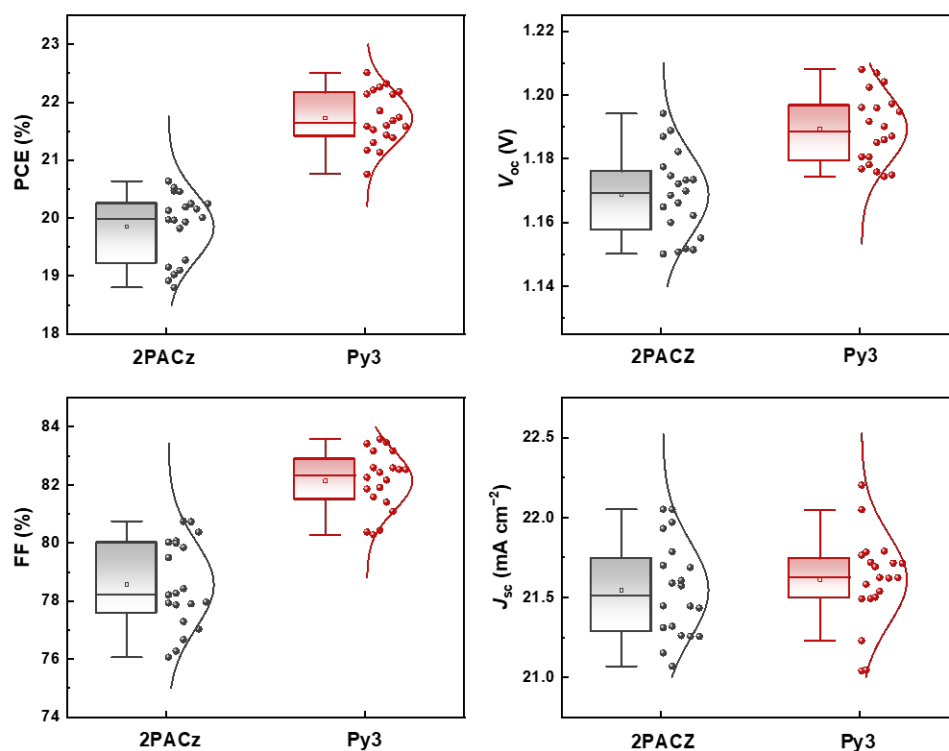
Supplementary Fig. 41 Statistical box charts of the photovoltaic parameters of devices with 2PACz and MeO-2PACz. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



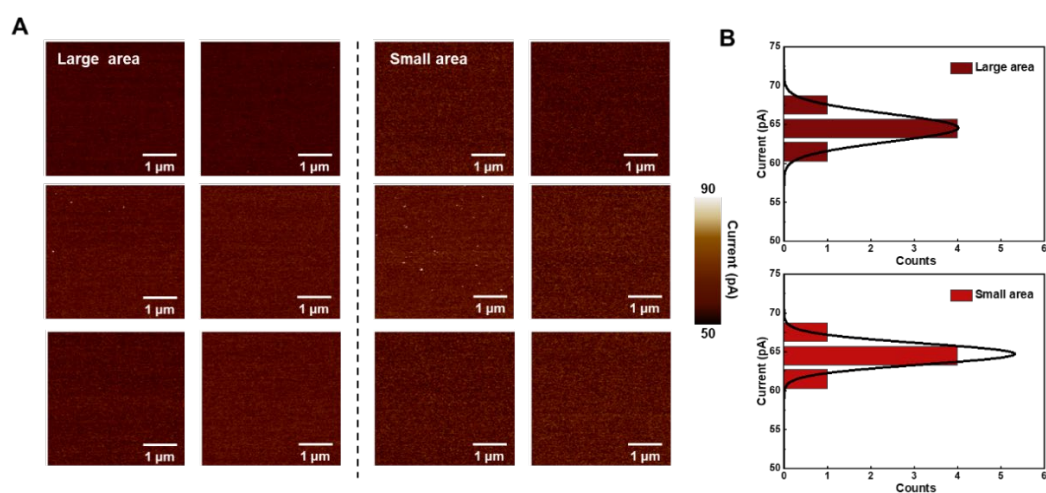
Supplementary Fig. 42 Maximum power point tracking of the perovskite devices under continuous 1-sun light soaking at ~ 55 °C. The initial PCEs of the 2PACz and MeO-2PACz-based devices are 22.9% and 23.8%, respectively.



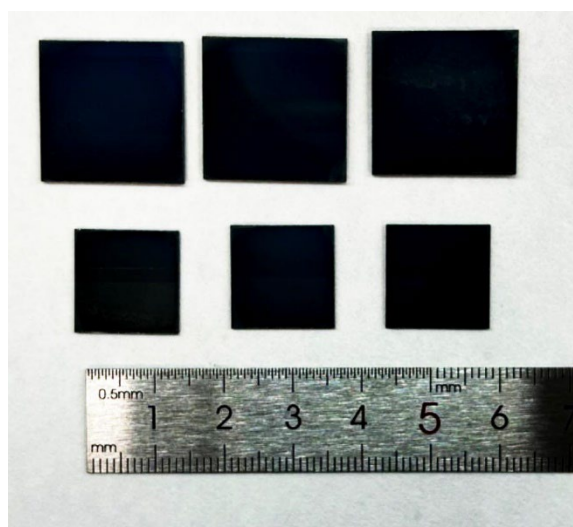
Supplementary Fig. 43 The EQE spectrum of the wide bandgap PSC based on Py3.



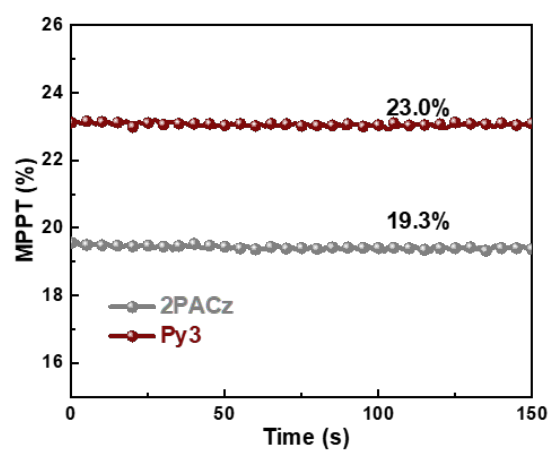
Supplementary Fig. 44 Statistics of photovoltaic parameters (PCE, V_{oc} , FF and J_{sc}) obtained from the J - V characteristic of 20 devices (wide bandgap PSCs). The average value and standard deviation of each parameter is summarized in Table S4. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



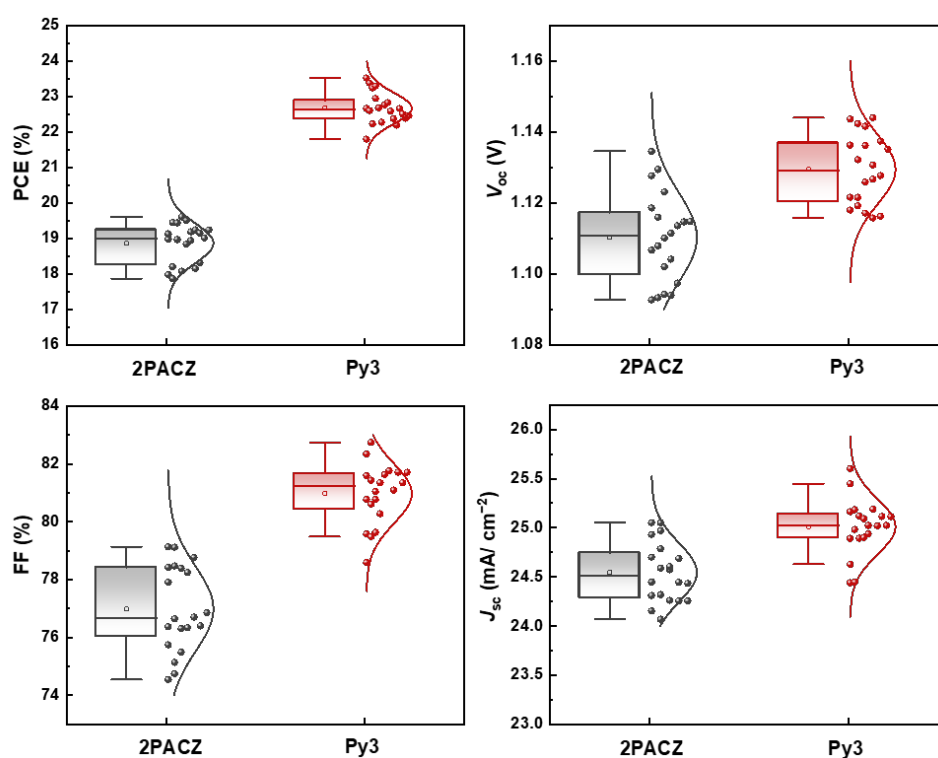
Supplementary Fig. 45 (A) Surface current signals captured at different regions of the Py3 films in small and large scales (B) Statistical plots of the average signal of each c-AFM image. Histograms showing the current signals of six samples per type, fitted with Gaussian distributions (solid lines).



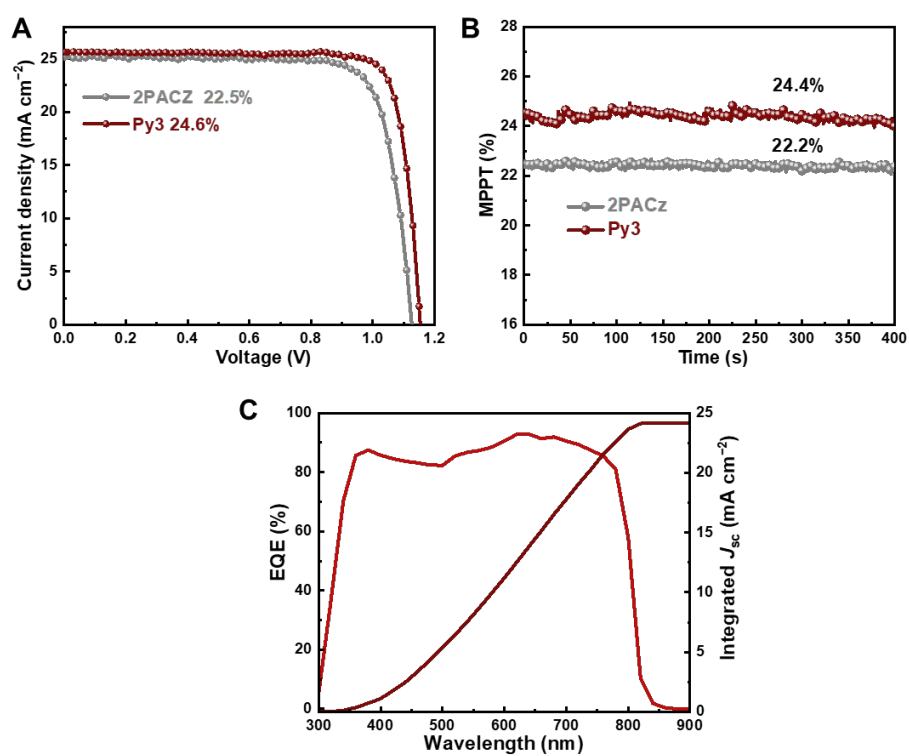
Supplementary Fig. 46 Photographs of perovskite layers deposited on ITO/Py3 substrates with different sizes.



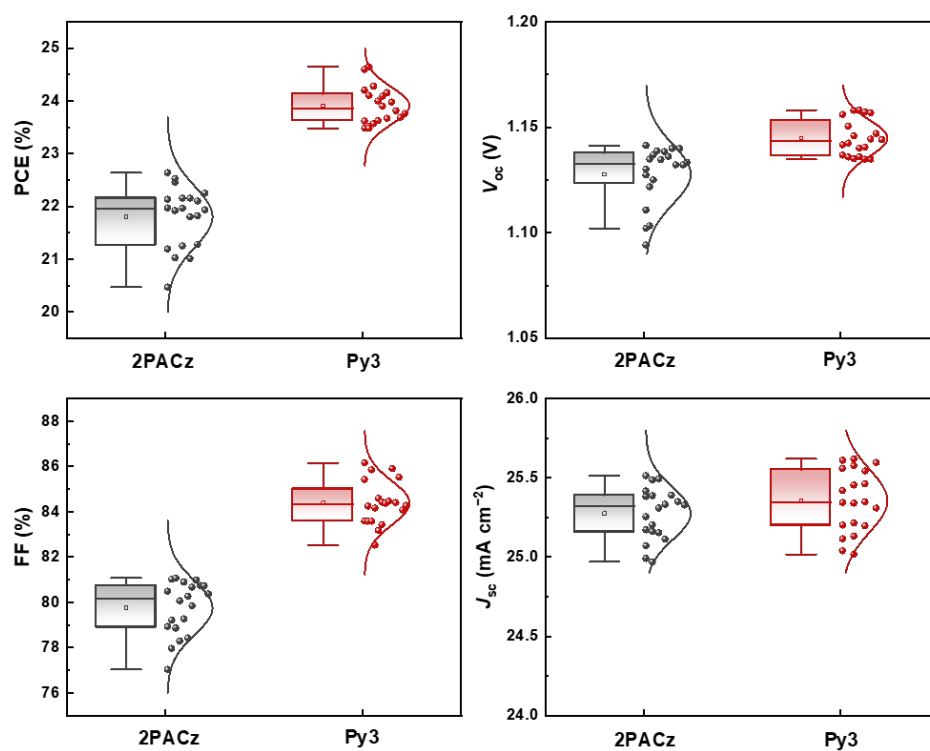
Supplementary Fig. 47 Stabilized PCE of the champion 1-cm² PSC based on 2PACz and Py3 determined by MPP tracking.



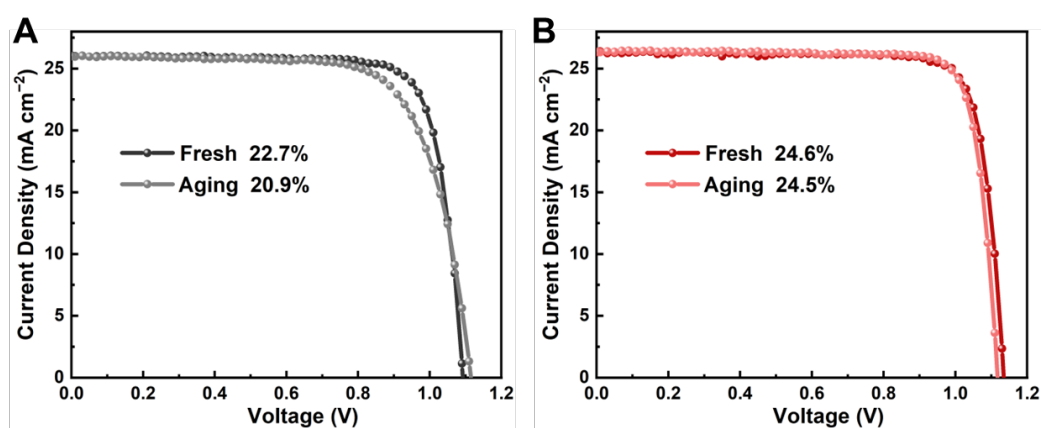
Supplementary Fig. 48 Statistics of photovoltaic parameters (PCE, V_{OC} , FF and J_{SC}) obtained from the J - V characteristic of 20 devices (1-cm² PSC). The average value and standard deviation of each parameter is summarized in Table S5. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



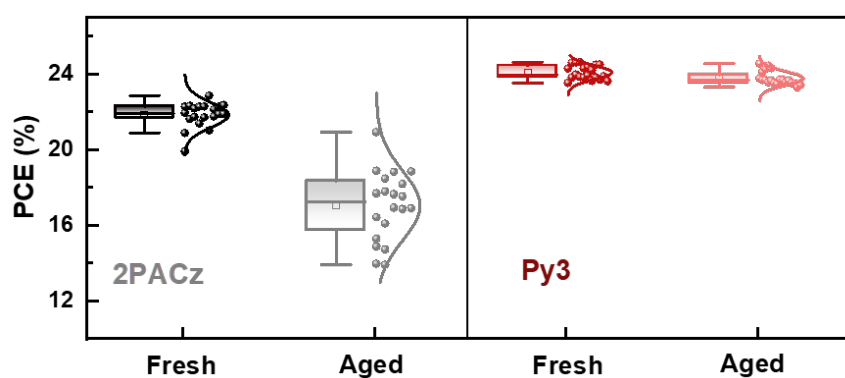
Supplementary Fig. 49 (A) J - V curves of the best performing FAPbI₃ PSCs based on 2PACz and Py3. The photovoltaic parameters are summarized in Table S6. (B) The stabilized efficiency measured by MPP tracking, and (C) the EQE spectrum of the devices.



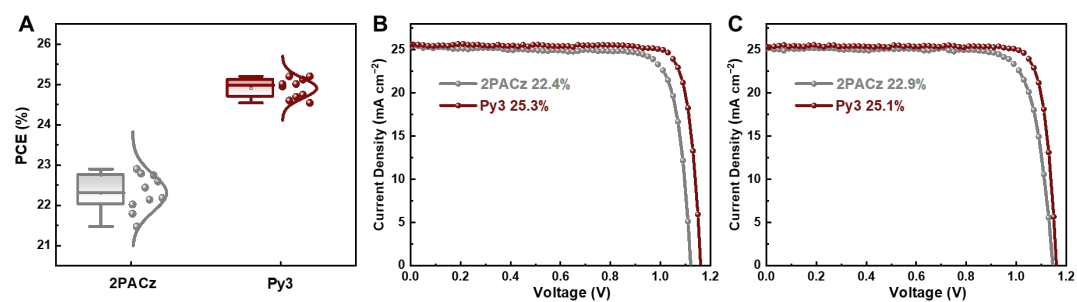
Supplementary Fig. 50 Statistics of photovoltaic parameters (PCE, V_{oc} , FF and J_{sc}) obtained from the J - V characteristic of 20 devices (FAPbI₃ PSCs). The average value and standard deviation of each parameter is summarized in Table S6. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



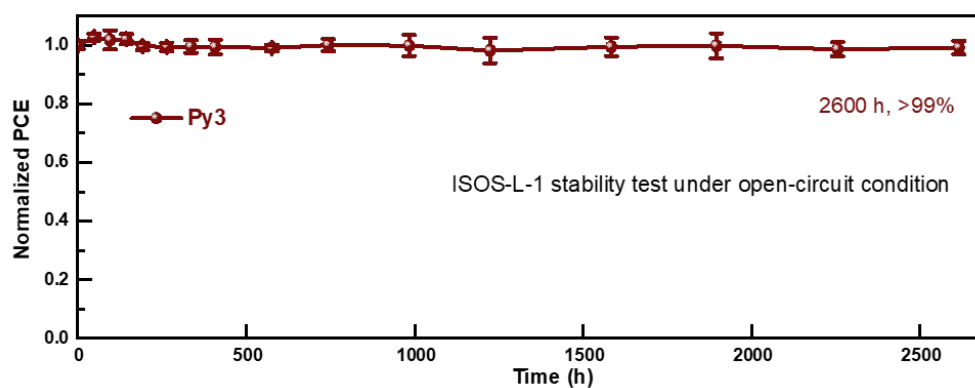
Supplementary Fig. 51 (A) Typical $J-V$ curves of PSC devices fabricated by the fresh 2PACz film and the aged 2PACz film. (B) Typical $J-V$ curves of PSC devices fabricated by the fresh Py3 film and the aged Py3 film-based devices. The aged films of 2PACz and Py3 were obtained by annealing the fresh films at 130 °C for 24 h before perovskite layer deposition.



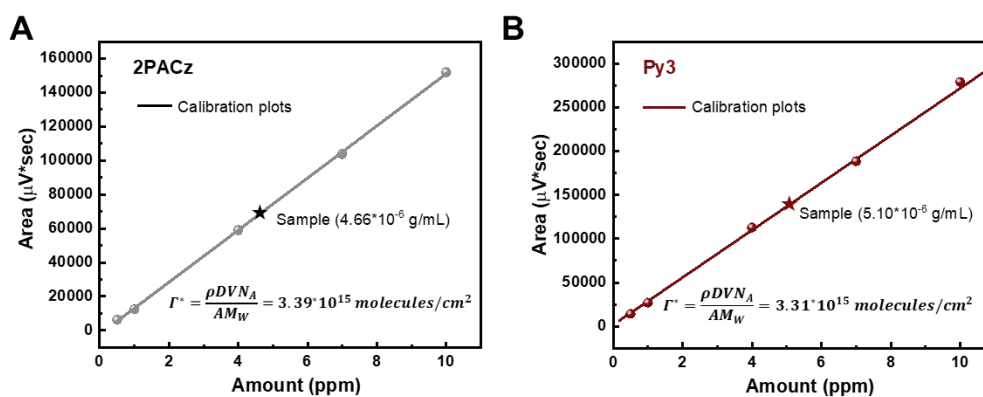
Supplementary Fig. 52 Statistical distribution of PCE values of PSCs based on fresh and aged 2PACz and Py3. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



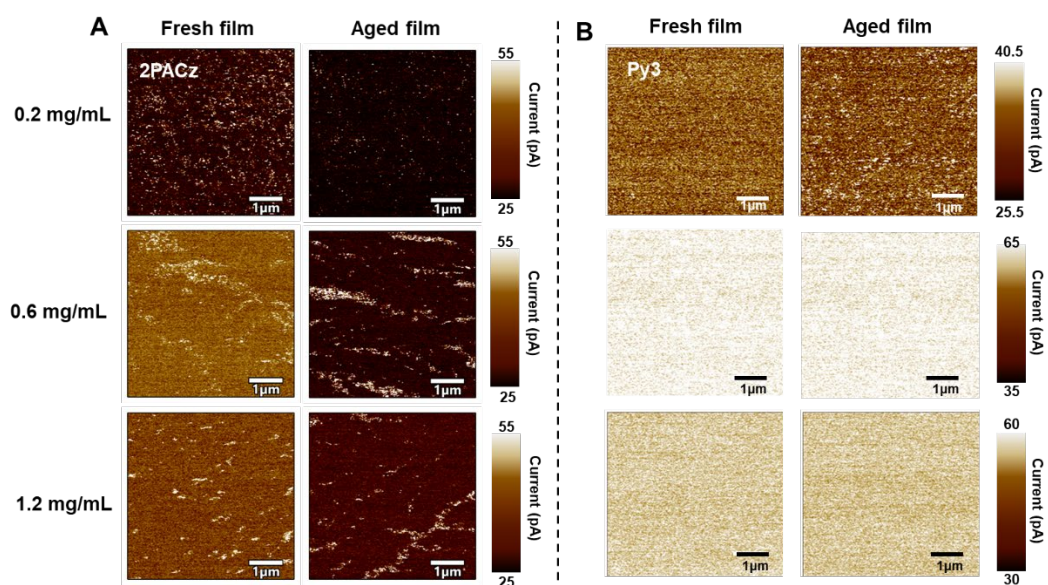
Supplementary Fig. 53 Initial PCEs of the devices for (A) the stability tests under open-circuit conditions; (B) MPP tracking at room temperature; (C) MPP tracking at an elevated temperature of ~55 °C.



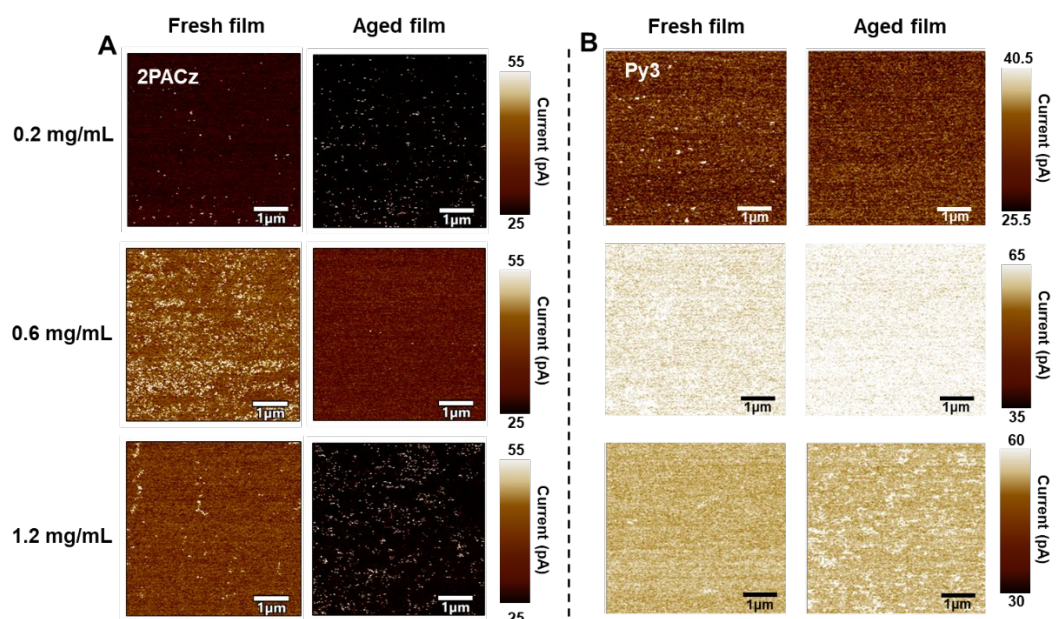
Supplementary Fig. 54 Operational stability test following the ISOS-L-1 protocol under continuous 1-sun illumination at open-circuit condition based on encapsulated devices with Py3 in ambient air ($\sim 30\%$ RH, $\sim 35^\circ\text{C}$). Error bars represent the standard deviation of five devices for each condition.



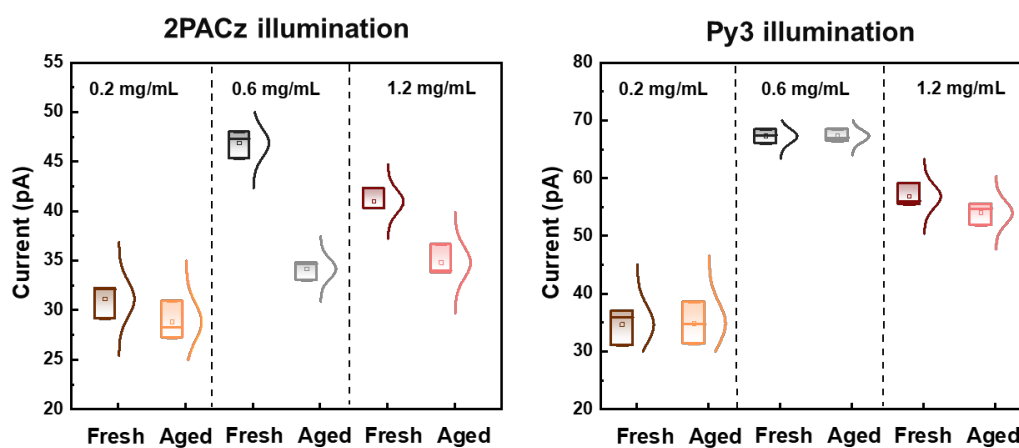
Supplementary Fig. 55 Calibration plots and the determined concentration of (A) 2PACz, and (B) Py3 in the UPLC measurements. The densities of the molecules were determined to be $3.31 \cdot 10^{15}/\text{cm}^2$ and $3.39 \cdot 10^{15}/\text{cm}^2$ in the samples of Py3 and 2PACz, respectively.



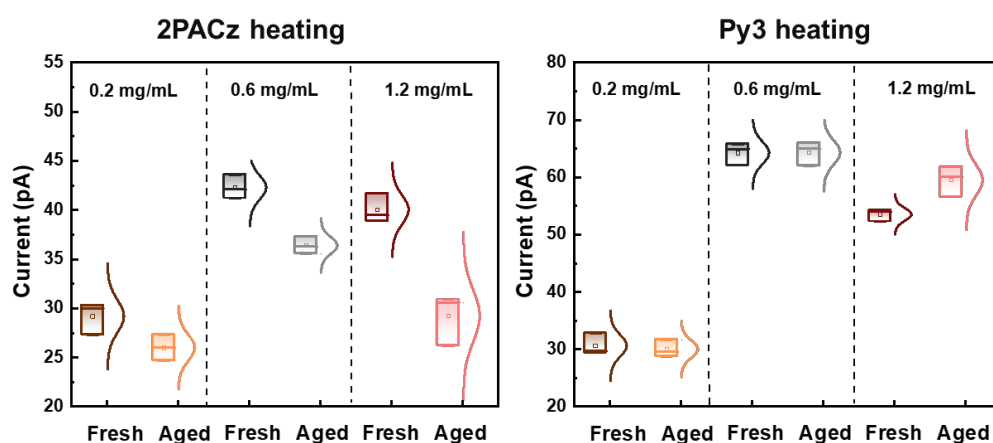
Supplementary Fig. 56 Conductive-AFM images of the 2PACz and Py3 samples with varied thicknesses before and after the aging test under continuous light illumination for three days.



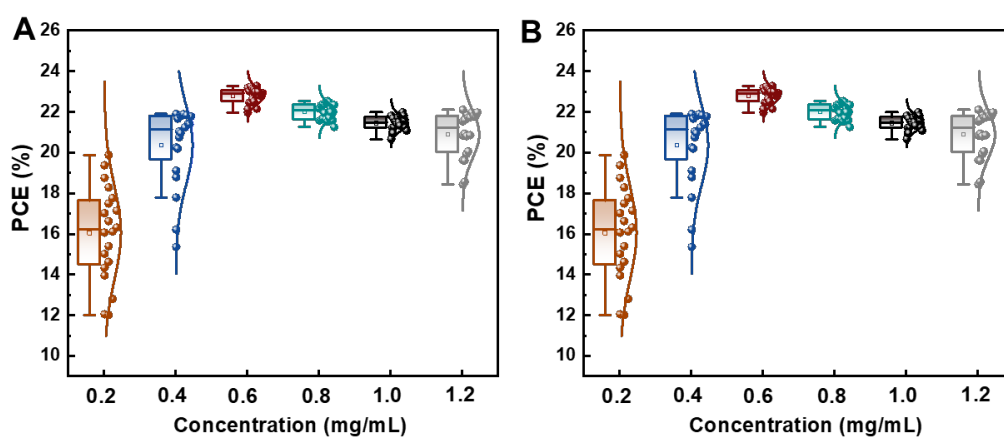
Supplementary Fig. 57 Conductive-AFM images of the (A) 2PACz and (B) Py3 samples with varied thicknesses before and after the aging test under continuous heating at 55 °C for three days.



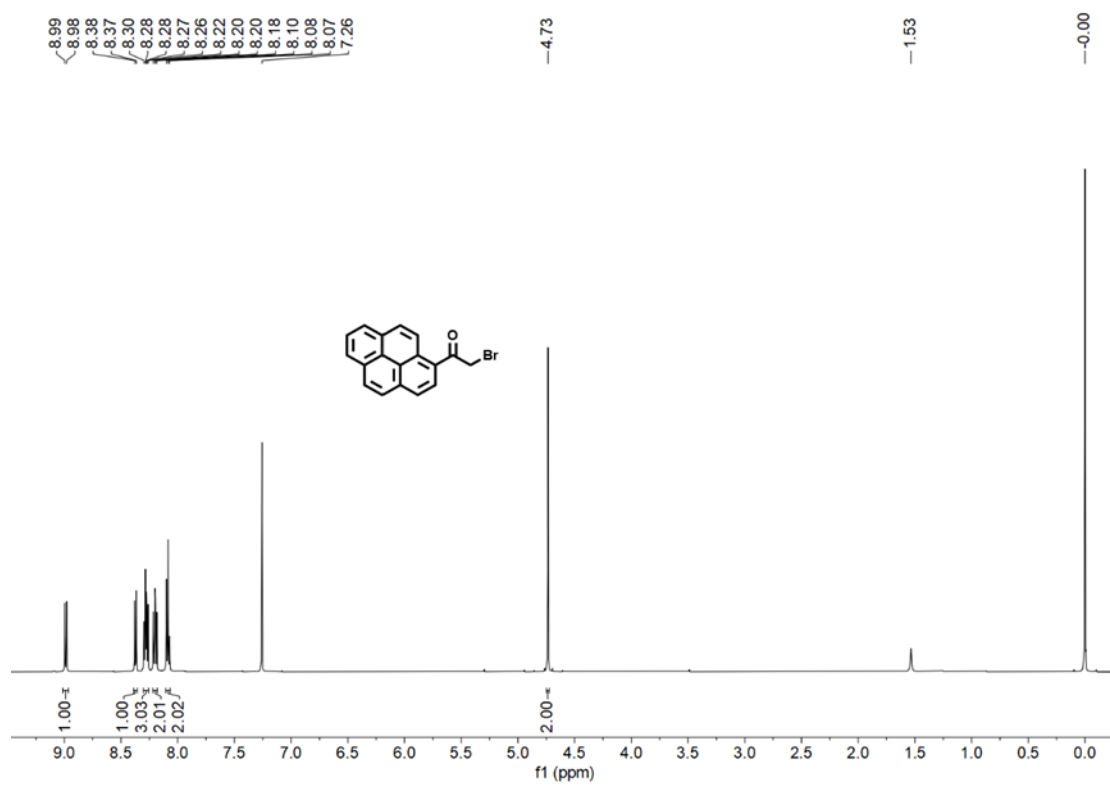
Supplementary Fig. 58 Statistical current signals obtained by conductive-AFM measurements on the 2PACz and Py3 samples with different thicknesses before and after the aging test under continuous light illumination for three days. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



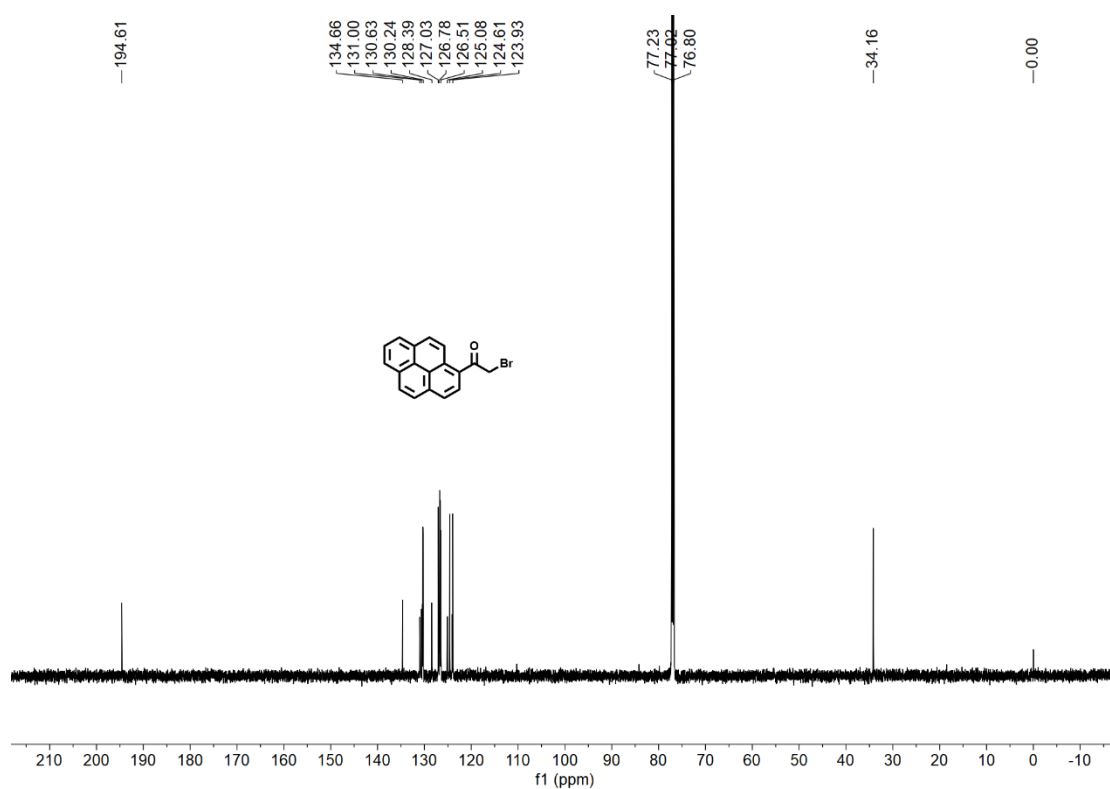
Supplementary Fig. 59 Statistical current signals obtained by conductive-AFM measurements on the 2PACz and Py3 samples with different thicknesses before and after the aging test under continuous heating at 55 °C for three days. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



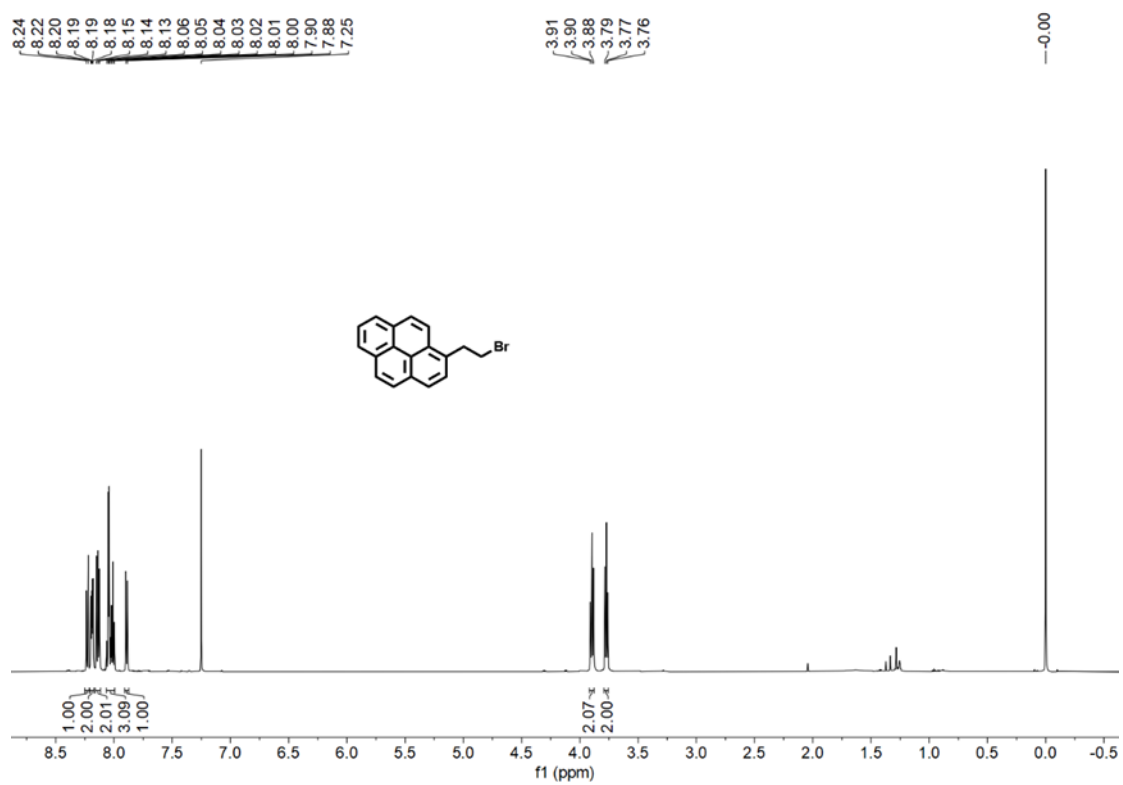
Supplementary Fig. 60 Statistical PCEs of perovskite devices fabricated using (A) 2PACz and (B) Py3 layers with different thicknesses, achieved by varying the concentration of the molecular precursors. For the statistical analysis, center line, median; box limits, 25th and 75th percentiles; curve, normal distribution curve; whiskers, outliers.



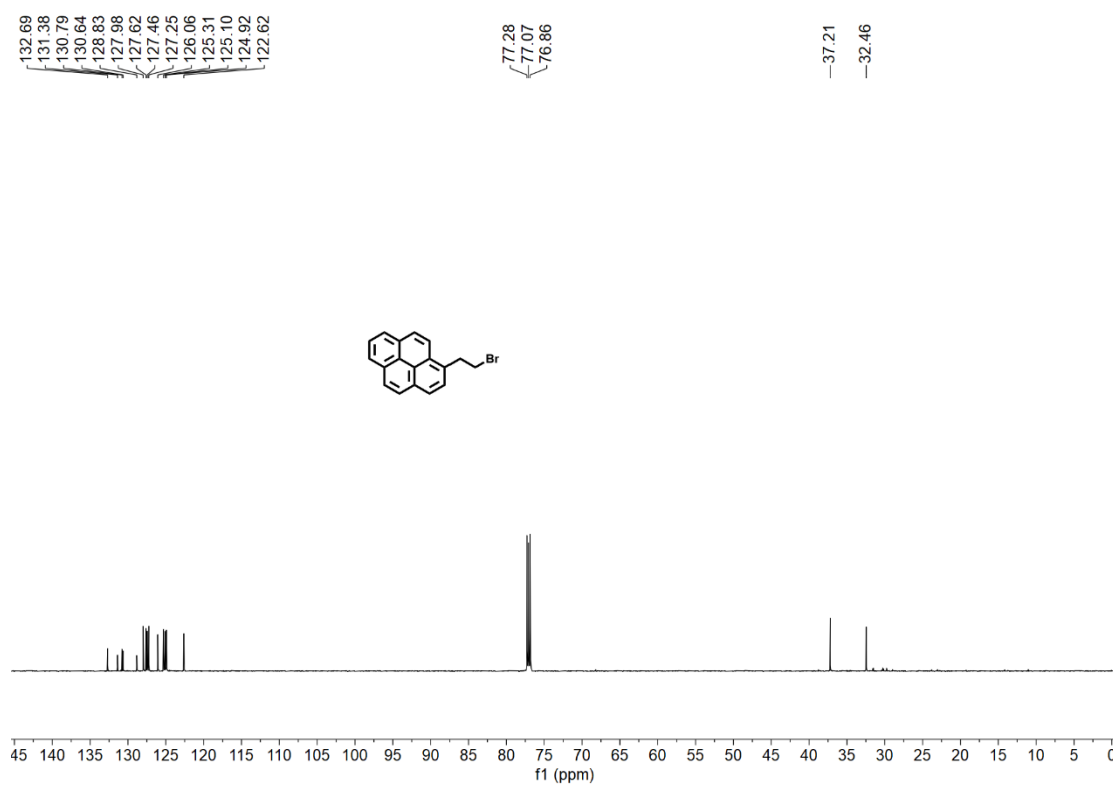
Supplementary Fig. 61 The ^1H NMR spectrum of M1 in CDCl_3 .



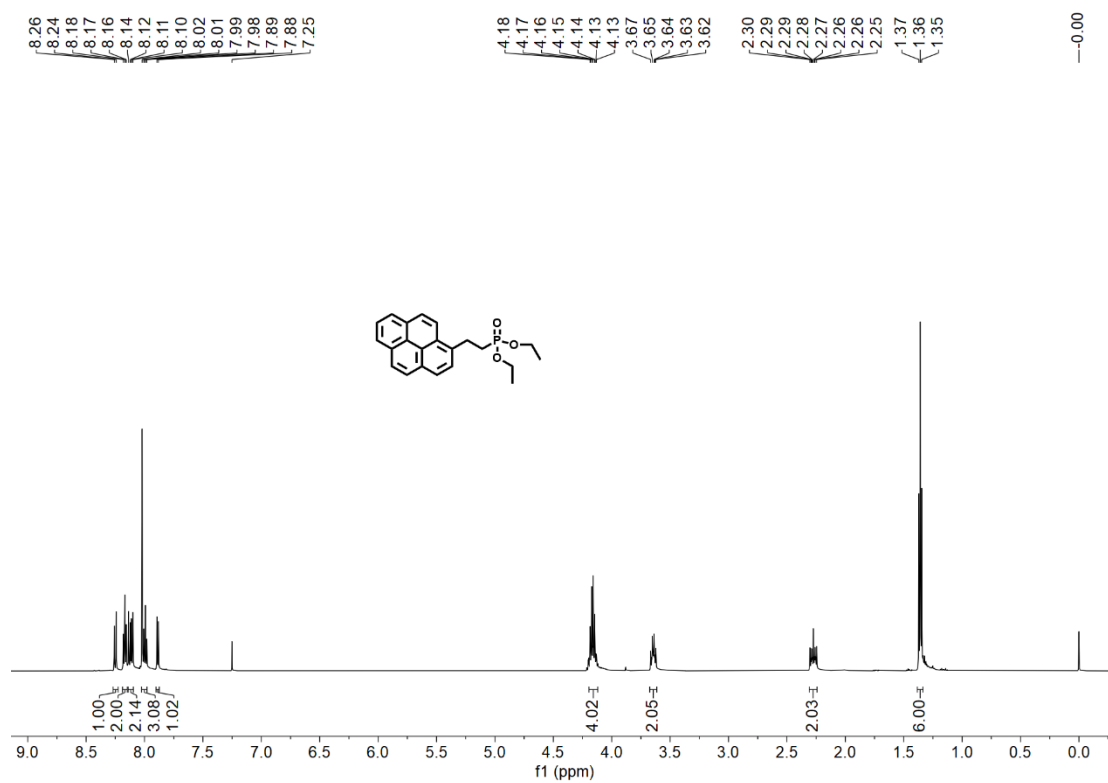
Supplementary Fig. 62 The ¹³C NMR spectrum of M1 in CDCl₃.



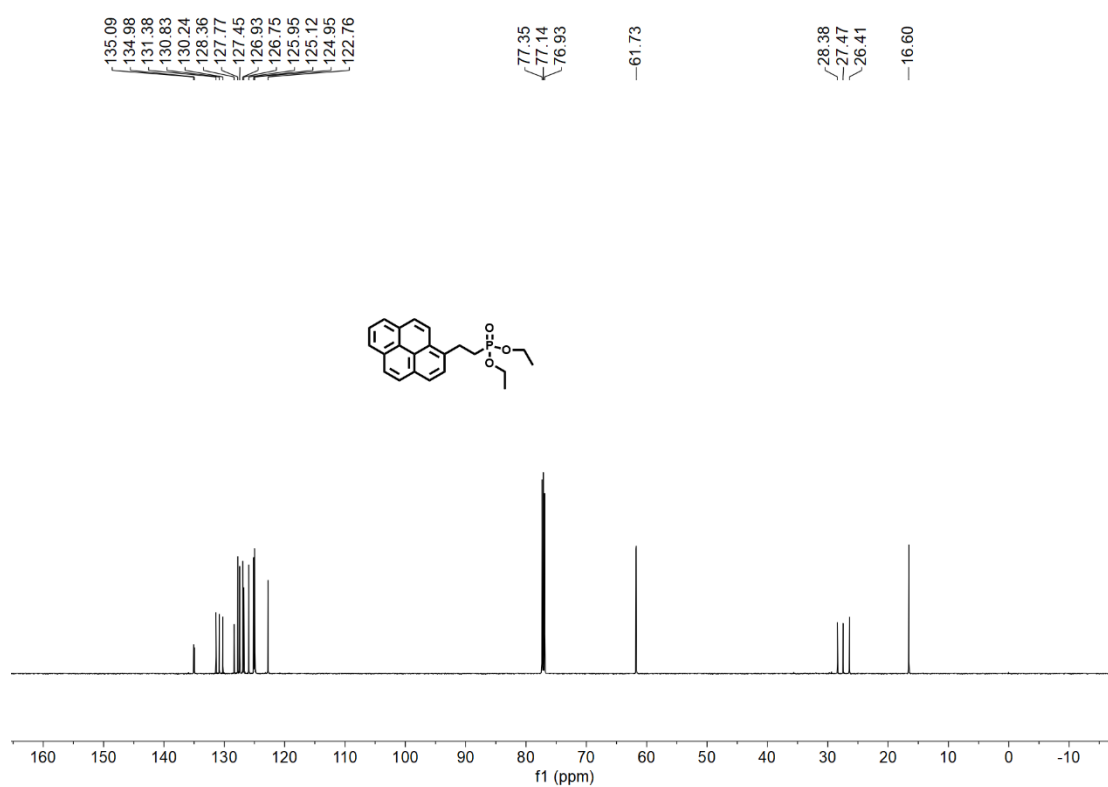
Supplementary Fig. 63 The ¹H NMR spectrum of M2 in CDCl₃.



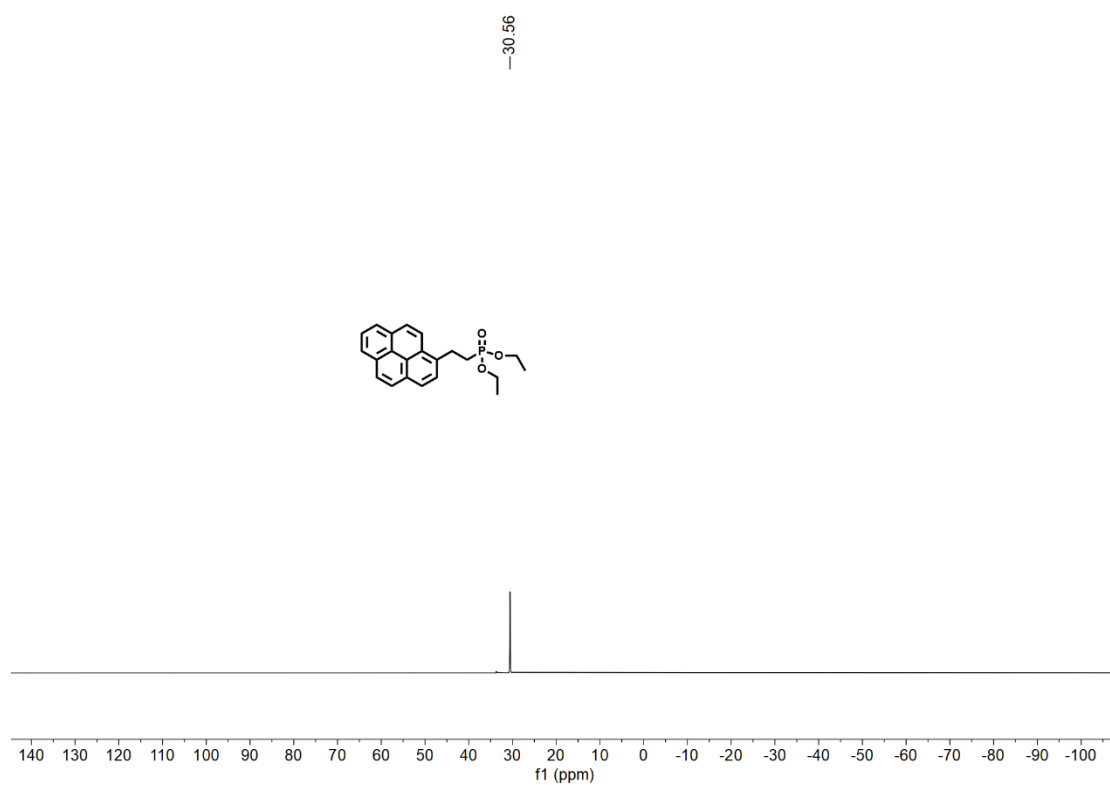
Supplementary Fig. 64 The ¹³C NMR spectrum of M2 in CDCl₃.



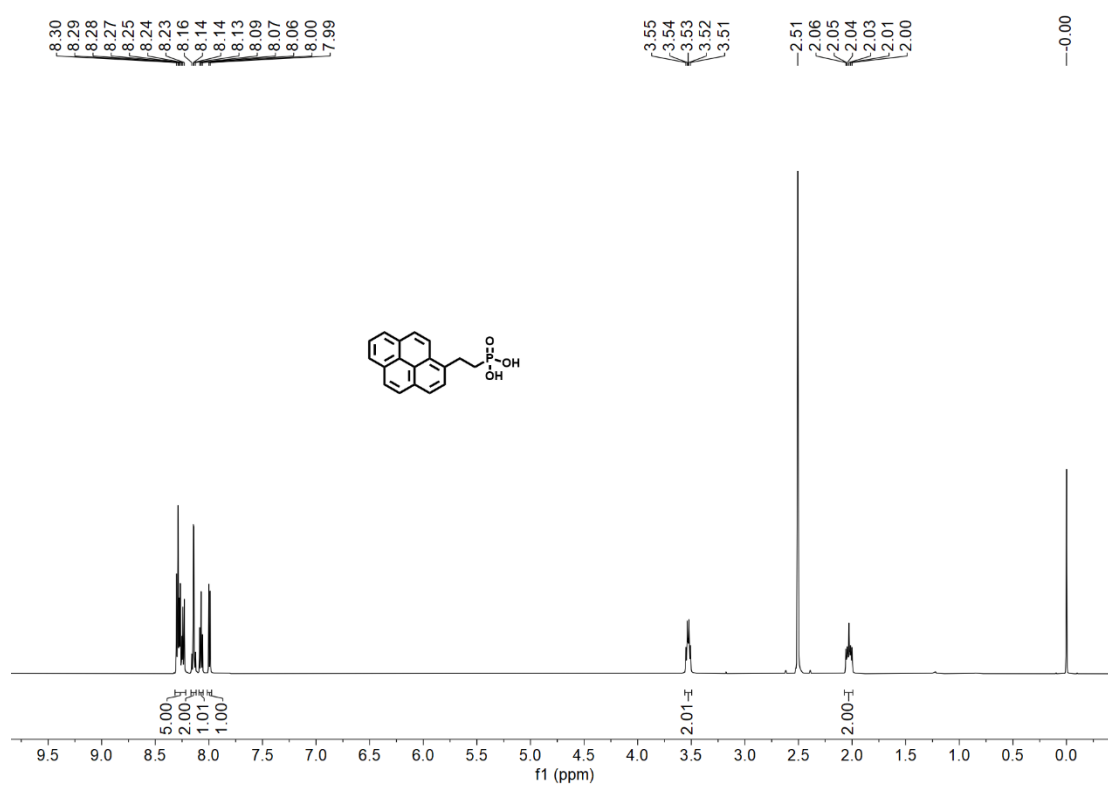
Supplementary Fig. 65 The ^1H NMR spectrum of M3 in CDCl_3 .



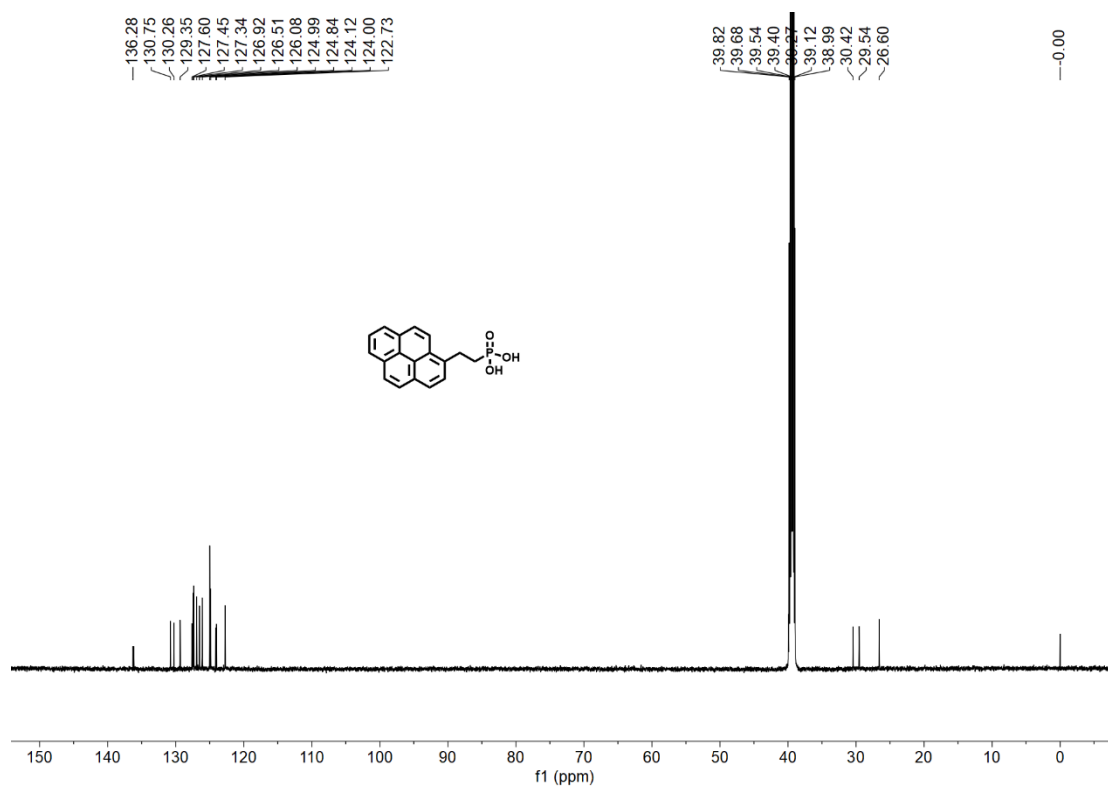
Supplementary Fig. 66 The ¹³C NMR spectrum of M3 in CDCl₃.



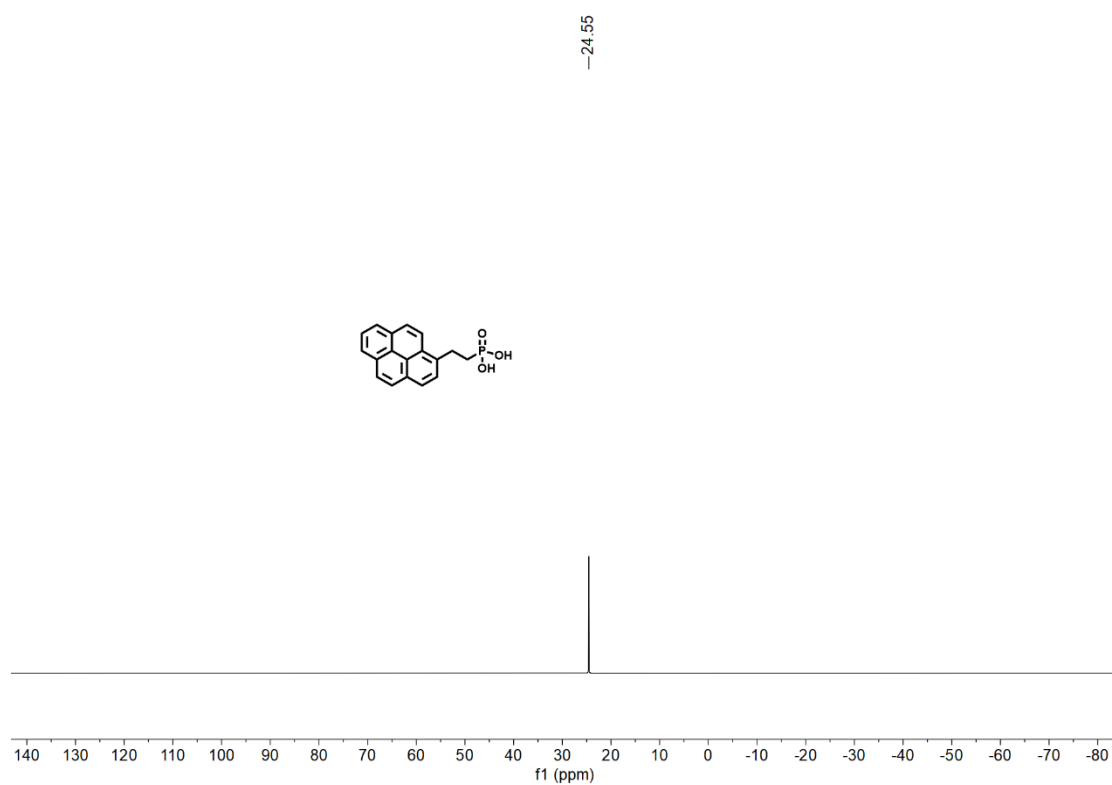
Supplementary Fig. 67 The ^{31}P NMR spectrum of M3 in CDCl_3 .



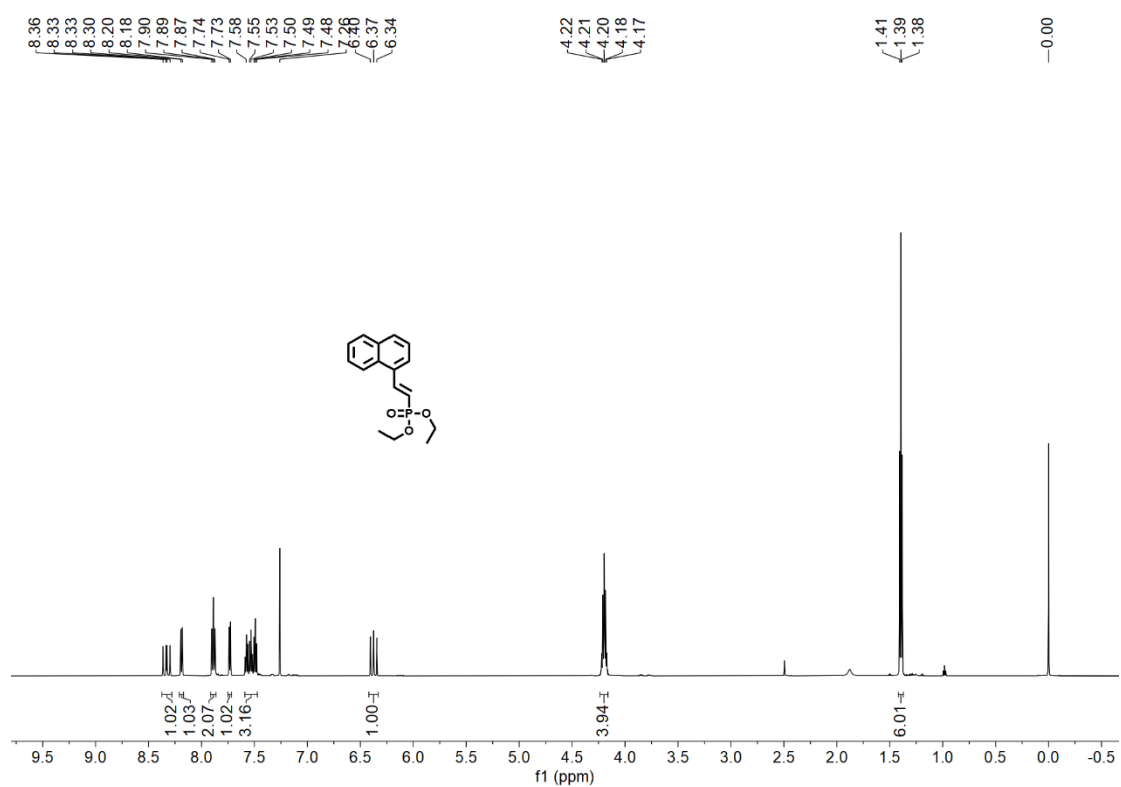
Supplementary Fig. 68 The ¹H NMR spectrum of Py3 in DMSO-*d*₆.



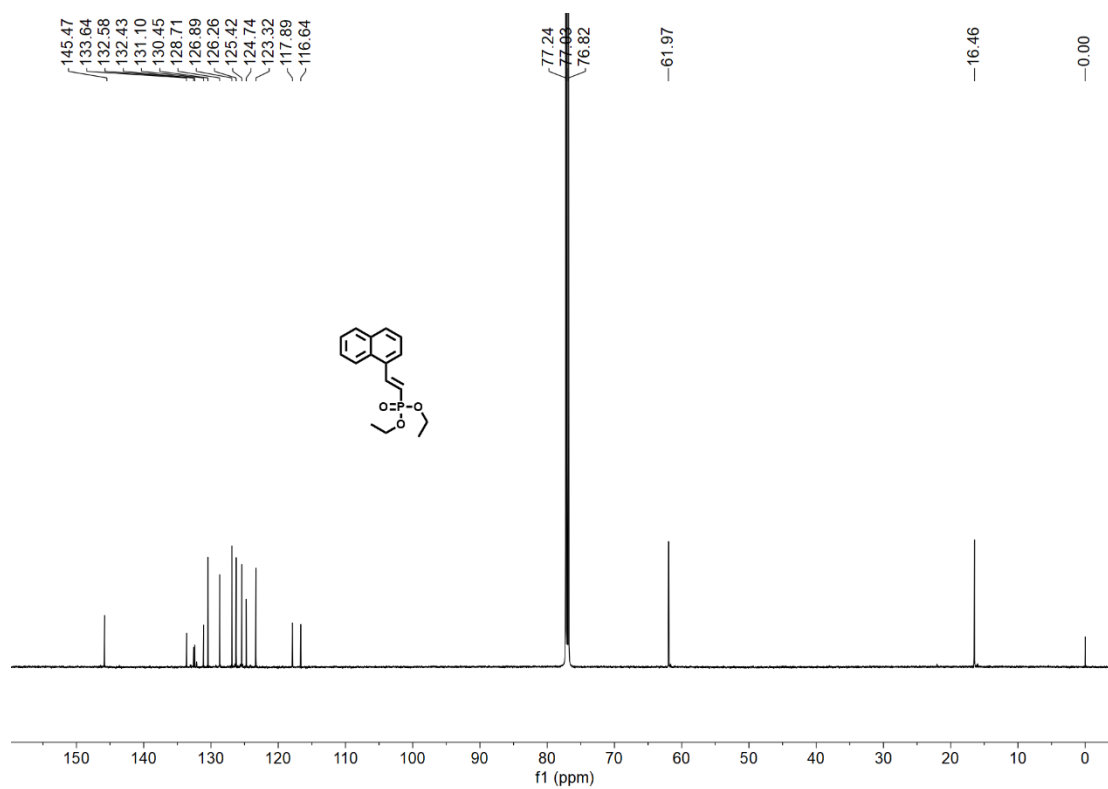
Supplementary Fig. 69 The ¹³C NMR spectrum of Py3 in DMSO-*d*₆.



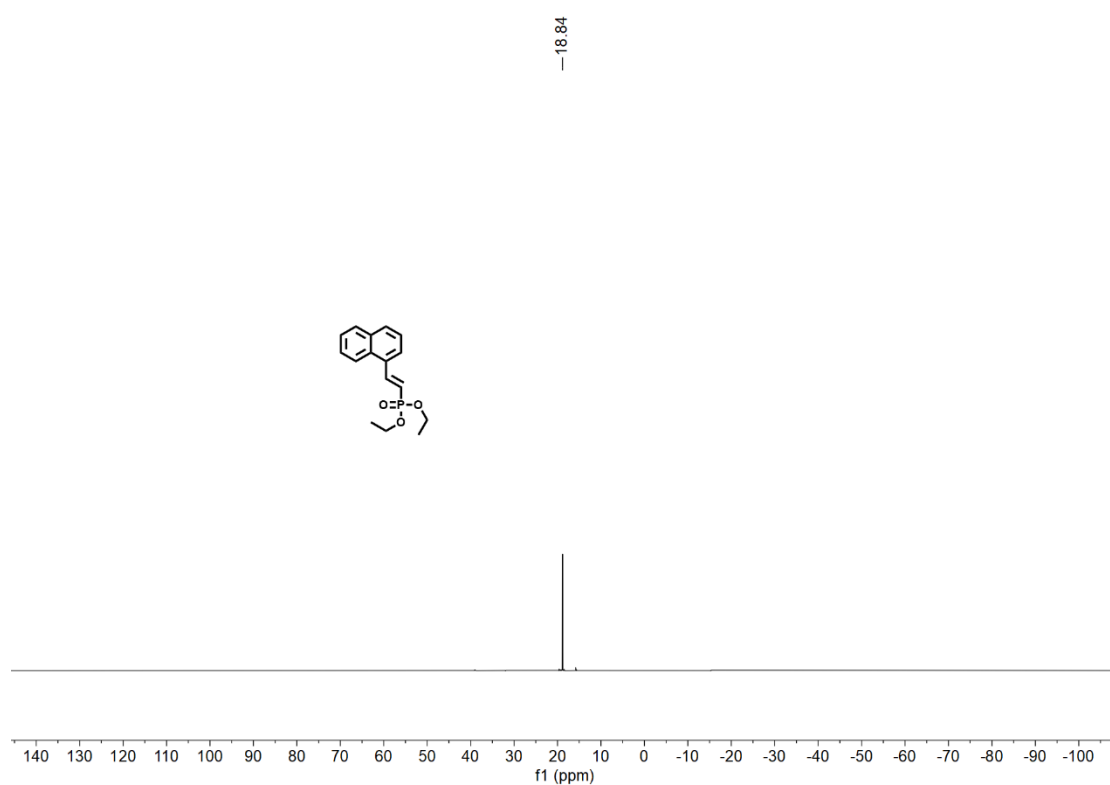
Supplementary Fig. 70 The ^{31}P NMR spectrum of Py3 in $\text{DMSO-}d_6$.



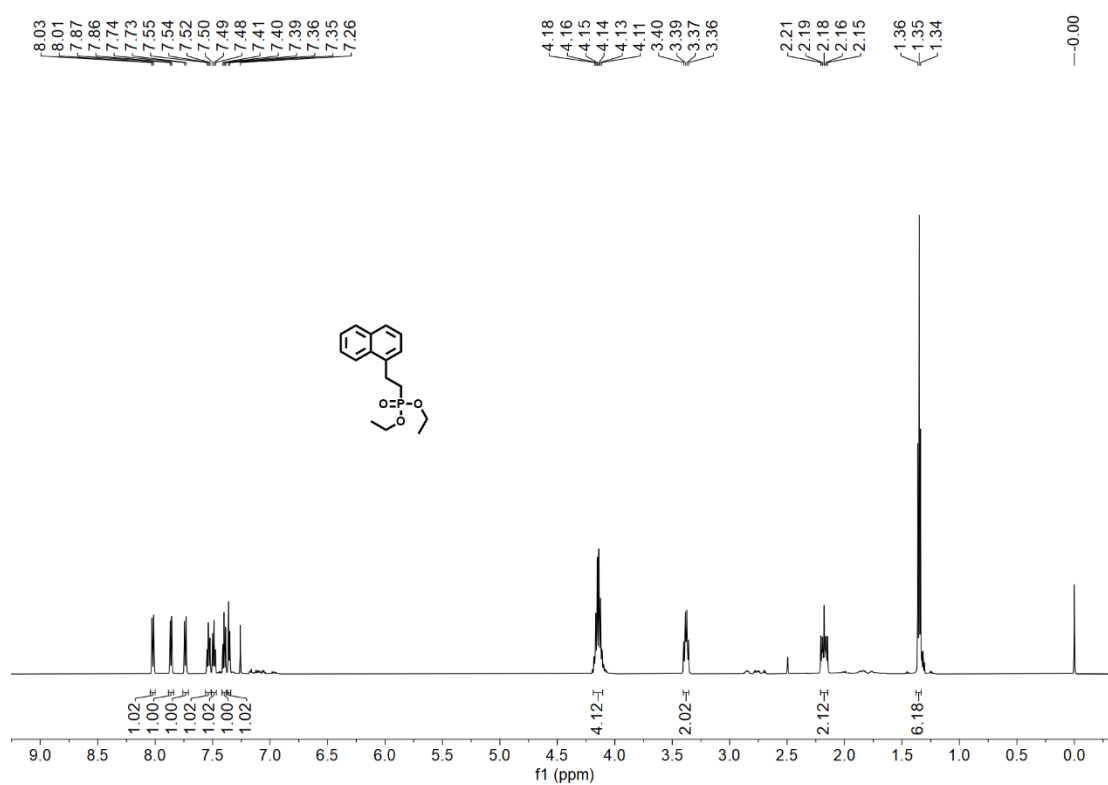
Supplementary Fig. 71 The ^1H NMR spectrum of M4 in CDCl_3 .



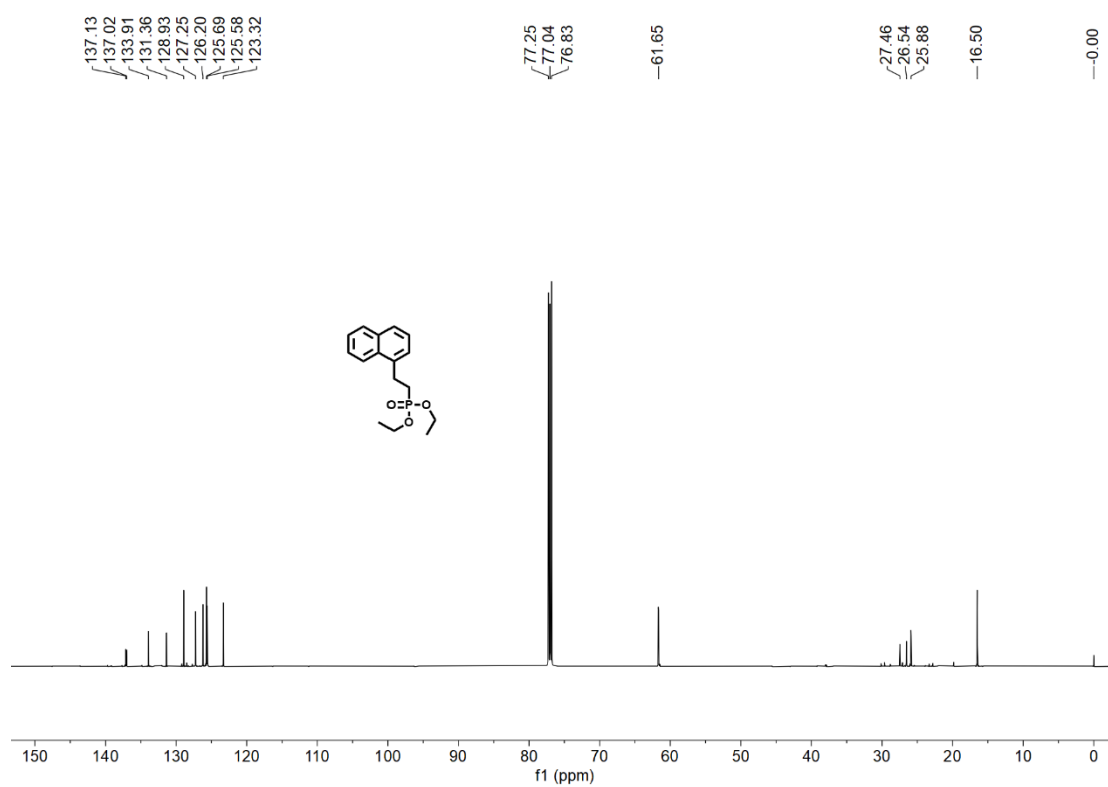
Supplementary Fig. 72 The ^{13}C NMR spectrum of M4 in CDCl_3 .



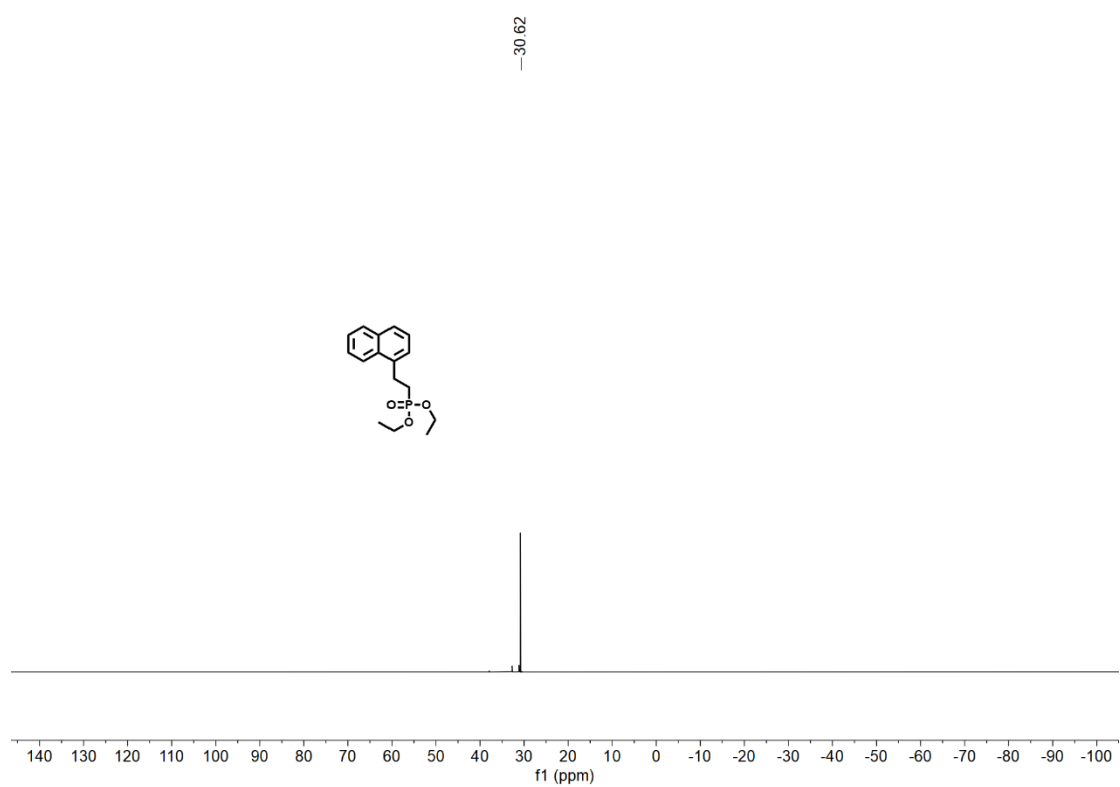
Supplementary Fig. 73 The ^{31}P NMR spectrum of M4 in CDCl_3 .



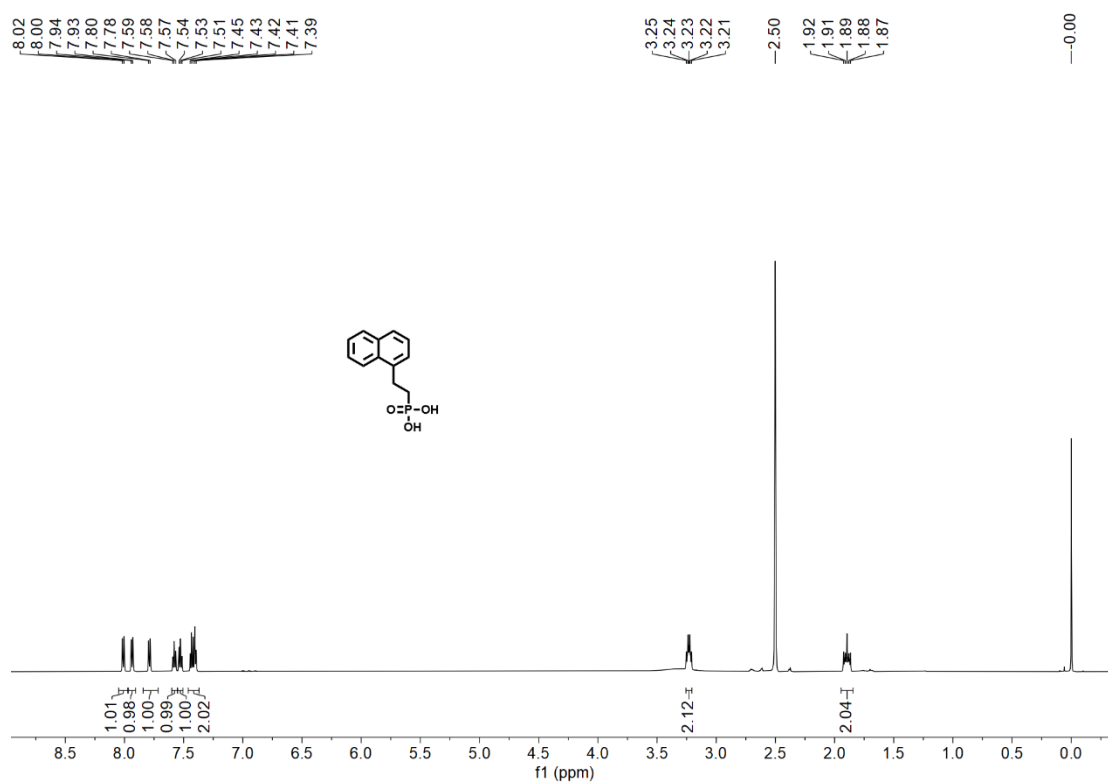
Supplementary Fig. 74 The ^1H NMR spectrum of M5 in CDCl_3 .



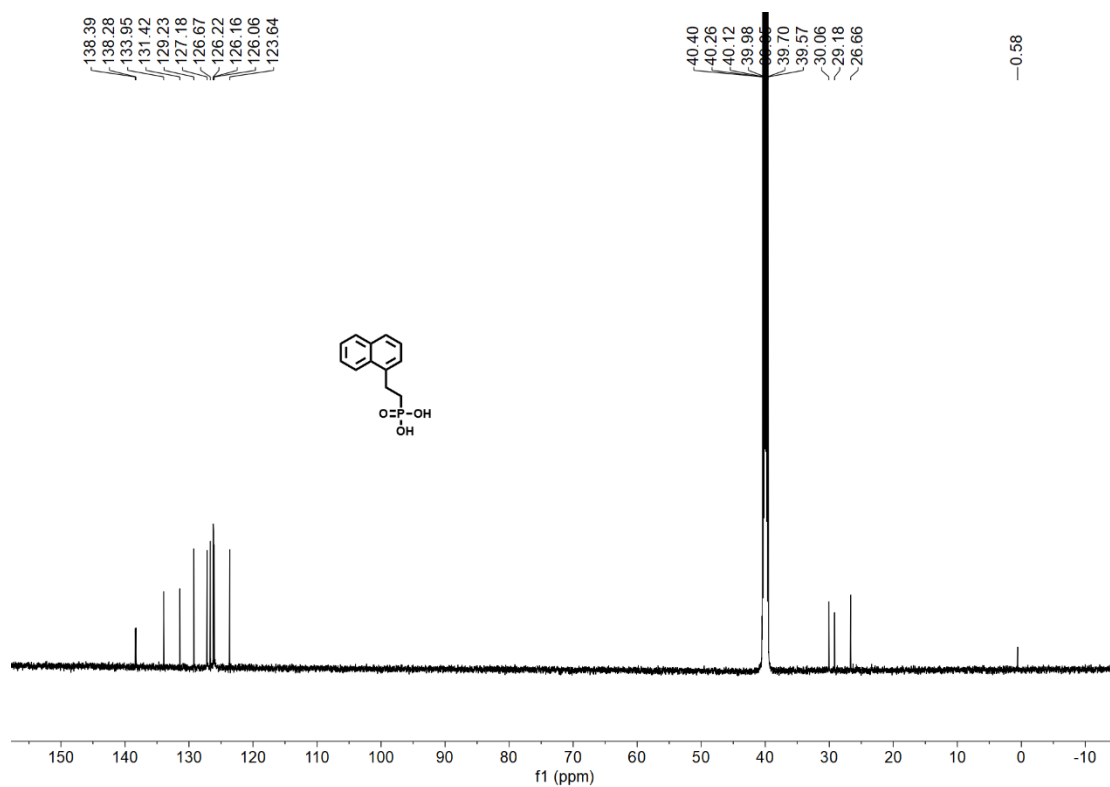
Supplementary Fig. 75 The ¹³C NMR spectrum of M5 in CDCl₃.



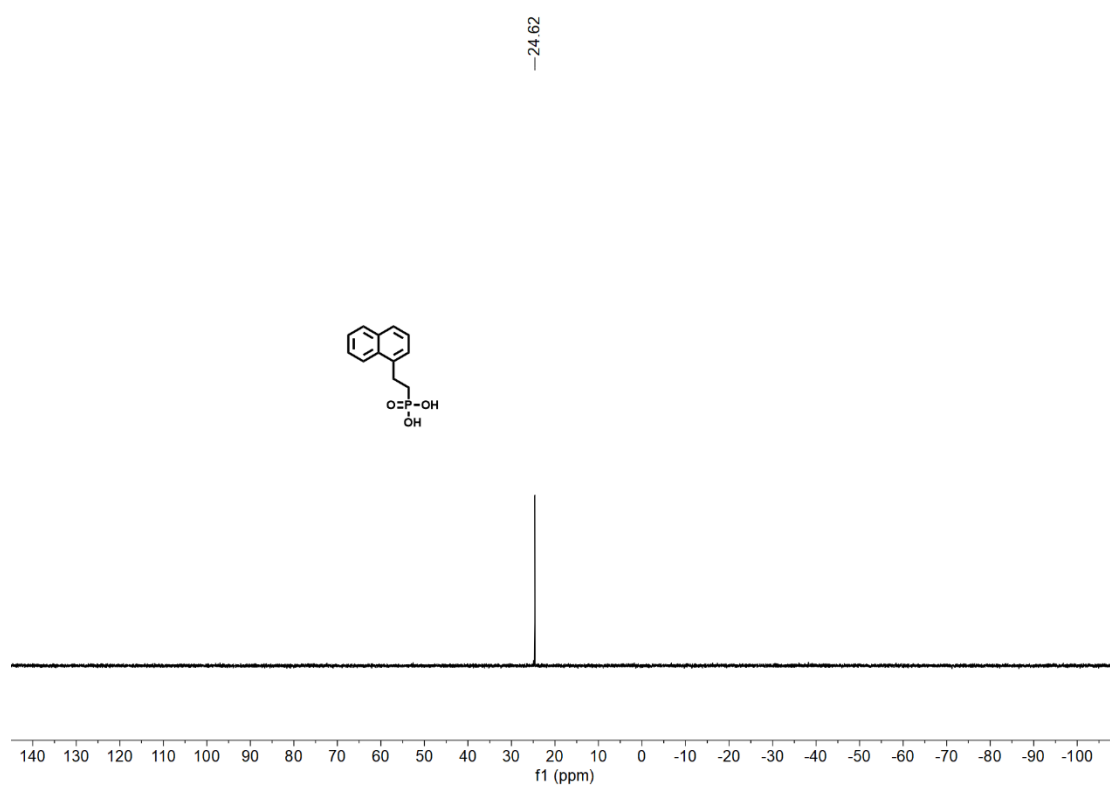
Supplementary Fig. 76 The ^{31}P NMR spectrum of M5 in CDCl_3 .



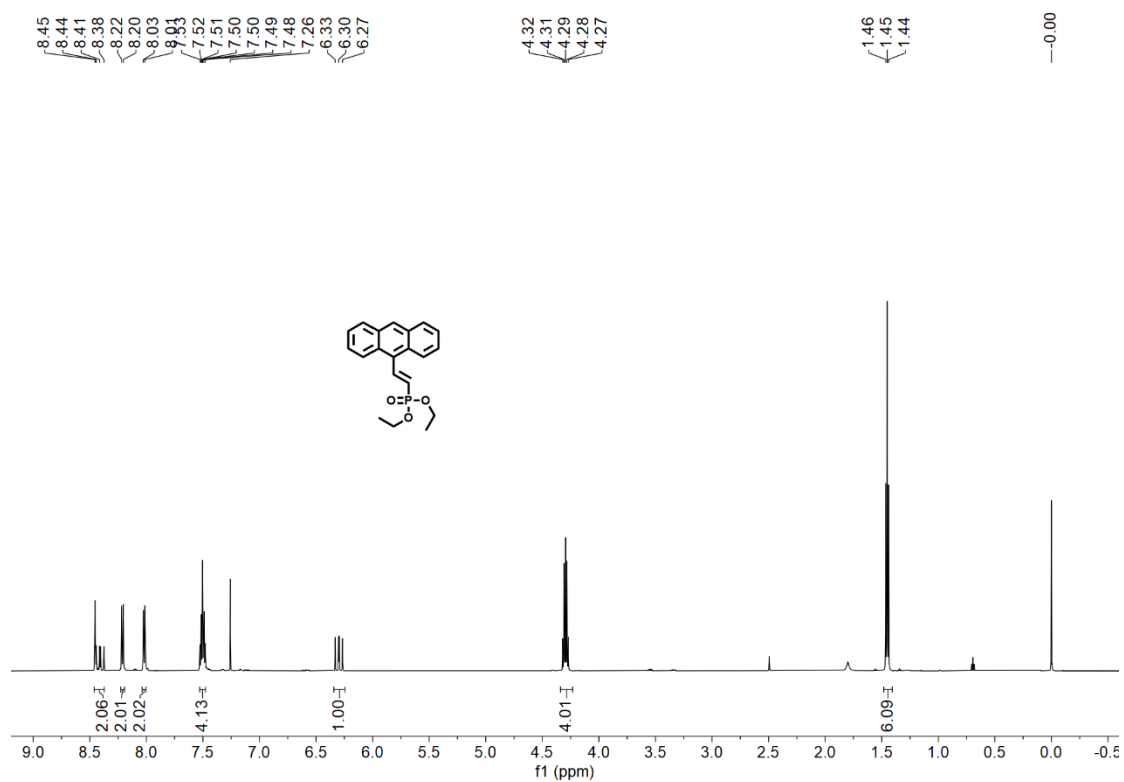
Supplementary Fig. 77 The ¹H NMR spectrum of naphthalene-based variant in DMSO-*d*₆.



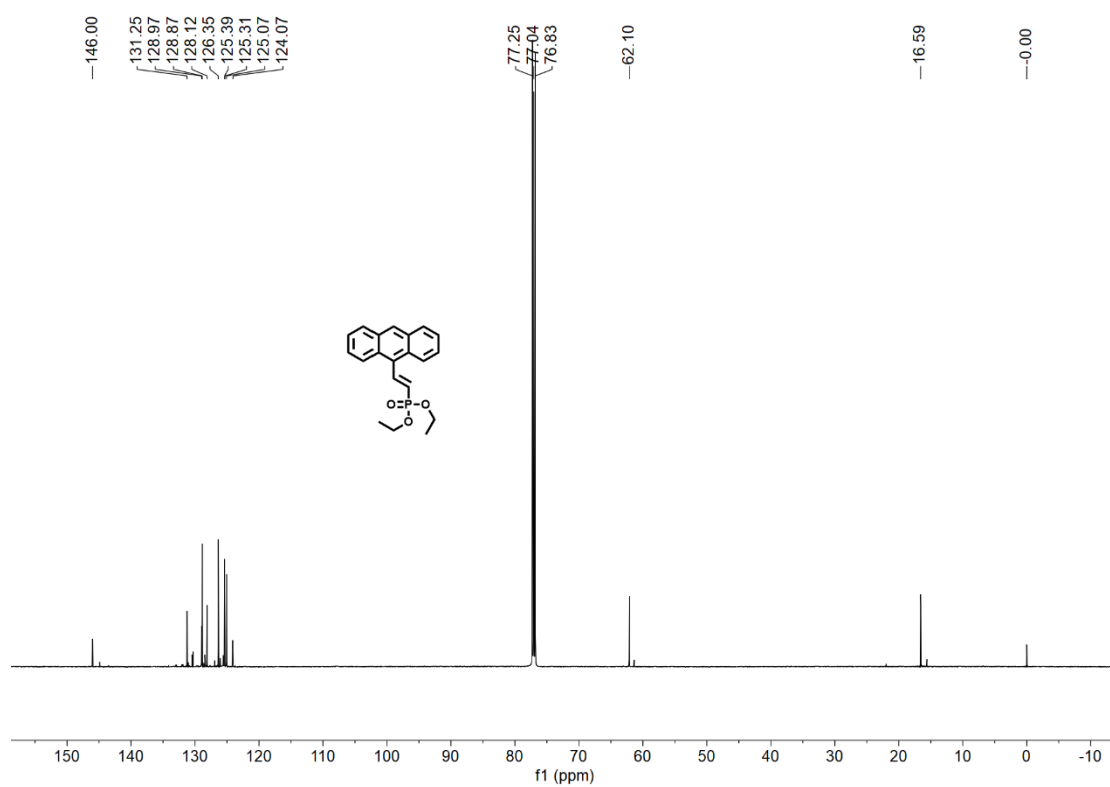
Supplementary Fig. 78 The ¹³C NMR spectrum of naphthalene-based variant in DMSO-*d*₆.



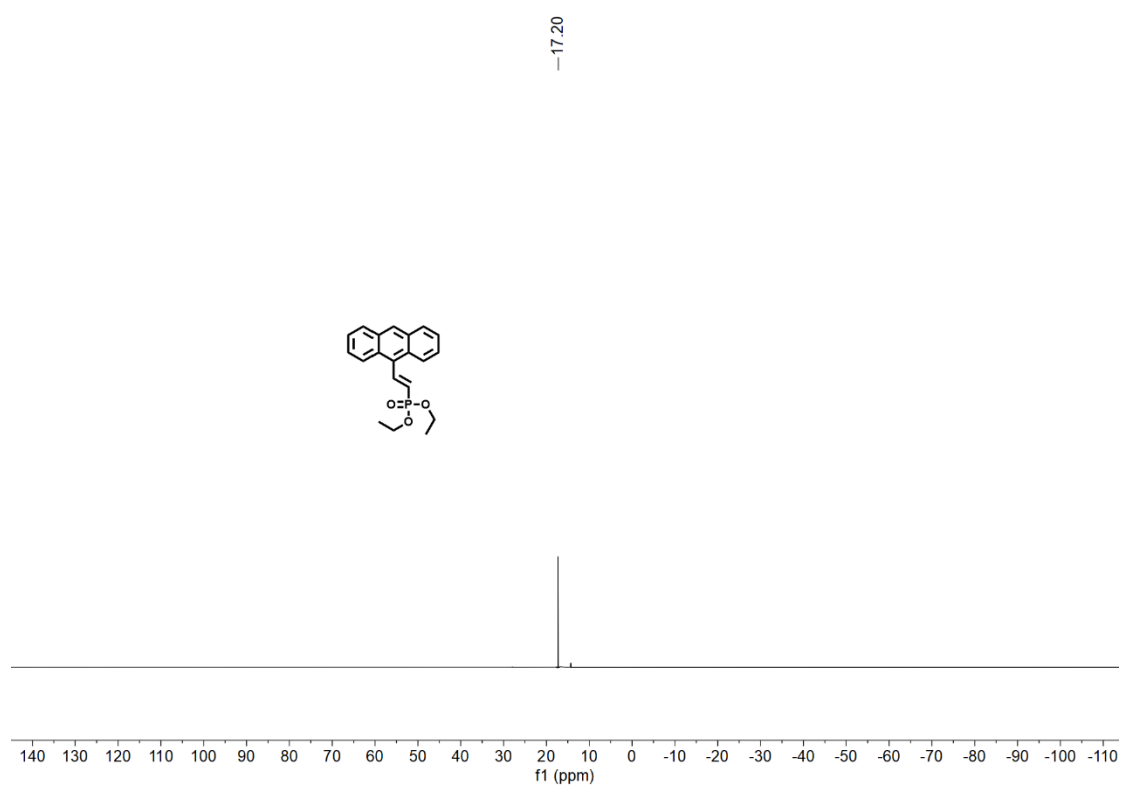
Supplementary Fig. 79 The ^{31}P NMR spectrum of naphthalene-based variant in $\text{DMSO-}d_6$.



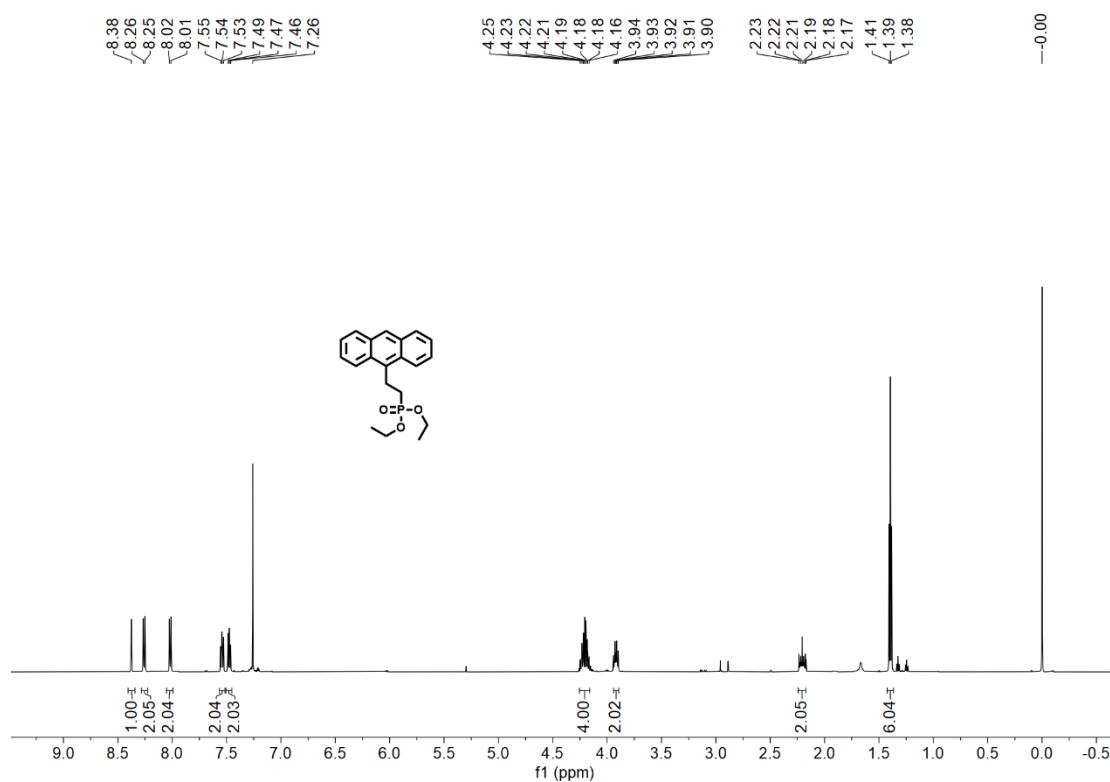
Supplementary Fig. 80 The ¹H NMR spectrum of M6 in CDCl₃.



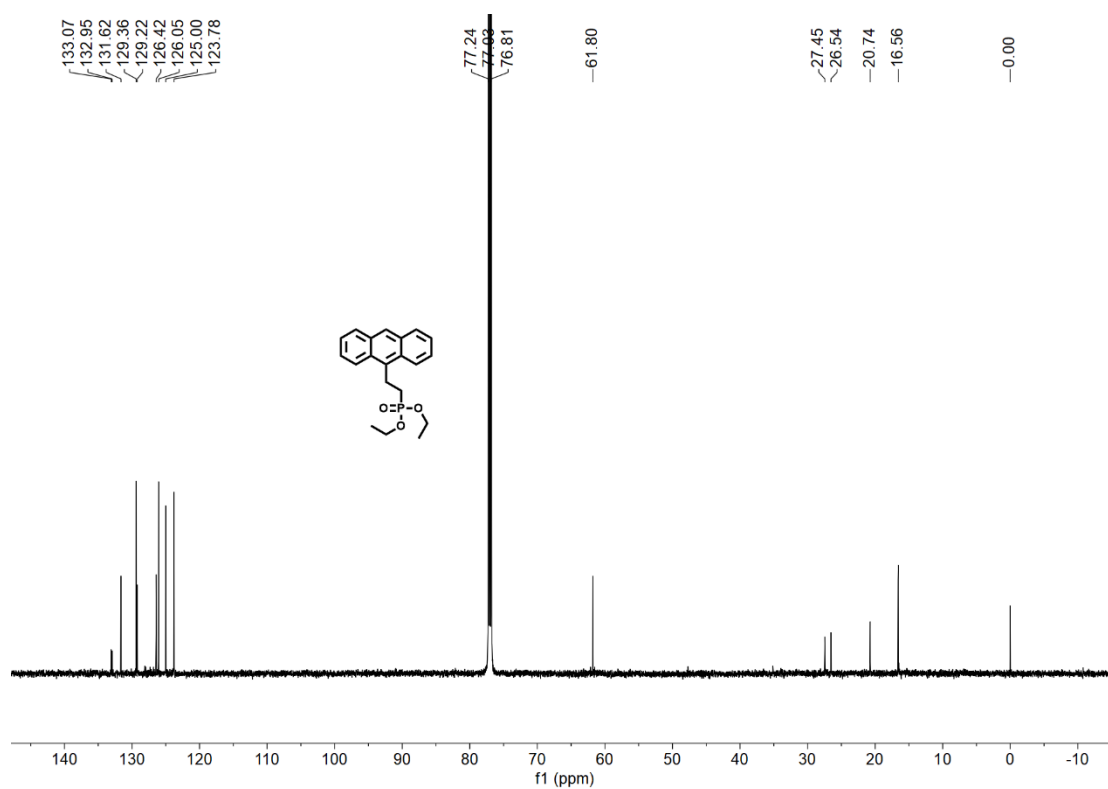
Supplementary Fig. 81 The ¹³C NMR spectrum of M6 in CDCl₃.



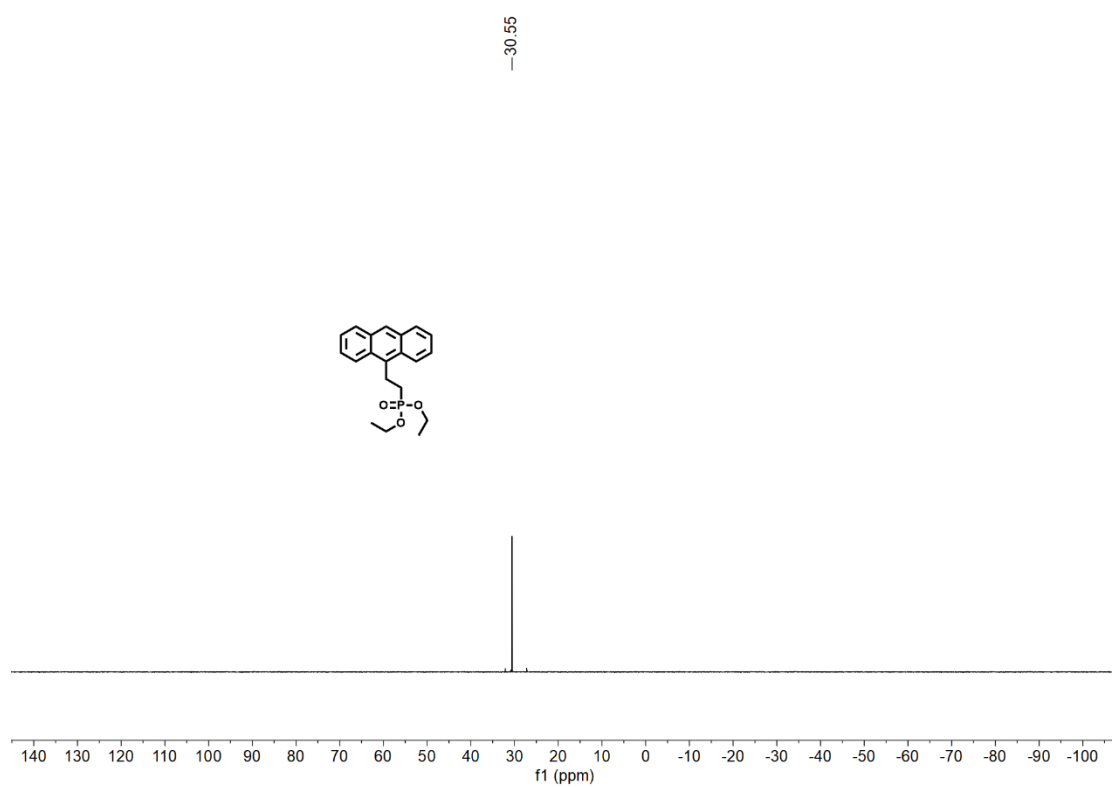
Supplementary Fig. 82 The ^{31}P NMR spectrum of M6 in CDCl_3 .



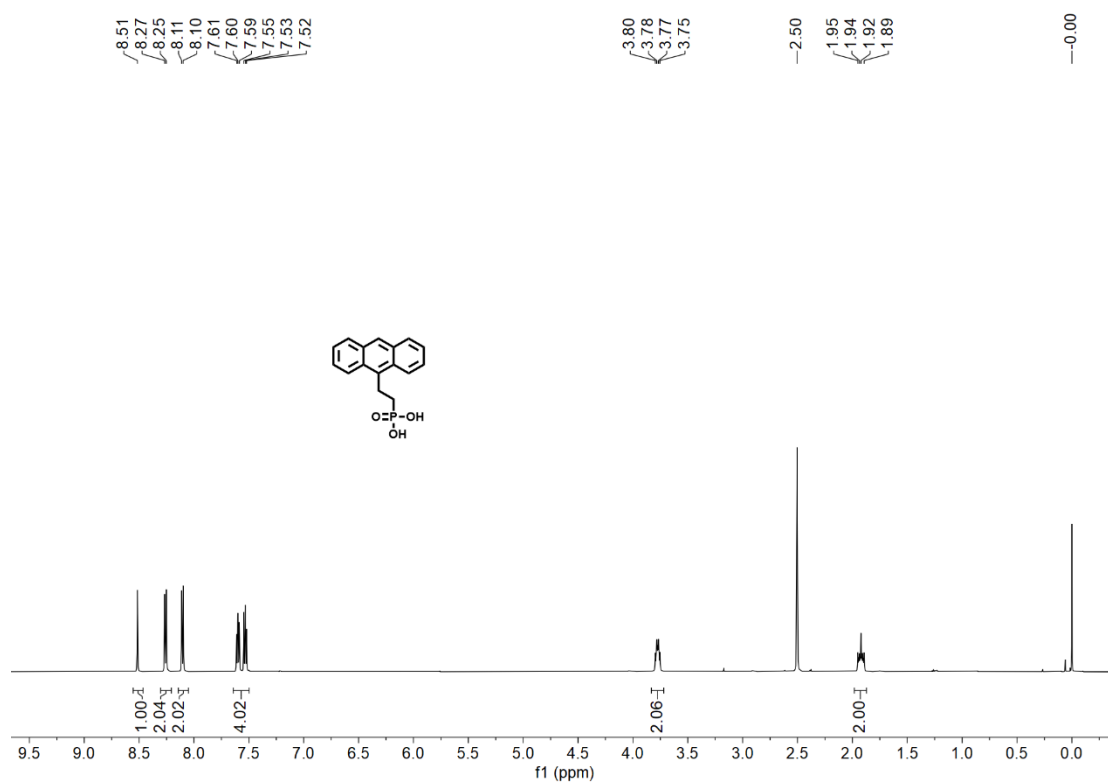
Supplementary Fig. 83 The ^1H NMR spectrum of M7 in CDCl_3 .



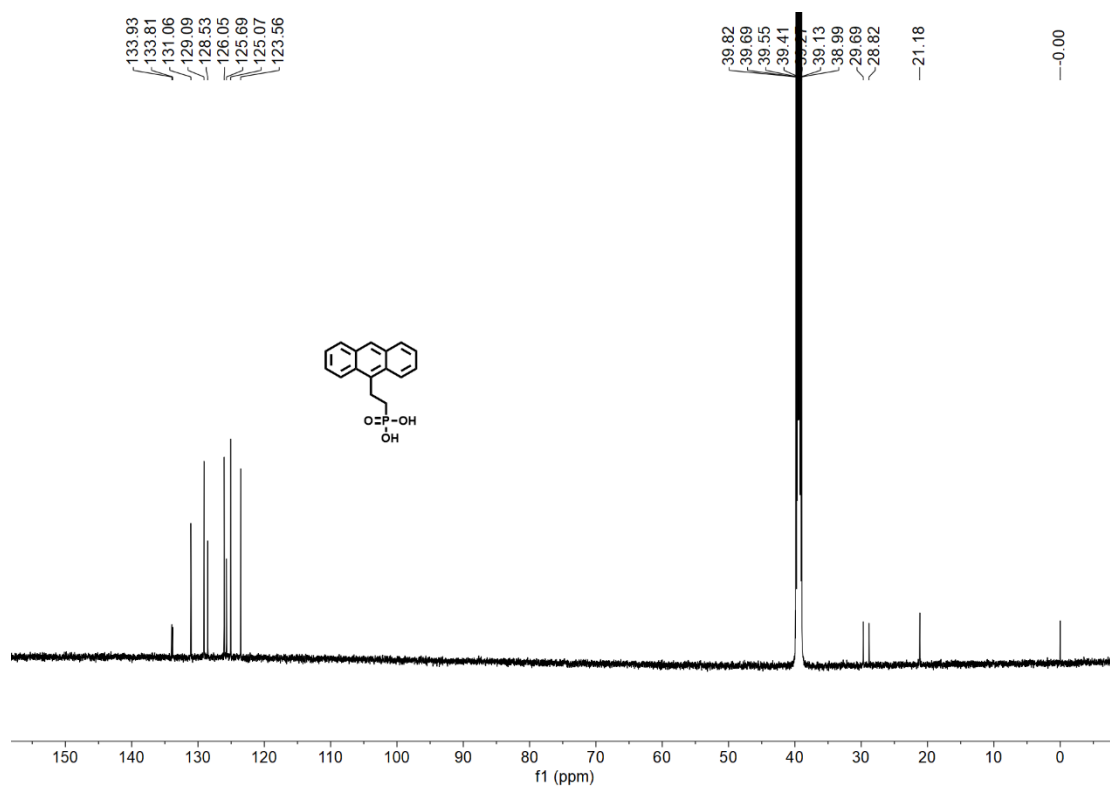
Supplementary Fig. 84 The ¹³C NMR spectrum of M7 in CDCl₃.



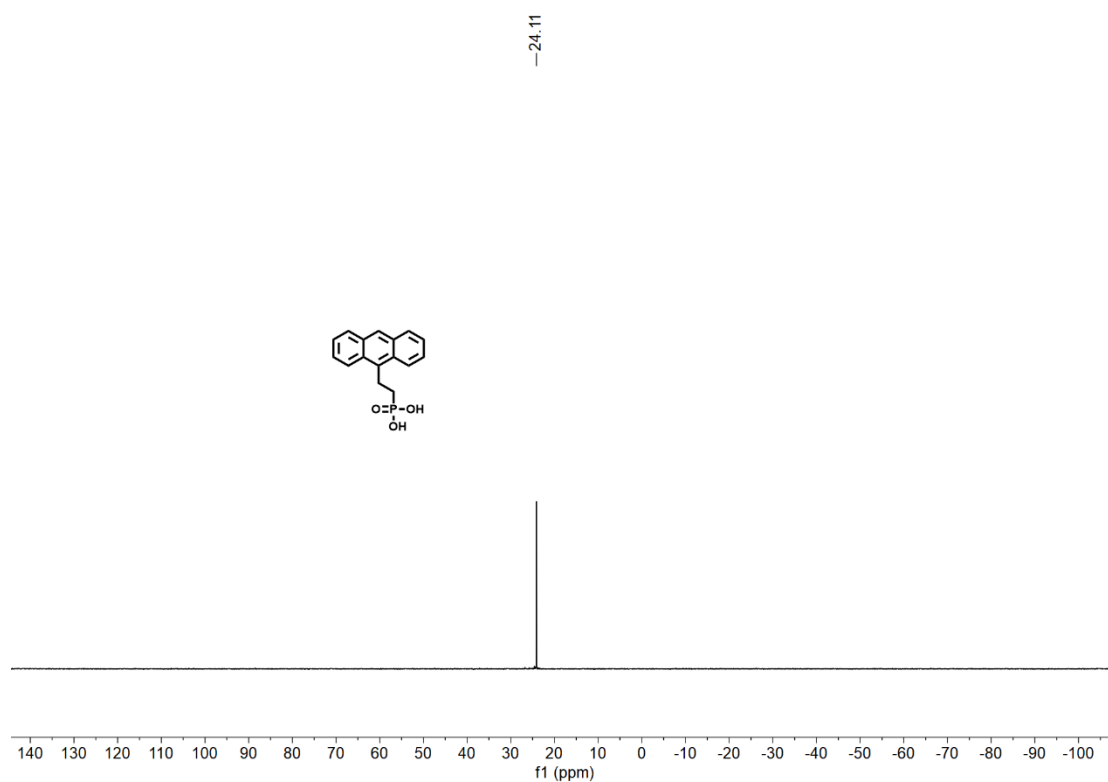
Supplementary Fig. 85 The ³¹P NMR spectrum of M7 in CDCl₃.



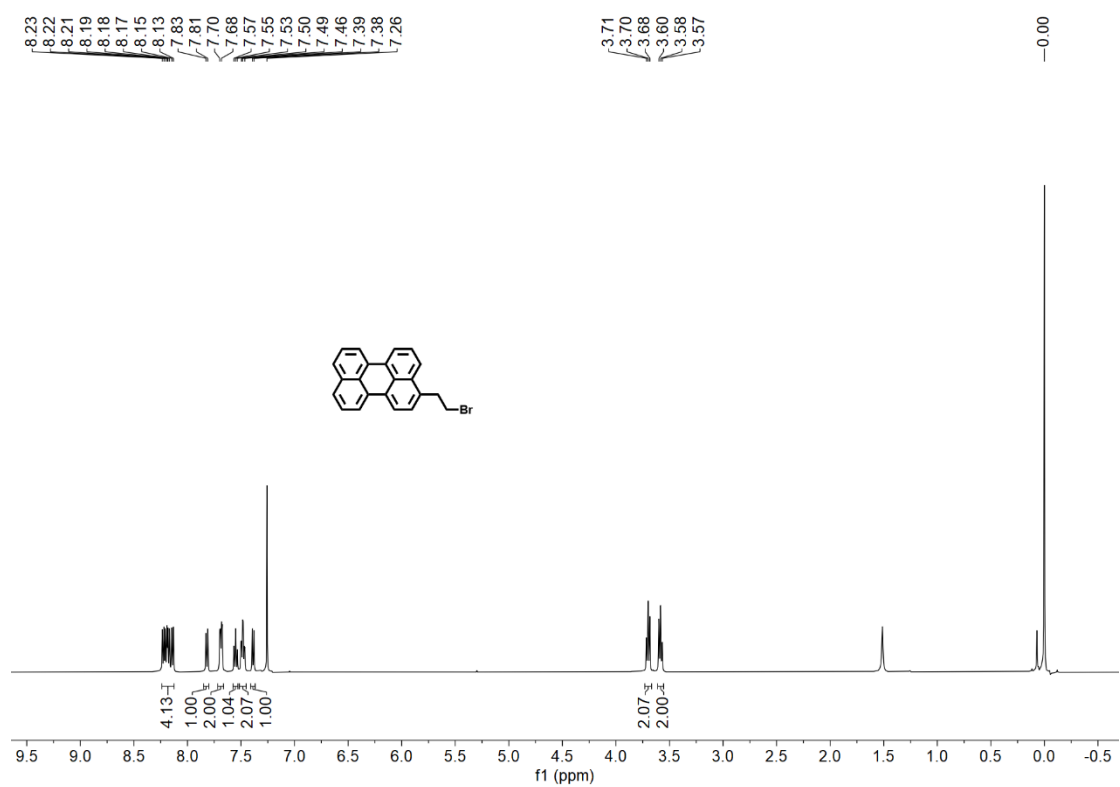
Supplementary Fig. 86 The ¹H NMR spectrum of anthracene-based variant in DMSO-*d*₆.



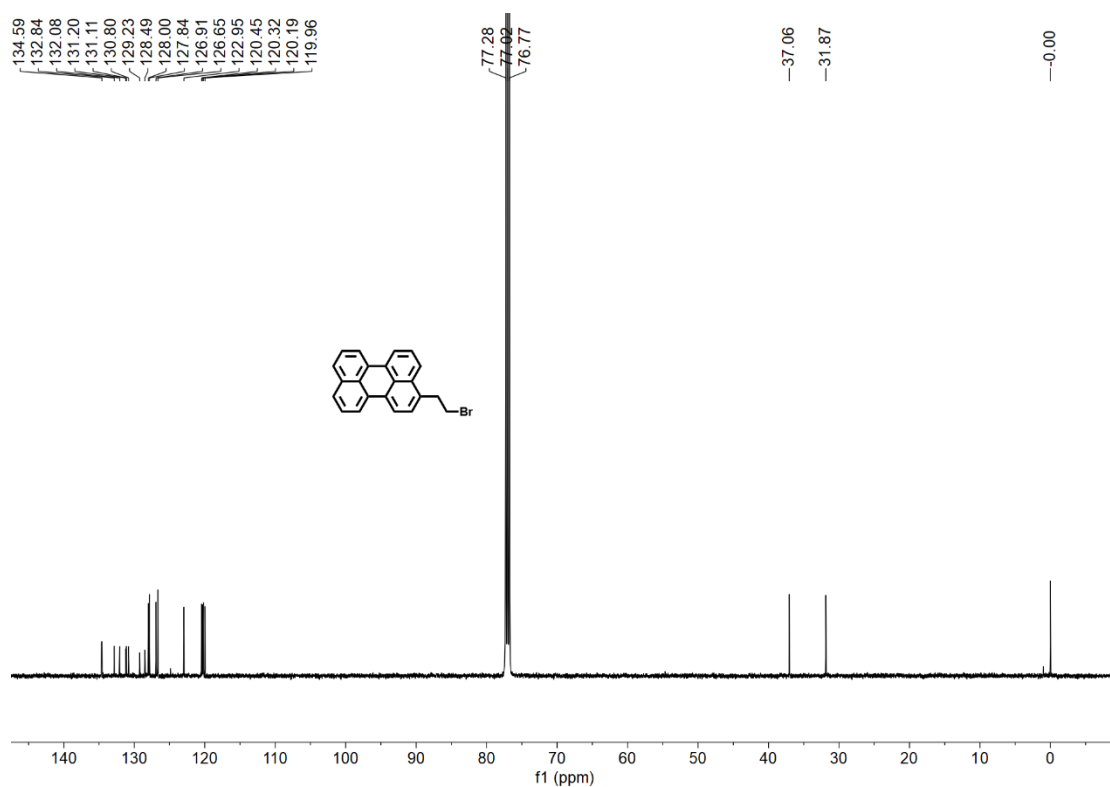
Supplementary Fig. 87 The ¹³C NMR spectrum of anthracene-based variant in DMSO-*d*₆.



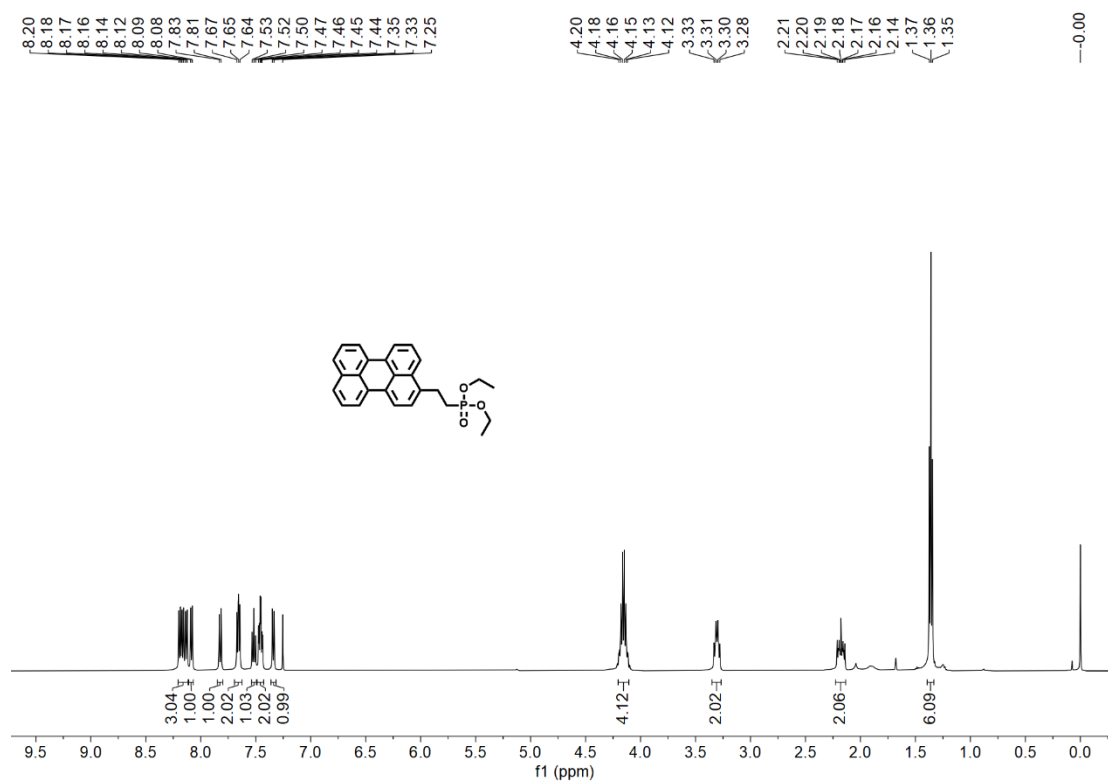
Supplementary Fig. 88 The ^{31}P NMR spectrum of anthracene-based variant in $\text{DMSO-}d_6$.



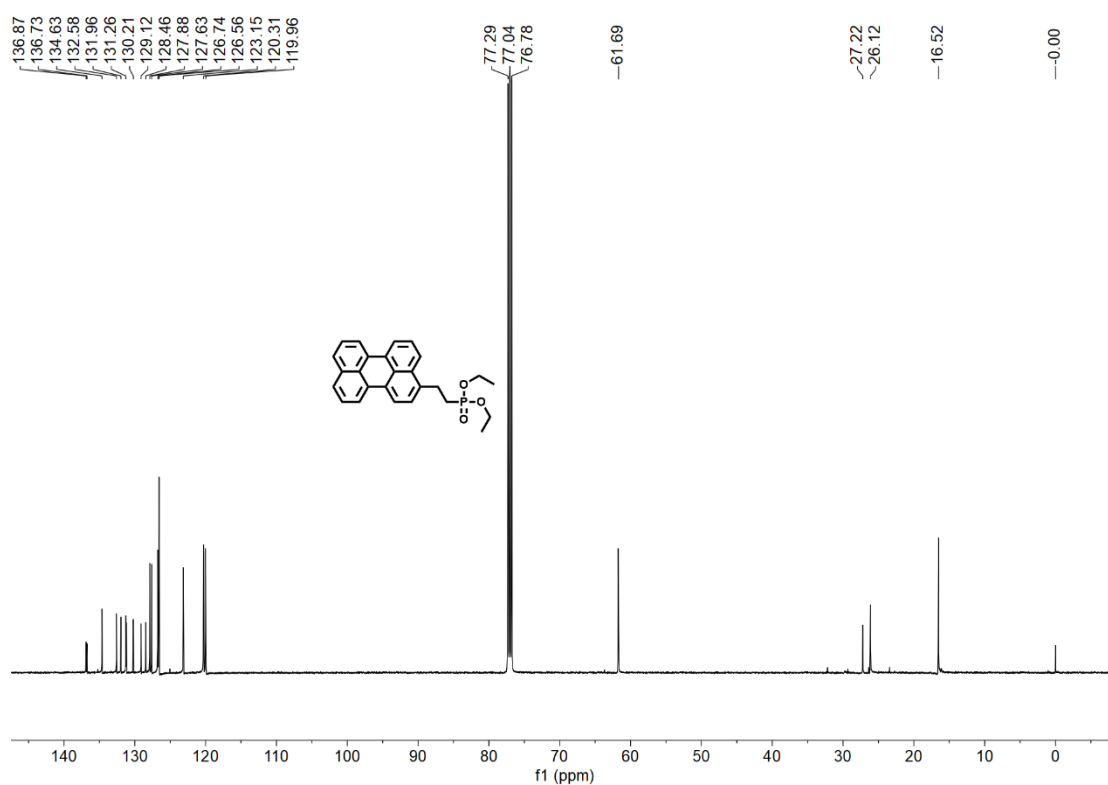
Supplementary Fig. 89 The ¹H NMR spectrum of M9 in CDCl₃.



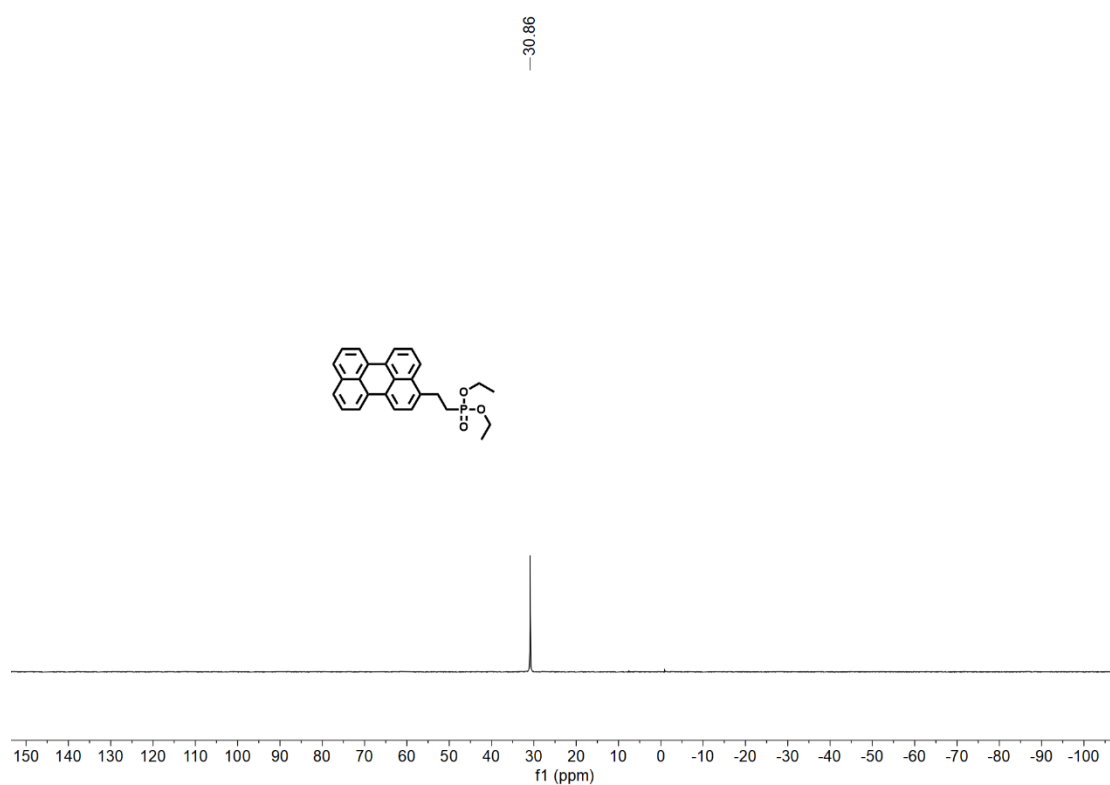
Supplementary Fig. 90 The ¹³C NMR spectrum of M9 in CDCl₃.



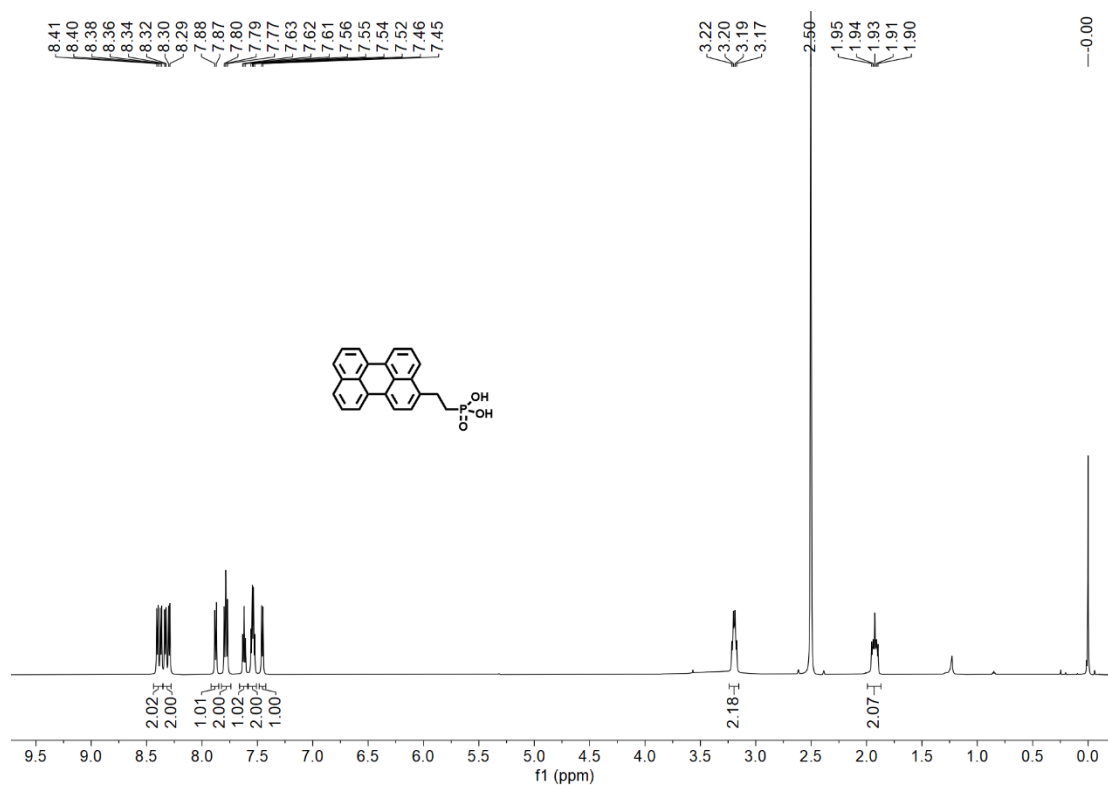
Supplementary Fig. 91 The ^1H NMR spectrum of M10 in CDCl_3 .



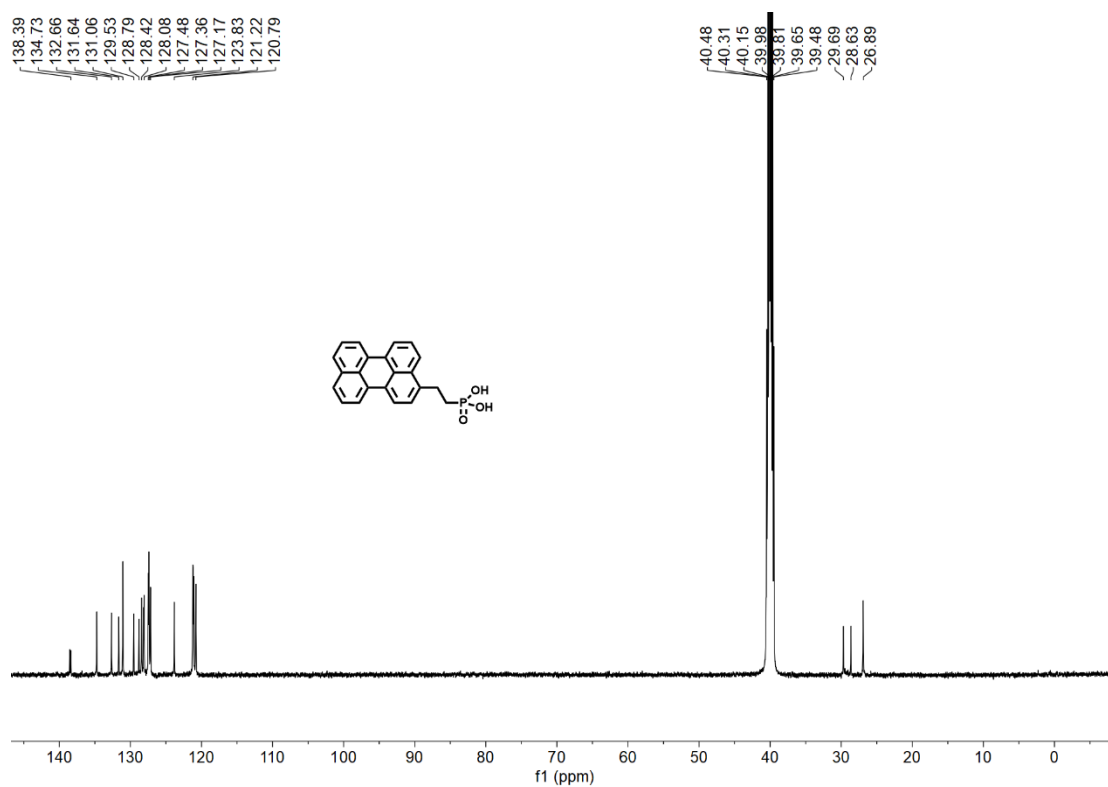
Supplementary Fig. 92 The ^{13}C NMR spectrum of M10 in CDCl_3 .



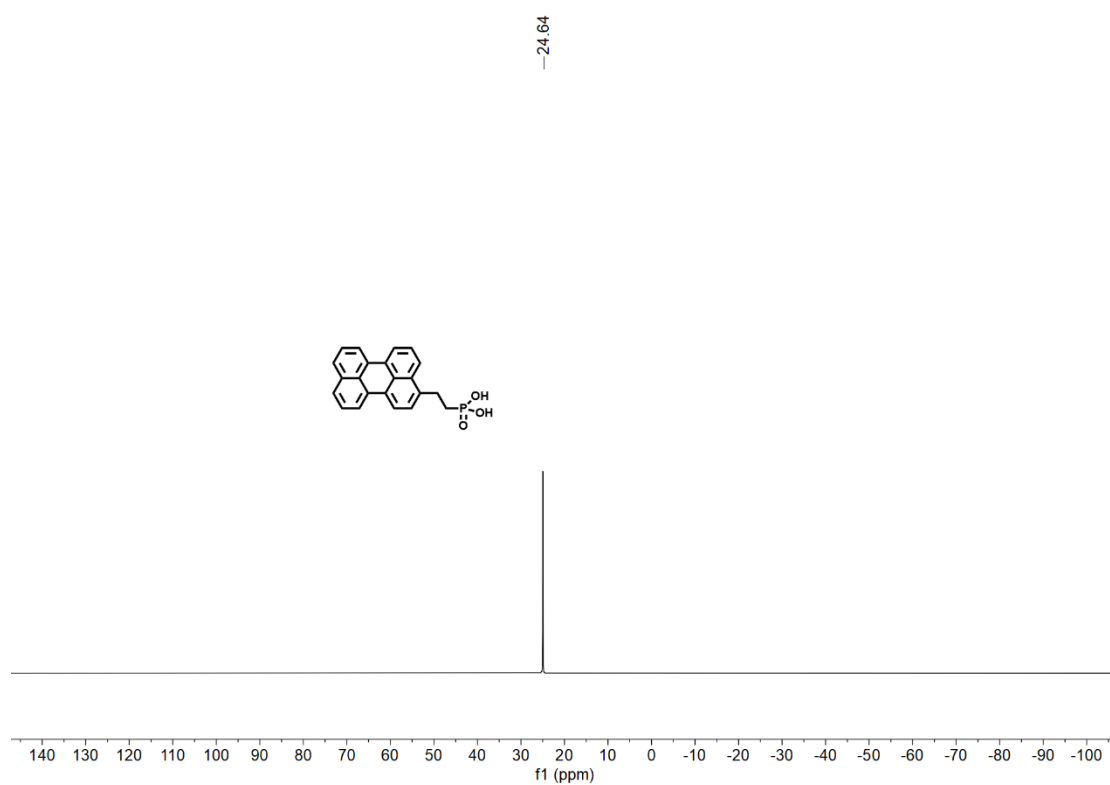
Supplementary Fig. 93 The ^{31}P NMR spectrum of M10 in CDCl_3 .



Supplementary Fig. 94 The ¹H NMR spectrum of perylene-based variant in DMSO-*d*₆.



Supplementary Fig. 95 The ¹³C NMR spectrum of perylene-based variant in DMSO-*d*₆.



Supplementary Fig. 96 The ^{31}P NMR spectrum of perylene-based variant in $\text{DMSO}-d_6$.

Table S1. Calculation of the synthetic cost of Py3.

Chemical (Brand)	Weight reagent (g/g)	Weight solvent (g/g)	Weight workup (g/g)	Price of chemical (US \$/kg)	Chemical cost (US\$)	Cost per step (US \$/step)
Pyrene (Psaitong)	3.03	53		211	0.64	1st step: 5.65 (3.38 g)
2-Bromoacetyl bromide (Adamas-Beta)	3.33			66	0.22	
AlCl ₃ (Macklin)	2.40			33	0.08	
CH ₂ Cl ₂ (ultra dry, Meryer)				21	1.1	
CH ₂ Cl ₂ (Greagent)			200	1.4	0.28	
Na ₂ SO ₄ (Psaitong)			2	6.9	0.01	
Petroleum ether (Greagent)			770	2.7	2.1	
CH ₂ Cl ₂ (Greagent)			399	1.4	0.56	
Silica gel (STEEMA)			150	4.4	0.66	
Product: M1						
M1 (used in the 2nd step)	3.38	40			5.65	2nd step: 10.52 (2.60)
Triethylsilane (Meryer)	2.56			94	0.24	
Trifluoroacetic acid (Anpel)				20	0.80	
CH ₂ Cl ₂ (Greagent)			200	1.4	0.28	
NaHCO ₃ (Bidepharm)			15	3.2	0.05	
Na ₂ SO ₄ (Psaitong)			2	6.9	0.01	
Petroleum ether (Greagent)			1000	2.7	2.7	
CH ₂ Cl ₂ (Greagent)			90	1.4	0.13	
Silica gel (STEEMA)			150	4.4	0.66	
Product: M2						

						g)
M2 (used in the 3rd step)	2.60				10.52	
Triethyl phosphite (Meryer)	10			15.8	0.16	
CH ₂ Cl ₂ (Greagent)			770	1.4	1.08	
MeOH (Meryer)			40	5.6	0.22	
Silica gel (STEEMA)			100	4.4	0.44	3rd step:
Product: M3						12.42 (2.77 g)
M3 (used in the 3rd step)	2.77				12.42	
Bromotrimethylsilane (Energy Chemical)	6			41	0.25	
1,4-dioxane (Adamas-Beta)		20		29	0.59	
MeOH (Macklin)			20	11	0.22	4th step:
Product: Py3						13.48 (1.88 g)
						=7.17 US\$/g

NOTE: The actual purchased price used here is usually 70% of the price shown in the website of vendors with 7.20 of the exchange rate of CNY against US dollar.

Table S2 Cost analysis of the first step of the synthetic route of MeO-2PACz.

Chemical (Brand)	Weight reagent (g/g)	Weight solvent (g/g)	Weight workup (g/g)	Price of chemical (US \$/kg)	Chemical cost (US\$)	Cost per step (US \$/step)
3,6-Dimethoxy-9H-carbazole (Macklin)	0.534			14500	7.74	
1,2-dibromoethane (TCI)	3.67			66.3	0.24	
Tetrabutylammonium bromide (Macklin)	0.08			19.1	0.002	
KOH (Macklin)	6			14.5	0.09	
Ethylacetate (Greagent)			80	2.4	0.19	
Na ₂ SO ₄ (Psaitong)			2	6.9	0.01	
Actone (SCRC)			10	6.7	0.07	
CH ₂ Cl ₂ (Greagent)			400	1.4	0.56	
Silica gel (STEEMA)			70	4.4	0.31	
Product: 9-(2-bromoethyl)-3,6-dimethoxy-9H-carbazole						1st step: 9.21 (0.352 g) =26.16 US\$/g

NOTE: The actual purchased price used here is usually 70% of the price shown in the website of vendors with 7.20 of the exchange rate of CNY against US dollar.

Table S3. Summary of photovoltaic parameters of (FA_{0.98}MA_{0.02})_{0.95}Cs_{0.05}Pb(I_{0.98}Br_{0.02})₃ PSCs based on 2PACz and Py3.

Molecule	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
2PACz	1.14±0.01	25.2±0.4	80.2±1.1	22.8±0.6
	1.14	25.3	80.0	23.1
Py3	1.17±0.02	25.7±0.3	84.5±1.0	25.4±0.9
	1.18	26.0	85.1	26.1

Note: The parameters listed in parentheses are the best performing devices. The plus-minus sign ($\pm$) represents the standard deviation.

Table S4. Summary of photovoltaic parameters of wide band PSCs based on 2PACz and Py3.

Molecule	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
2PACz	1.17±0.02	21.5±0.5	78.5±1.5	19.8±0.8
	1.19	21.9	79.8	20.6
Py3	1.19±0.04	21.6±0.3	82.9±1.3	21.6±0.9
	1.23	22.3	83.2	22.8

Table S5. Summary of photovoltaic parameters of 1-cm² PSCs based on 2PACz and Py3.

Molecule	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
2PACz	1.05±0.02	24.5±0.7	76.8±2.5	18.8±0.9
	1.04	24.8	75.9	19.6
Py3	1.13±0.02	24.9±0.4	81.1±1.7	22.7±0.8
	1.14	25.0	82.9	23.5

Table S6. Summary of photovoltaic parameters of FAPbI₃ PSCs based on 2PACz and Py3.

Molecule	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
2PACz	1.12±0.02	25.3±0.3	80.1±2.3	22.0±0.7
	1.13	25.3	79.1	22.5
Py3	1.14±0.02	25.4±0.3	84.2±1.6	23.8±0.8
	1.16	25.5	83.6	24.6

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